



Lebanese University
Faculty of Sciences



M1104

Analysis II

Department of Mathematics



Spring Semester 2022 ~ 2023

INTEGRAL CALCULUS AND SERIES

LEBANESE UNIVERSITY

February 27, 2020

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CHAPTER 1

INDEFINITE INTEGRAL

Integration is the inverse process of differentiation. In differentiation we are given a function and we are required to find its derivative, while in integration the derivative of some function is given and we are required to find that function. For example $F'(x) = 2x$, can we discover what the function $F(x)$ is? One function with this property is $F(x) = x^2$, or more generally $F(x) = x^2 + c$ where c is any constant.

In this chapter, I and J unless otherwise stated will denote non empty intervals of $\mathbb{R}$.

1.1 Primitive, Antiderivative and Indefinite Integral.

1.1 Definition. Let $F, f : I \mapsto \mathbb{R}$ be two functions defined on an interval I of $\mathbb{R}$. We say that F is a primitive of f on I if F is differentiable on I and $F'(x) = f(x)$ for all $x \in I$.

1.2 Remark. If I is any subset of $\mathbb{R}$, not necessarily interval, and $F'(x) = f(x)$ for all $x \in I$, then F is said to be an antiderivative of f on I .

1.3 Example. Let $F(x) = \ln|x|$ and $f(x) = \frac{1}{x}$. Then; (a) F is a primitive of f on $]0, \infty[$, (b) F is a primitive of f on $] - \infty, 0[$, (c) F is an antiderivative of f on $] - \infty, 0[\cup]0, \infty[$.

1.4 Remark. If $F(x)$ is a primitive of f on I , then for any constant $c \in \mathbb{R}$, we have $F(x) + c$ is also a primitive of f on I .

Proof. Since $(F(x) + c)' = (F(x))' = f(x)$ for all $x \in I$, then $F(x) + c$ is a primitive of f on I . $\square$

1.5 Property. If F and G are two primitives of f on I , then there exists $c \in \mathbb{R}$ such that $F(x) - G(x) = c$ for all $x \in I$.

Proof. We have $F'(x) = G'(x) = f(x)$ for all $x \in I$. Then $(F(x) - G(x))' = 0$ for all $x \in I$. Since I is an interval, then there exists $c \in \mathbb{R}$ such that $F(x) - G(x) = c$. $\square$

From this property one can deduce that; If F is a primitive of f on I , then the set of all primitives of f on I is $\{F(x) + c : c \in \mathbb{R}\}$.

1.6 Remark. The difference between two antiderivatives is not necessarily a constant. $F_1(x) = \arctan x$ and $F_2(x) = -\arctan \frac{1}{x}$ are two antiderivatives of $f(x) = \frac{1}{1+x^2}$ on $\mathbb{R}^*$, but $F_1(x) = F_2(x) + \frac{\pi}{2}$ if $x > 0$ and $F_1(x) = F_2(x) - \frac{\pi}{2}$ if $x < 0$.

1.7 Definition. The set $\{F(x) + c : c \in \mathbb{R}\}$ of all primitives of f on I is said to be an indefinite integral of the function f w.r.t x on I and it is denoted by $\int f(x) dx$ and we write

$$\int f(x) dx = F(x) + c.$$

The symbol $\int$ is the integral sign, $f(x)$ is called the integrand, x stands for variable of integration, c is called constant of integration and $\int f(x) dx$ is called integral (indefinite integral) of f w.r.t x . The presence of indefinite constant c justifies the name indefinite integral.

1.8 Property. Let $\lambda, \mu \in \mathbb{R}$. If f and g admit primitives F and G on I respectively, then $\lambda f + \mu g$ admits a primitive $\lambda F + \mu G$ on I and we write

$$\int [\lambda f(x) + \mu g(x)] dx = \lambda \int f(x) dx + \mu \int g(x) dx.$$

Proof. Since $[\lambda \int f(x) dx + \mu \int g(x) dx]' = [\lambda F(x) + \mu G(x)]' = \lambda f(x) + \mu g(x)$ for all $x \in I$, then $\int [\lambda f(x) + \mu g(x)] dx = \lambda \int f(x) dx + \mu \int g(x) dx$. $\square$

This property can be generalized to the form:

$$\int [\lambda_1 f_1(x) \pm \lambda_2 f_2(x) \pm \cdots \pm \lambda_n f_n(x)] dx = \lambda_1 \int f_1(x) dx \pm \lambda_2 \int f_2(x) dx \pm \cdots \pm \lambda_n \int f_n(x) dx.$$

1.9 Theorem. If $f : I \mapsto \mathbb{R}$ is a continuous function, then f admits a primitive on I .

Proof. We will see the proof in the next chapter. $\square$

Note that, there are non continuous functions which admit primitives. For example

$$F(x) = \begin{cases} x^2 \sin \frac{1}{x}, & \text{if } x \neq 0; \\ 0, & \text{if } x = 0. \end{cases} \text{ is differentiable on } \mathbb{R} \text{ but it is not of class } \mathcal{C}^1.$$

Also note that, even if a primitive for some function f exists, it may not be possible to find a primitive for f explicitly. For example $f(x) = e^{x^2}$.

1.2 Table of Elementary Integrals

Based on differentiation and definition of primitives, the table of derivatives of the main elementary functions can be read backwards as a list of primitives. The student is strongly advised to commit these results to memory, because no further progress is otherwise possible.

Table of primitives of: usual functions(1 – 9), fractional rational functions (10 – 11) and fractional irrational functions (12 – 14). Here $a \in \mathbb{R}^*$ and the case $a = 1$ is considerable.

$\int f(x) dx = F(x)$	conditions on x	I
1) $\int x^{-1} dx = \int \frac{1}{x} dx = \ln x $	$x \neq 0$	$] - \infty, 0[$ or $] 0, \infty[$
2) $\int x^\alpha dx = \frac{x^{\alpha+1}}{\alpha+1}$ i) $\alpha \in \mathbb{R} \setminus \{-1\}$ ii) $\alpha \in \mathbb{N}$ iii) $\alpha \in \{-2, -3, \dots\}$	$x > 0$ $\dots$ $x \neq 0$	$] 0, \infty[$ $\mathbb{R}$ $] - \infty, 0[$ or $] 0, \infty[$
3) $\int \cos x dx = \sin x$	$\dots$	$\mathbb{R}$
4) $\int \sin x dx = -\cos x$	$\dots$	$\mathbb{R}$
5) $\int \frac{1}{\cos^2 x} dx = \int (1 + \tan^2 x) dx = \tan x$	$x \in \mathbb{R} \setminus \{\frac{\pi}{2} + k\pi; k \in \mathbb{Z}\}$	$] - \frac{\pi}{2} + k\pi, \frac{\pi}{2} + k\pi[, k \in \mathbb{Z}$
6) $\int e^x dx = e^x$	$\dots$	$\mathbb{R}$
7) $\int \cosh x dx = \sinh x$	$\dots$	$\mathbb{R}$
8) $\int \sinh x dx = \cosh x$	$\dots$	$\mathbb{R}$
9) $\int \frac{1}{\cosh^2 x} dx = \int (1 - \tanh^2 x) dx = \tanh x$	$\dots$	$\mathbb{R}$
10) $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \arctan \frac{x}{a}$	$\dots$	$\mathbb{R}$
11) $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left \frac{x-a}{x+a} \right $	$x \neq \pm a$	$] \infty, - a [$ or $] - a , a [$ or $] a , \infty[$
12) $\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin \frac{x}{ a }$	$ x < a $	$] - a , a [$
13) $\int \frac{dx}{\sqrt{a^2 + x^2}} = \arg \sinh \frac{x}{a} = \ln(x + \sqrt{a^2 + x^2})$	$\dots$	$\mathbb{R}$
14) $\int \frac{dx}{\sqrt{x^2 - a^2}} = \ln x + \sqrt{x^2 - a^2} $	$ x > a $	$] \infty, - a [$ or $] a , \infty[$

1.3 Integration by Substitution

1.10 Theorem. Assume $f : I \mapsto \mathbb{R}$ admits a primitive (continuous) on I and $u : J \mapsto \mathbb{R}$ is differentiable on J such that $u(J) \subseteq I$.

If the variable t in the integral $\int f(u(t))u'(t) dt$ be changed to x by putting $u(t) = x$, then

$$\int f(u(t))u'(t) dt = \int f(x) dx \Big|_{x=u(t)}.$$

If in addition u is strictly monotone, then

$$\int f(x) dx = \int f(u(t))u'(t) dt \Big|_{t=u^{-1}(x)}.$$

Proof. Let F be a primitive of f on I . Then $\int f(x) dx \Big|_{x=u(t)} = F(u(t))$. Hence $\frac{d}{dt} \left[\int f(x) dx \Big|_{x=u(t)} \right] = \frac{d}{dt} F(u(t)) = F'(u(t))u'(t) = f(u(t))u'(t)$. Thus $\int f(u(t))u'(t) dt = \int f(x) dx \Big|_{x=u(t)}$. If in addition

u is strictly monotone, then

$$\int f(u(t))u'(t) dt \Big|_{t=u^{-1}(x)} = \left[\int f(x) dx \Big|_{x=u(t)} \right]_{t=u^{-1}(x)} = F(u(t)) \Big|_{t=u^{-1}(x)} = F(x) = \int f(x) dx.$$

□

1.11 Examples. To calculate

1. $\int \frac{\sin(\ln t)}{t} dt$, we put $\ln t = x$ and $\frac{1}{t} dt = dx$. Hence

$$\int \frac{\sin(\ln t)}{t} dt = \int \sin x dx = -\cos x + c = -\cos(\ln t) + c.$$

2. $\int \frac{2t+5}{t^2+5t+3} dt$, we put $t^2+5t+3 = x$ and $(2t+5)dt = dx$. Hence

$$\int \frac{2t+5}{t^2+5t+3} dt = \int \frac{1}{x} dx = \ln|x| + c = \ln|t^2+5t+3| + c.$$

3. $\int (6t+15)(t^2+5t+3) dt$, we put $t^2+5t+3 = x$ and $(2t+5)dt = dx$. Hence

$$\int (6t+15)(t^2+5t+3) dt = 3 \int (2t+5)(t^2+5t+3) dt = 3 \int x dx = \frac{3x^2}{2} + c = \frac{3(t^2+5t+3)^2}{2} + c.$$

4. $\int te^{1-t^2} dt$, we put $1-t^2 = x$ and $-2tdt = dx$. Hence

$$\int te^{1-t^2} dt = \frac{-1}{2} \int -2te^{1-t^2} dt = \frac{-1}{2} \int e^x dx = \frac{-1}{2} e^x + c = \frac{-1}{2} e^{1-t^2} + c.$$

5. $\int \sin^3 t \cos t dt$, we put $\sin t = x$ and $\cos t dt = dx$. Hence

$$\int \sin^3 t \cos t dt = \int x^3 dx = \frac{x^4}{4} + c = \frac{\sin^4 t}{4} + c.$$

6. $\int \sqrt{1-x^2} dx$, we put $x = \sin t = u(t)$ and $dx = \cos t dt$. We have $f(x) = \sqrt{1-x^2}$ is continuous on $I = [-1, 1]$ and u is differentiable and strictly monotone on $J = [-\frac{\pi}{2}, \frac{\pi}{2}]$ such that $u(J) = [-1, 1] \subseteq I$. Hence

$$\int \sqrt{1-x^2} dx = \int \sqrt{1-\sin^2 t} \cos t dt = \int |\cos t| \cos t dt = \int \cos^2 t dt = \frac{1}{2} \int (1+\cos(2t)) dt = \frac{1}{2} \left[t + \frac{1}{2} \sin(2t) \right] + c = \frac{1}{2} [t + \sin t \cos t] + c = \frac{1}{2} [\arcsin x + x\sqrt{1-x^2}] + c.$$

7. $\int \frac{1}{(1+x^2)\sqrt{1+x^2}} dx$, we put $x = \tan t = u(t)$ and $dx = (1+\tan^2 t)dt$. We have $f(x) = \frac{1}{(1+x^2)\sqrt{1+x^2}}$ is continuous on $I = \mathbb{R}$ and u is differentiable and strictly monotone on $J =]-\frac{\pi}{2}, \frac{\pi}{2}[$ such that $u(J) = \mathbb{R} \subseteq I$. Hence

$$\begin{aligned} \int \frac{1}{(1+x^2)\sqrt{1+x^2}} dx &= \int \frac{1+\tan^2 t}{(1+\tan^2 t)\sqrt{1+\tan^2 t}} dt = \int \frac{1}{\sqrt{1+\tan^2 t}} dt = \int \sqrt{\cos^2 t} dt = \\ &= \int |\cos t| dt = \int \cos t dt = \sin t + c = \sin(\arctan x) + c = \dots = \frac{x}{\sqrt{1+x^2}} + c. \end{aligned}$$

1.4 Integration by Parts

1.12 Theorem. Let u and v be two differentiable functions over I . If $u'v$ admits a primitive on I , then so is uv' and

$$\int u(x)v'(x) dx = u(x)v(x) - \int u'(x)v(x) dx.$$

Proof. Let H be any primitive of $u'v$ on I . Then

$$[u(x)v(x) - H(x)]' = u'(x)v(x) + u(x)v'(x) - u'(x)v(x) = u(x)v'(x).$$

Thus $\int u(x)v'(x) dx = u(x)v(x) - \int u'(x)v(x) dx$. □

We can apply the integration by parts formula if one of the functions is differentiable and the other one is of class C^1 on I .

How to choose u and v' :

- Whatever you let v' be, you need to be able to find v .
- It helps if $u'v$ is simpler than uv' .

1.13 Examples. To calculate

1. $\int \ln x dx$, we put $u(x) = \ln x$ and $v'(x) = 1$. Then $u'(x) = \frac{1}{x}$ and $v(x) = x$. Hence

$$\int \ln x dx = x \ln x - \int \frac{1}{x} x dx = x \ln x - x + c.$$

2. $\int x^2 \ln x dx$, we put $u(x) = \ln x$ and $v'(x) = x^2$. Then $u'(x) = \frac{1}{x}$ and $v(x) = \frac{x^3}{3}$. Hence

$$\int x^2 \ln x dx = \frac{x^3}{3} \ln x - \int \frac{1}{x} \frac{x^3}{3} = \frac{x^3}{3} \ln x - \frac{x^3}{9} + c.$$

3. $\int xe^x dx$, we put $u(x) = x$ and $v'(x) = e^x$. Then $u'(x) = 1$ and $v(x) = e^x$. Hence

$$\int xe^x dx = xe^x - \int e^x dx = xe^x - e^x + c.$$

Tabular Method: If $f(x)$ is $n + 1$ times differentiable (polynomial of degree n) and $g(x)$ is one of the following functions e^{ax} , $\sin(ax)$, $\cos(ax)$, $\sinh(ax)$, $\cosh(ax)$ where $a \in \mathbb{R}^*$ or any function has known $n + 1$ successive primitives $g_1(x), \dots, g_{n+1}(x)$, then

$$\int f(x)g(x) dx = f(x)g_1(x) - f'(x)g_2(x) + \dots + (-1)^n f^{(n)}(x)g_{n+1}(x) + (-1)^{n+1} \int f^{(n+1)}(x)g_{n+1}(x) dx.$$

1.5 Integration of Rational Functions

If the integrand $f(x) = \frac{N(x)}{D(x)}$ is a rational function whose numerator $N(x)$ and denominator $D(x)$ are polynomials and degree of numerator is greater than or equal degree of denominator, then divide the numerator by the denominator and then use the rule:

$$\frac{\text{Numerator}}{\text{Denominator}} = \text{Quotient} + \frac{\text{Remainder}}{\text{Denominator}} \Leftrightarrow \frac{N(x)}{D(x)} = Q(x) + \frac{R(x)}{D(x)}$$

with $R(x)$ a polynomial of degree less than degree of $D(x)$. Therefore

$$\int \frac{N(x)}{D(x)} dx = \int Q(x) dx + \int \frac{R(x)}{D(x)} dx.$$

The problem boils down to integrating a rational map $\frac{R(x)}{D(x)}$ in which the numerator's degree is less than the denominator's.

1.14 Example. $\frac{4x^2 + 4x + 1}{x - 2} = (4x + 12) + \frac{25}{x - 2}.$

From Algebra, $\frac{R(x)}{D(x)}$ can be decomposed uniquely as linear combinations of simple elements of the form $\frac{1}{(x - a)^n}$ and/or $\frac{ax + b}{(x^2 + px + q)^n}$ with $\Delta = p^2 - 4q < 0$. We discuss the possible situations which will turn out to be fundamental for treating the generic integrand.

(a) **Primitive of simple elements of first kind:** $\frac{1}{(x - a)^n}.$

If $n = 1$, then $\int \frac{1}{(x - a)} dx = \ln |x - a| + c.$

If $n \geq 2$, then $\int \frac{1}{(x - a)^n} dx = \int (x - a)^{-n} d(x - a) = \frac{1}{(1 - n)(x - a)^{n-1}} + c.$

(b) **Primitive of simple elements of second kind:** $\frac{ax + b}{(x^2 + px + q)^n}$ with $\Delta = p^2 - 4q < 0$.

Note that $\frac{ax + b}{(x^2 + px + q)^n} = \frac{a}{2} \frac{2x + p}{(x^2 + px + q)^n} + \frac{b - \frac{ap}{2}}{(x^2 + px + q)^n}$. For the first fraction

$$\int \frac{2x + p}{(x^2 + px + q)^n} dx = \int \frac{d(x^2 + px + q)}{(x^2 + px + q)^n} = \begin{cases} \ln(x^2 + px + q) + c & \text{If } n = 1, \\ \frac{(x^2 + px + q)^{-n+1}}{-n+1} + c & \text{if } n \geq 2. \end{cases} \quad \text{For the second}$$

one, let $I_n = \int \frac{1}{(x^2 + px + q)^n} dx$. Note that

$$x^2 + px + q = x^2 + 2\frac{p}{2}x + \frac{p^2}{4} - \frac{p^2}{4} + q = (x + \frac{p}{2})^2 + \frac{4q - p^2}{4} = t^2 + h^2,$$

where $t = x + \frac{p}{2}$ and $h = \sqrt{\frac{4q - p^2}{4}}$. Hence $I_n = \int \frac{1}{(t^2 + h^2)^n} dt$. We can evaluate I_n by two ways:

First way: we put $t = h \tan \theta$ and $dt = h(1 + \tan^2 \theta) d\theta$. Then

$$I_n = \frac{1}{h^{2n-1}} \int \frac{1}{(1 + \tan^2 \theta)^{n-1}} d\theta = \int \cos^{2n-2} \theta d\theta.$$

We compute $\int \cos^{2n-2} \theta d\theta$ and we replace θ by $\arctan \frac{t}{h} = \arctan \frac{x + \frac{p}{2}}{\sqrt{\frac{4q - p^2}{4}}}$.

Second way: We compute I_1 , then we find a recurrent relation between I_n and I_{n+1} for $n \geq 2$.

- $I_1 = \int \frac{dt}{t^2 + h^2} = \frac{1}{h} \arctan \frac{t}{h} + c = \frac{1}{\sqrt{\frac{4q-p^2}{4}}} \arctan \frac{x + \frac{p}{2}}{\sqrt{\frac{4q-p^2}{4}}} + c.$
- To find a relation between I_n and I_{n+1} , we integrate I_n by parts.

$$\begin{aligned}
I_n &= \int \frac{1}{(t^2 + h^2)^n} dt \quad (\text{put } u = \frac{1}{(t^2 + h^2)^n} \text{ and } v' = 1) \\
&= \frac{t}{(t^2 + h^2)^n} + 2n \int \frac{t^2}{(t^2 + h^2)^{n+1}} dt \\
&= \frac{t}{(t^2 + h^2)^n} + 2n \int \frac{t^2 + h^2 - h^2}{(t^2 + h^2)^{n+1}} dt \\
&= \frac{t}{(t^2 + h^2)^n} + 2n(I_n - h^2 I_{n+1}).
\end{aligned}$$

Hence, $I_{n+1} = \frac{1}{2nh^2} \left[(2n-1)I_n + \frac{t}{(t^2 + h^2)^n} \right]$. Finally, replace t by $x + \frac{p}{2}$ and h by $\sqrt{\frac{4q-p^2}{4}}$.

Here is a version of the Fundamental Theorem of Algebra and the partial fraction decomposition.

1.15 Theorem.

1. A polynomial $D(x)$ of degree d with real coefficients decomposes uniquely as a product

$$D(x) = \alpha(x - \alpha_1)^{r_1} \cdots (x - \alpha_m)^{r_m} (x^2 + p_1x + q_1)^{s_1} \cdots (x^2 + p_nx + q_n)^{s_n}$$

where $\alpha, \alpha_i, p_j, q_j$ are real and r_i, s_j are integers such that $r_1 + \cdots + r_m + 2s_1 + \cdots + 2s_n = d$. The real roots α_i of $D(x)$ are distinct and counted with multiplicity r^i . The factors $x^2 + p_jx + q_j$ are pairwise distinct and irreducible over $\mathbb{R}$ i.e. $\Delta = p_j^2 - 4q_j < 0$.

2. $\frac{R(x)}{D(x)} = \frac{R(x)}{\alpha(x - \alpha_1)^{r_1} \cdots (x - \alpha_m)^{r_m} (x^2 + p_1x + q_1)^{s_1} \cdots (x^2 + p_nx + q_n)^{s_n}}$ can be decomposed uniquely as sum of partial fractions

$$\frac{R(x)}{D(x)} = \frac{1}{\alpha} [L_1(x) + \cdots + L_m(x) + Q_1(x) + \cdots + Q_n(x)] \quad (1.1)$$

where each L_i takes the form

$$L_i(x) = \frac{A_{i1}}{(x - \alpha_i)} + \frac{A_{i2}}{(x - \alpha_i)^2} + \cdots + \frac{A_{ir_i}}{(x - \alpha_i)^{r_i}}$$

and each Q_j takes the form

$$Q_j(x) = \frac{B_{j1}x + C_{j1}}{(x^2 + p_jx + q_j)} + \frac{B_{j2}x + C_{j2}}{(x^2 + p_jx + q_j)^2} + \cdots + \frac{B_{js_j}x + C_{js_j}}{(x^2 + p_jx + q_j)^{s_j}}$$

Proof. admitted. □

To recover the undetermined coefficients $A_{i\ell}, B_{j\mu}$ and $C_{j\mu}$ we multiply equation (1.1) by $D(x)$ both sides and we take $x = \alpha_i$ to obtain A_{ir_i} , for the other coefficients we use the following rules: Two polynomials of degree d coincide if and only if one of the following conditions holds

- The coefficients of corresponding monomials coincide.
- The polynomials assume the same values at $d + 1$ distinct points.

Once these coefficients have been determined, we can start integrating the right-hand side of (1.1) and rely on the fundamental cases above. We illustrate this technique by a few examples.

1.16 Examples.

1. (**Non repeated linear factors in $D(x)$**) Let us compute $\int \frac{2x^3 + x^2 - 4x + 7}{x^2 + x - 2} dx$.

The numerator has greater degree than the denominator, so by Euclidean division we obtain

$$\frac{2x^3 + x^2 - 4x + 7}{x^2 + x - 2} = (2x - 1) + \frac{x + 5}{x^2 + x - 2}.$$

The denominator factorizes as $x^2 + x - 2 = (x - 1)(x + 2)$ non repeated linear factors, hence

$$\frac{x + 5}{x^2 + x - 2} = \frac{x + 5}{(x - 1)(x + 2)} = \frac{a}{x - 1} + \frac{b}{x + 2}.$$

That is,

$$x + 5 = a(x + 2) + b(x - 1). \quad (1.2)$$

To find the coefficients a and b , we compute (1.2) at the zero's $x = 1, x = -2$ of $D(x)$. For $x = 1$ we obtain $a = 2$ and for $x = -2$ we obtain $b = -1$.

A general way to find the coefficients, we compare the coefficients of corresponding monomials. Equation (1.2) can be written

$$x + 5 = (a + b)x + (2a - b).$$

Comparing coefficients yields the linear system

$$\begin{cases} a + b = 1, \\ 2a - b = 5, \end{cases}$$

solved by $a = 2$ and $b = -1$. Therefore

$$\int \frac{2x^3 + x^2 - 4x + 7}{x^2 + x - 2} dx = \int \left[(2x - 1) + \frac{2}{x - 1} - \frac{1}{x + 2} \right] dx = x^2 - x + 2 \ln |x - 1| - \ln |x + 2| + c.$$

2. (**Repeated linear factors in $D(x)$**) Let us determine a primitive of $f(x) = \frac{x^2 + 2x}{(x - 1)^3(x - 2)^2}$.

The numerator has less degree than the denominator. We must search for the reals a, b, c, d and e such that

$$\frac{x^2 + 2x}{(x - 1)^3(x - 2)^2} = \frac{a}{x - 1} + \frac{b}{(x - 1)^2} + \frac{c}{(x - 1)^3} + \frac{d}{x - 2} + \frac{e}{(x - 2)^2},$$

or,

$$x^2 + 2x = a(x - 1)^2(x - 2)^2 + b(x - 1)(x - 2)^2 + c(x - 2)^2 + d(x - 1)^3(x - 2) + e(x - 1)^3. \quad (1.3)$$

Now, $x = 1$ implies $c = 3$ and $x = 2$ implies $e = 8$. To find a, b and d we take three values for $x \neq 1, 2$. For instance $x = 0, 3, -1$ gives the linear system

$$\begin{cases} 4a - 4b + 2d = -4, \\ 4a + 2b + 8d = -52, \\ 36a - 18b + 24d = 36, \end{cases}$$

solved by $a = 18, b = 10$ and $d = -18$. Therefore

$$\begin{aligned}\int f(x) dx &= \int \left[\frac{18}{x-1} + \frac{10}{(x-1)^2} + \frac{3}{(x-1)^3} + \frac{-18}{x-2} + \frac{8}{(x-2)^2} \right] dx \\ &= 18 \ln |x-1| - \frac{10}{(x-1)} - \frac{3}{2(x-1)^2} - 18 \ln |x-2| - \frac{8}{(x-2)} + c.\end{aligned}$$

Another possibility to find a, b and d , we compare the coefficients of corresponding monomials in (1.3) to obtain a linear system in three equations with three unknowns.

3. **(Irreducible quadratic factors in $D(x)$)** Let us determine the partial fraction decomposition of $\frac{x^4 + 2x^2}{(x^2 + x + 1)^3}$.

The numerator has less degree than the denominator. We must search for the reals a, b, c, d, e and f such that

$$\frac{x^4 + 2x^2}{(x^2 + x + 1)^3} = \frac{ax + b}{(x^2 + x + 1)} + \frac{cx + d}{(x^2 + x + 1)^2} + \frac{ex + f}{(x^2 + x + 1)^3}.$$

To find the reals a, b, c, d, e and f , we take six different values for x to obtain a linear system in six equations with six unknowns, or by comparing the coefficients of corresponding monomials.

4. **(Mixed factors in $D(x)$)** Let us determine the partial fraction decomposition of

$$\frac{5x^2}{(x+1)(x-2)(x+3)^2(x^2+4)^2}.$$

The numerator has less degree than the denominator. We must search for the reals a, b, c, d, e, f, g and h such that

$$\frac{5x^2}{\underbrace{(x+1)}_{\text{non rep}} \underbrace{(x-2)}_{\text{non rep}} \underbrace{(x+3)^2}_{\text{repeated}} \underbrace{(x^2+4)^2}_{\text{quadratic}}} = \frac{a}{x+1} + \frac{b}{x-2} + \frac{c}{x+3} + \frac{d}{(x+3)^2} + \frac{ex+f}{x^2+4} + \frac{gx+h}{(x^2+4)^2},$$

or,

$$\begin{aligned}5x^2 &= a(x-2)(x+3)^2(x^2+4)^2 + b(x+1)(x+3)^2(x^2+4)^2 \\ &\quad + c(x+1)(x-2)(x+3)(x^2+4)^2 + d(x+1)(x-2)(x^2+4)^2 \\ &\quad + (ex+f)(x+1)(x-2)(x+3)^2(x^2+4) + (gx+h)(x+1)(x-2)(x+3)^2.\end{aligned}\quad (1.4)$$

If we compute (1.4) at $x = -1, 2, -3$, we obtain a, b and d . The other coefficients c, e, f, g and h can be determined by comparing the coefficients of corresponding monomials or by taking five different values of x other than $-1, 2, -3$.

Note that many functions that are not rational can be transformed into a rational function in a new variable. The table below included some special cases: Let F be a rational function in many variables,

$\int F dx$	new variable t
$\int F(x, x^{\frac{m}{n}}, \dots, x^{\frac{r}{s}}) dx$	$x = t^k, k = LCM\{n, \dots, s\}$
$\int F(x, (\frac{ax+b}{cx+d})^{\frac{m}{n}}, \dots, (\frac{ax+b}{cx+d})^{\frac{r}{s}}) dx$ $(a, b, c, d) \in \mathbb{R}^4$ with $ad - bc \neq 0$	$(\frac{ax+b}{cx+d}) = t^k, k = LCM\{n, \dots, s\}$ i.e k is a common denominator of $\frac{m}{n}, \dots, \frac{r}{s}$
$\int F(e^{ax}) dx, a \in \mathbb{R}^* \text{ (even } F \text{ is not rational)}$	$t = e^{ax}, x = \frac{1}{a} \ln t \text{ and } dx = \frac{1}{at} dt$
$\int F(\sin x, \cos x) dx$	$t = \tan \frac{x}{2}, x = 2 \arctan t \text{ and } dx = \frac{2}{1+t^2} dt$ $\sin x = \frac{2t}{1+t^2} \text{ and } \cos x = \frac{1-t^2}{1+t^2}$
$\int F(\sinh x, \cosh x) dx$	$t = \tanh \frac{x}{2}, x = 2 \arg \tanh t \text{ and } dx = \frac{2}{1-t^2} dt$ $\sinh x = \frac{2t}{1-t^2} \text{ and } \cosh x = \frac{1+t^2}{1-t^2}$
$\int F(\sin^2 x, \cos^2 x, \tan x) dx$	$t = \tan x, x = \arctan t \text{ and } dx = \frac{1}{1+t^2} dt$ $\sin^2 x = \frac{t^2}{1+t^2} \text{ and } \cos x = \frac{1}{1+t^2}$
$\int F(x, \sqrt{ax^2 + bx + c}) dx, a \neq 0 \text{ and } b^2 - 4ac \neq 0$ First case $a > 0$ Second case $a < 0$ then $b^2 - 4ac > 0$, roots: α, β	$\sqrt{ax^2 + bx + c} = x\sqrt{a} + t$ $\sqrt{ax^2 + bx + c} = \sqrt{a(x - \alpha)(x - \beta)} = t(x - \alpha)$

1.6 Integrating of rational function F in some usual functions

1.6.1 Rational function in sine and cosine $\int F(\sin x, \cos x) dx$.

- (1) **Particular case.** If F is a polynomial, then by linearity of integral it is sufficient to know how to integrate

$$\int \cos^m x \sin^n x dx.$$

If m is odd, use the substitution $t = \sin x$.

If n is odd, use the substitution $t = \cos x$.

If $m, n \in \mathbb{Q}$ and $m + n$ is a negative even integer, use the substitution $t = \tan x$.

If m and n are even, or one is even and the other is zero, we transform $\cos^m x \sin^n x$ into linear combinations of $\sin(\alpha x)$ and $\cos(\beta x)$ by using Euler's formulae

$$\cos x = \frac{e^{ix} + e^{-ix}}{2} \text{ and } \sin x = \frac{e^{ix} - e^{-ix}}{2i},$$

or by using repeatedly the following formulae

- (i) $\cos(2a) = 2 \cos^2 a - 1.$
- (ii) $\cos(2a) = 1 - 2 \sin^2 a.$
- (iii) $\cos(3a) = 4 \cos^3 a - 3 \cos a.$
- (iv) $\sin(3a) = 3 \sin a - 4 \sin^3 a.$
- (v) $\sin a \cos b = \frac{1}{2} (\sin(a+b) + \sin(a-b)).$
- (vi) $\sin a \cos a = \frac{1}{2} \sin(2a).$
- (vii) $\cos a \cos b = \frac{1}{2} (\cos(a+b) + \cos(a-b)).$
- (viii) $\sin a \sin b = \frac{1}{2} (\cos(a-b) - \cos(a+b)).$

- (2) **Bioche rule.** Let $W(x) = F(\sin x, \cos x) dx$.

If $W(-x) = W(x)$, use the substitution $t = \cos x$.

If $W(\pi - x) = W(x)$, use the substitution $t = \sin x$.

If $W(\pi + x) = W(x)$, use the substitution $t = \tan x$.

- (3) **General case.** Always the change of variable $t = \tan \frac{x}{2}$ transform $\int F(\sin x, \cos x) dx$ into integral of rational function in t (see previous table).

1.6.2 Rational function in sine hyperbolic and cosine hyperbolic $\int F(\sinh x, \cosh x) dx$.

- (1) **Bioche rule.** Let $W(x) = F(\sinh x, \cosh x) dx$.

If $W(-x) = W(x)$, use the substitution $t = \cosh x$.

If $W(\pi - x) = W(x)$, use the substitution $t = \sinh x$.

If $W(\pi + x) = W(x)$, use the substitution $t = \tanh x$.

- (2) **General case.** Always the change of variable $t = \tanh \frac{x}{2}$ or $t = e^x$ transform $\int F(\sinh x, \cosh x) dx$ into integral of rational function in t (see previous table).

1.6.3 Rational function in x and $\sqrt{ax^2 + bx + c}$ i.e $\int F(x, \sqrt{ax^2 + bx + c}) dx$.

If $a \neq 0$ and $\Delta = b^2 - 4ac \neq 0$, by completing square method the term $\sqrt{ax^2 + bx + c}$ can be transformed to one of the following general forms $\sqrt{\alpha^2 - t^2}$, $\sqrt{\alpha^2 + t^2}$ or $\sqrt{t^2 - \alpha^2}$.

For $\sqrt{\alpha^2 - t^2}$, use the substitution $t = \alpha \sin \theta$ or $t = \alpha \cos \theta$ or $t = \alpha \tanh \theta$.

For $\sqrt{t^2 - \alpha^2}$, use the substitution $\epsilon t = \alpha \cosh \theta$ where $\epsilon = \text{sign}(\alpha t)$ or $t = \alpha \frac{1}{\cosh \theta}$.

For $\sqrt{\alpha^2 + t^2}$, use the substitution $t = \alpha \sinh \theta$ or $t = \alpha \tan \theta$.

1.7 Exercises

Exercise 1. Compute the indefinite integrals:

$$\begin{aligned}
 &1) \int \frac{e^x}{e^x + 1} dx, \quad 2) \int \frac{1}{e^x + 1} dx, \quad 3) \int \frac{1}{(1 + x^2) \arctan x} dx, \quad 4) \int \tan x dx, \\
 &5) \int \tan^2 x dx, \quad 6) \int \tan^4 x dx, \quad 7) \int \frac{1}{\tan^4 x} dx, \quad 8) \int \tan^3 x dx, \\
 &9) \int \frac{\sin^4 x}{\cos^6 x} dx, \quad 10) \int \frac{x}{\sqrt{1 + x^2}} dx, \quad 11) \int \frac{1}{x \ln |x|} dx, \quad 12) \int \frac{1}{x \ln x [\ln(|\ln x|)]} dx.
 \end{aligned}$$

Exercise 2. Compute the indefinite integrals:

$$\begin{aligned}
 &1) \int \arctan x dx, \quad 2) \int x \arctan x dx, \quad 3) \int (1 + x^2) e^{-2x} dx, \quad 4) \int (1 - x^3) \cos(-2x) dx, \\
 &5) \int (x^2 - 1) \sqrt{1 - x} dx, \quad 6) \int (x^{10} + x^4) \ln x dx, \quad 7) \int e^x \sin x dx, \quad 8) \int e^x \cos x dx, \\
 &9) \int x e^x \sin x dx, \quad 10) \int x \frac{\cos x}{\sin x^3} dx, \quad 11) \int \sin \sqrt{x} dx, \quad 12) \int \sin(\ln x) dx, \\
 &13) \int \sin 2x \cos 3x dx, \quad 14) \int x^3 \sin x dx, \quad 15) \int \arccos^3 x dx, \quad 16) \int \sin(2x) (\ln(1 + \cos^2 x))^2 dx.
 \end{aligned}$$

Exercise 3. Assume that $n \geq 2$ is an integer.

1. Prove the following reduction formulae:

$$\begin{aligned}
 (a) \quad &\int \sin^n x dx = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x dx. \\
 (b) \quad &\int \cos^n x dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x dx. \\
 (c) \quad &\int \tan^n x dx = \frac{1}{n-1} \tan^{n-1} x - \int \tan^{n-2} x dx. \\
 (d) \quad &\int \cot^n x dx = \frac{-1}{n-1} \cot^{n-1} x - \int \cot^{n-2} x dx. \\
 (e) \quad &\int \sec^n x dx = \frac{1}{n-1} \sec^{n-2} x \tan x + \frac{n-2}{n-1} \int \sec^{n-2} x dx. \\
 (f) \quad &\int \csc^n x dx = \frac{-1}{n-1} \csc^{n-2} x \cot x + \frac{n-2}{n-1} \int \csc^{n-2} x dx.
 \end{aligned}$$

You may deduce (b), (d), and (f) from (a), (c) and (e) respectively.

2. Evaluate: $\int \sin^4 x dx$, $\int \cos^4 x dx$, $\int \tan^4 x dx$, $\int \cot^4 x dx$, $\int \sec^4 x dx$ and $\int \csc^4 x dx$.

Exercise 4. Let $n \in \mathbb{N}$ and $I_n = \int \frac{dx}{(x^2 + 1)^n}$.

Find a relation between I_{n+1} and I_n , then evaluate I_2 .

Exercise 5. Let a, b, c and d be non zero real numbers. Assume that $I = \int \frac{\cos x}{c \cos x + d \sin x} dx$ and $J = \int \frac{\sin x}{c \cos x + d \sin x} dx$

1. Calculate $cI + dJ$ and $dI - cJ$, then deduce the values of I and J .

2. Deduce $\int \frac{a \cos x + b \sin x}{c \cos x + d \sin x} dx$.

Exercise 6. Consider the integrals $I = \int \sin x \sinh x \, dx$ and $J = \int \cos x \cosh x \, dx$

1. Find two relation between I and J , then deduce the values of I and J .
2. Deduce $\int \sin^2 x \sinh^2 x \, dx$.

Exercise 7. Give the general form of the partial fraction decomposition for the following rational functions.

1. $\frac{x^5 + 3x}{(x^2 - 2x)(x + 1)^2}$,
2. $\frac{1 + x}{(x^3 + 1)^3}$,
3. $\frac{1 + x}{(x^4 + 1)^2}$.

Exercise 8. Let $f(x) = \frac{x + 2}{(|x| + 3)(x - 3)}$.

1. Justify the existence of a primitive of f on $] - \infty, 3[$.
2. Give a primitive of f on $] - \infty, 3[$.
3. What is the primitive that vanishes at $x = 0$.

Exercise 9. Let $n \in \mathbb{N}^*$. Compute the indefinite integrals:

- 1) $\int \frac{1}{(x + 2)(x^2 - 1)} \, dx$,
- 2) $\int \frac{1}{(x - 1)^2(x^2 + 1)} \, dx$,
- 3) $\int \frac{x^2}{x^4 - 1} \, dx$,
- 4) $\int \frac{x - 1}{x^3 + 1} \, dx$,
- 5) $\int \frac{1}{x^3 - 1} \, dx$,
- 6) $\int \frac{x^4 + 1}{x^4 - 1} \, dx$,
- 7) $\int \frac{1}{x^4 + 1} \, dx$,
- 8) $\int \frac{x + 1}{x^2 + 4x + 8} \, dx$,
- 9) $\int \frac{e^{2x} + e^x}{e^{2x} + 4e^x + 8} \, dx$,
- 10) $\int \frac{x + 1}{1 - 6x - 9x^2} \, dx$,
- 11) $\int \frac{1}{x^4 + x^2 + 1} \, dx$,
- 12) $\int \frac{1}{x(x^n + 1)} \, dx$.

Exercise 10. Compute the indefinite integrals:

- 1) $\int \sin^5 x \cos^4 x \, dx$,
- 2) $\int \sin^4 x \cos^5 x \, dx$,
- 3) $\int \sin^3 x \cos^3 x \, dx$,
- 4) $\int \sin^2 x \cos^2 x \, dx$,
- 5) $\int \sin^2 x \cos^4 x \, dx$,
- 6) $\int \sin^2 x \cos^6 x \, dx$,
- 7) $\int \frac{1}{\sin^{\frac{3}{2}} x \cos^{\frac{5}{2}} x} \, dx$,
- 8) $\int \frac{1}{\sin x} \, dx$,
- 9) $\int \frac{1}{\sin 2x} \, dx$,
- 10) $\int \frac{1}{\cos^3 x} \, dx$,
- 11) $\int \frac{\sin^3 x}{2 + \cos x} \, dx$,
- 12) $\int \frac{1}{\cos x (\sin x - \cos x)} \, dx$,
- 13) $\int \frac{1}{2 + \sin x} \, dx$,
- 14) $\int \frac{1}{1 + \cos x} \, dx$,
- 15) $\int \frac{x \sin x}{(1 + \cos x)^2} \, dx$,
- 16) $\int \frac{\sinh^3 x}{2 + \cosh x} \, dx$,
- 17) $\int \frac{\tanh x \sinh x}{\sinh^3 x + \cosh^3 x} \, dx$,
- 18) $\int \frac{1}{(e^x + 1)^2} \, dx$,
- 19) $\int \sqrt{e^x + 1} \, dx$,
- 20) $\int \frac{x^2}{(x \sin x + \cos x)^2} \, dx$.

Exercise 11. Compute the indefinite integrals:

- 1) $\int \sqrt{x - x^2} \, dx$,
- 2) $\int \frac{3x + 1}{\sqrt{x^2 + x + 1}} \, dx$,
- 3) $\int \frac{x + 1}{\sqrt{x^2 - 3x + 2}} \, dx$,
- 4) $\int \frac{1}{x\sqrt{1 - x^2}} \, dx$,
- 5) $\int \frac{2x + 1}{\sqrt{-2x^2 + 4x + 8}} \, dx$,
- 6) $\int \frac{1}{(2x + 1)\sqrt{x^2 + x + 1}} \, dx$.

Exercise 12. Compute the indefinite integrals:

- 1) $\int \frac{1}{\sqrt{1 + x}} \, dx$,
- 2) $\int \frac{x}{\sqrt[4]{x^3 + 1}} \, dx$,
- 3) $\int \frac{\sqrt{x + 1} - \sqrt[3]{x + 1}}{\sqrt{x + 1} + \sqrt[3]{x + 1}} \, dx$,
- 4) $\int \sqrt{\frac{x - 1}{x - 2}} \, dx$,
- 5) $\int \frac{1}{x} \sqrt{\frac{1 + x}{1 - x}} \, dx$,
- 6) $\int \frac{1}{\sqrt{x + 1} + \sqrt{x}} \, dx$.

The Riemann idea of integration for a bounded function f over a closed bounded intervals $[a, b]$ is to assign a real number A , called the area to the region R bounded by the curve $x = a, x = b, y = 0$ and $y = f(x)$, where we assume that f is non-negative. The number A , the area under the curve, is called the integral of f over $[a, b]$ and denoted by the symbol $\int_a^b f(x) dx$. In this chapter we will answer the following questions:

1. How do Bernhard Riemann and Jean-Gaston Darboux define the area under the curve?
2. What conditions imposed on f to obtain it Riemann integrable over $[a, b]$?
3. What are the basic properties of Riemann integral?
4. How to calculate Riemann integral?

2.1 The Riemann Integral.

2.1 Definition. (Partitions, Upper And Lower Sums)

Let f be a bounded function on $[a, b]$.

- A **partition** P of $[a, b]$ is a set of points $\{x_k\}_{k=0}^n$ such that $a = x_0 < x_1 < x_2 < \cdots < x_n = b$.
- For each subinterval $[x_{k-1}, x_k]$ of $[a, b]$, set

$$m_k = \inf_{x \in [x_{k-1}, x_k]} f(x), \text{ and } M_k = \sup_{x \in [x_{k-1}, x_k]} f(x).$$
 - The **lower Darboux** sum of f with respect to P is the sum $L(f; P) = \sum_{k=1}^n m_k(x_k - x_{k-1})$.
 - The **upper Darboux** sum of f with respect to P is $U(f; P) = \sum_{k=1}^n M_k(x_k - x_{k-1})$.

Let $m = \inf_{x \in [a, b]} f(x)$ and $M = \sup_{x \in [a, b]} f(x)$. It is clear that, for any partition P of $[a, b]$ we have

$$m(b - a) \leq L(f; P) \leq U(f; P) \leq M(b - a).$$

Also, if $f \geq 0$, then the lower and the upper sums approximate the area under the curve from below and from above.

2.2 Examples.

1. $P = \{a = x_0, x_1 = b\}$ is the trivial partition of $[a, b]$

2. $P_n = \{x_i = a + \frac{i}{n}(b-a); 0 \leq i \leq n\}$ is the partition of $[a, b]$ that divides it into n subintervals with equal lengths.

2.3 Definition. (Refinement)

A partition Q is a **refinement** of a partition p if Q contains all of the points of P . In this case we write $P \subseteq Q$.

- 2.4 Property.** 1. If $P \subseteq Q$, then $L(f; P) \leq L(f; Q)$ and $U(f; P) \geq U(f; Q)$.
2. $L(f; P) \leq U(f; Q)$ for P, Q any partitions of $[a, b]$.

Proof. 1. Let $P = \{x_0, x_1, \dots, x_n\}$. Assume $Q = \{x_0, \dots, x_{k-1}, t, x_k, \dots, x_n\}$ is a refinement of P by adding a single point t to some subinterval $[x_{k-1}, x_k]$ of P . For $m'_k = \inf_{x \in [x_{k-1}, t]} f(x)$ and $m''_k = \inf_{x \in [t, x_k]} f(x)$ we have

$$\begin{aligned} m_k(x_k - x_{k-1}) &= m_k(x_k - t) + m_k(t - x_{k-1}) \\ &\leq m'_k(x_k - t) + m''_k(t - x_{k-1}). \end{aligned}$$

Then by induction, we obtain $L(f; P) \leq L(f; Q)$, and analogous argument holds for the other inequality.

2. Let P, Q be any partitions of $[a, b]$. Then $L(f; P) \leq L(f; P \cup Q) \leq U(f; P \cup Q) \leq U(f; Q)$. □

Note that $L(f; Q)$ and $U(f; Q)$ produces a better approximation to the area under the curve. This suggests that to get the real area we look at the **upper integral** $U(f) = \inf\{U(f; P); P \text{ is a partition of } [a, b]\}$ and the **lower integral** $L(f) = \sup\{L(f; P); P \text{ is a partition of } [a, b]\}$.

2.5 Corollary. If f is a bonded function on $[a, b]$, then $L(f) \leq U(f)$.

Proof. Let Q be a partition of $[a, b]$, then $L(f) \leq U(f; Q)$. It follows that $L(f) \leq U(f)$. □

Property 2.4 suggests the following definition for integrability

2.6 Definition. (Riemann Integrability In Sense Of Darboux)

Let $f : [a, b] \mapsto \mathbb{R}$ be a bounded function. We say that f is **Riemann integrable** on $[a, b]$ and we write $f \in \mathcal{R}([a, b])$ if $L(f) = U(f)$. In this case we define $\int_a^b f$ or $\int_a^b f(x) dx$ to be this common value, that is $\int_a^b f(x) dx = L(f) = U(f)$.

2.7 Examples.

1. Let $f : [a, b] \mapsto \mathbb{R}$ be a constant function with $f(x) = c$ for all x . We claim that $f \in \mathcal{R}([a, b])$ and $\int_a^b f(x) dx = c(b - a)$. Let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$, then $L(f; P) = \sum_{k=1}^n m_k(x_k - x_{k-1}) = \sum_{k=1}^n c(x_k - x_{k-1}) = c(b - a)$ and $U(f; P) = \sum_{k=1}^n M_k(x_k - x_{k-1}) = \sum_{k=1}^n c(x_k - x_{k-1}) = c(b - a)$. Hence $L(f) = U(f) = c(b - a)$. Thus $f \in \mathcal{R}([a, b])$ and we have $\int_a^b f(x) dx = L(f) = U(f) = c(b - a)$.
2. Let $f : [0, 1] \mapsto \mathbb{R}$ be defined by $f(x) = 1$ if $0 \leq x < 1$ and $f(1) = 2$. We claim that $f \in \mathcal{R}([0, 1])$ and $\int_0^1 f(x) dx = 1$. Let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$, then $m_k = 1$ for all $1 \leq k \leq n$ and $M_k = 1$ for $1 \leq k < n$ and $M_n = 2$. Hence $L(f; P) = \sum_{k=1}^n m_k(x_k - x_{k-1}) = \sum_{k=1}^n (x_k - x_{k-1}) = 1$ and $U(f; P) = \sum_{k=1}^n M_k(x_k - x_{k-1}) = \sum_{k=1}^{n-1} M_k(x_k - x_{k-1}) + M_n(x_n - x_{n-1}) = \sum_{k=1}^{n-1} (x_k - x_{k-1}) + 2(1 - x_{n-1}) = x_{n-1} + 2(1 - x_{n-1})$. It follows that $L(f) = U(f) = 1$, and $f \in \mathcal{R}([0, 1])$ and we have $\int_0^1 f(x) dx = 1$.
3. Let $f : [0, 1] \mapsto \mathbb{R}$ be defined by $f(x) = 1$ if $x \in \mathbb{Q}$ and zero otherwise. We claim that $f \notin \mathcal{R}([0, 1])$. By the density of rationales and irrationals in $\mathbb{R}$, we obtain $m_k = 0$ and $M_k = 1$ for all $1 \leq k \leq n$. It follows that $L(f; P) = 0$, $U(f; P) = 1$ and $L(f) = 0 \neq U(f) = 1$. Thus f is not integrable on $[0, 1]$.

2.8 Theorem. (Criteria For Integrability)

A bounded function f is integrable on $[a, b]$ if and only if for every $\varepsilon > 0$, there exists a partition P_ε such that $U(f; P_\varepsilon) - L(f; P_\varepsilon) < \varepsilon$.

Proof. Let $\varepsilon > 0$ and P_ε be a partition of $[a, b]$ such that $U(f; P_\varepsilon) - L(f; P_\varepsilon) < \varepsilon$. Then $U(f) - L(f) \leq U(f; P_\varepsilon) - L(f; P_\varepsilon) < \varepsilon$. It follows $U(f) = L(f)$ and $f \in \mathcal{R}([a, b])$.

For the proof of the converse statement: Assume that $f \in \mathcal{R}([a, b])$ and let $\varepsilon > 0$, then by definitions of inf and sup there exist two partitions P_1 and P_2 of $[a, b]$ such that

$$U(f; P_1) < U(f) + \frac{\varepsilon}{2}, \text{ and } L(f; P_2) > L(f) - \frac{\varepsilon}{2}.$$

Let $P_\varepsilon = P_1 \cup P_2$ be the common refinement and since $L(f) = U(f)$, we obtain

$$U(f; P_\varepsilon) - L(f; P_\varepsilon) \leq U(f; P_1) - L(f; P_2) \leq (U(f; P_1) - U(f)) + (L(f) - L(f; P_2)) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

2.9 Remark. Refining a partition can bring the upper and lower sums closer together. That is for $P_\varepsilon \subseteq Q_\varepsilon$, we have $U(f; Q_\varepsilon) - L(f; Q_\varepsilon) \leq U(f; P_\varepsilon) - L(f; P_\varepsilon) < \varepsilon$.

2.10 Corollary. (Sequential Criterion For Integrability)

A bounded function f is integrable on $[a, b]$ if and only if there exists a sequence of partitions (P_n) satisfying $\lim_{n \rightarrow \infty} [U(f; P_n) - L(f; P_n)] = 0$.

2.11 Property. (Sequential Criterion For Integrability)

A bounded function f is integrable on $[a, b]$ if and only if there exists a sequence of partitions (P_n) such that $L(f; P_n)$ and $U(f; P_n)$ are convergent and $\lim_{n \rightarrow \infty} L(f; P_n) = \lim_{n \rightarrow \infty} U(f; P_n)$. In this case

$$\int_a^b f = \lim_{n \rightarrow \infty} L(f; P_n) = \lim_{n \rightarrow \infty} U(f; P_n) = \lim_{n \rightarrow \infty} L(f; Q_n) = \lim_{n \rightarrow \infty} U(f; Q_n) \text{ for any } Q_n \supseteq P_n.$$

Proof. Assume $f \in \mathcal{R}([a, b])$ and let $I = \int_a^b f(x) dx$. Then by sequential criterion, there exists a sequence of partitions (P_n) satisfying $\lim_{n \rightarrow \infty} [U(f; P_n) - L(f; P_n)] = 0$. Since $L(f; P_n) \leq I \leq U(f; P_n)$, then $0 \leq I - L(f; P_n) \leq U(f; P_n) - L(f; P_n)$ and $L(f; P_n) - U(f; P_n) \leq I - U(f; P_n) \leq 0$. By passing to the limit, we obtain $\lim_{n \rightarrow \infty} L(f; P_n) = \lim_{n \rightarrow \infty} U(f; P_n) = I$. Also, by passing to the limit in $L(f; P_n) \leq L(f; Q_n) \leq I$ and $I \leq U(f; Q_n) \leq U(f; P_n)$ we obtain $I = \lim_{n \rightarrow \infty} L(f; Q_n) = \lim_{n \rightarrow \infty} U(f; Q_n)$. The proof of the converse statement is a consequence of sequential criterion. $\square$

2.2 Properties Of The Riemann Integral.

2.12 Theorem. Let $f : [a, b] \mapsto \mathbb{R}$ be a bounded function and $c \in (a, b)$. Then, f is integrable on $[a, b]$ if and only if f is integrable on $[a, c]$ and $[c, b]$. In this case we have $\int_a^b f = \int_a^c f + \int_c^b f$.

Proof. If f is integrable on $[a, b]$, then for $\varepsilon > 0$ there exists a partition P satisfying $U(f; P) - L(f; P) < \varepsilon$. Since refining a partition can bring the upper and lower sums closer together, we can add c to P if it is not already there and take $P_1 = P \cap [a, c]$ and $P_2 = P \cap [c, b]$ as partitions of $[a, c]$ and $[c, b]$. We have

$$U(f; P_1) - L(f; P_1) \leq U(f; P) - L(f; P) < \varepsilon \text{ and } U(f; P_2) - L(f; P_2) \leq U(f; P) - L(f; P) < \varepsilon.$$

Thus f is integrable on $[a, c]$ and $[c, b]$.

Conversely, if f is integrable on $[a, c]$ and $[c, b]$, then for $\varepsilon > 0$ there exists a partition P_1 of $[a, c]$ satisfying $U(f; P_1) - L(f; P_1) < \frac{\varepsilon}{2}$ and a partition P_2 of $[c, b]$ satisfying $U(f; P_2) - L(f; P_2) < \frac{\varepsilon}{2}$. Then $P = P_1 \cup P_2$ produces a partition of $[a, b]$ satisfying $U(f; P) - L(f; P) < \varepsilon$. Thus f is integrable on $[a, b]$ and we have

$$\int_a^b f \leq U(f; P) < L(f; P) + \varepsilon = L(f; P_1) + L(f; P_2) + \varepsilon \leq \int_a^c f + \int_c^b f + \varepsilon,$$

$$\int_a^c f + \int_c^b f \leq U(f; P_1) + U(f; P_2) < L(f; P_1) + L(f; P_2) + \varepsilon = L(f; P) + \varepsilon \leq \int_a^b f + \varepsilon.$$

Since $\varepsilon > 0$ arbitrary, the above inequalities imply $\int_a^b f \leq \int_a^c f + \int_c^b f$ and $\int_a^c f + \int_c^b f \leq \int_a^b f$.

consequently, we obtain $\int_a^b f = \int_a^c f + \int_c^b f$. $\square$

2.13 Definition. Let $f : [a, b] \mapsto \mathbb{R}$ be a bounded integrable function on $[a, b]$ and $c \in [a, b]$. Define

$$\int_b^a f = - \int_a^b f \text{ and } \int_c^c f = 0.$$

2.14 Remark. If f is an integrable function on some interval I , and a, b, c any three points chosen in any order from I , then it is straightforward to verify that $\int_a^b f = \int_a^c f + \int_c^b f$.

2.15 Theorem. If f is integrable on $[a, b]$ and $[c, d] \subseteq [a, b]$, then f is integrable on $[c, d]$.

Proof. The proof is similar to Theorem 2.12. □

2.16 Theorem. Let $f, g \in \mathcal{R}([a, b])$ and $\lambda, m, M \in \mathbb{R}$. Then

1. $f + g \in \mathcal{R}([a, b])$ and $\int_a^b (f + g) = \int_a^b f + \int_a^b g$.
2. $\lambda f \in \mathcal{R}([a, b])$ and $\int_a^b \lambda f = \lambda \int_a^b f$.
3. If $m \leq f \leq M$, then $m(b - a) \leq \int_a^b f \leq M(b - a)$.
4. If $f \leq g$, then $\int_a^b f \leq \int_a^b g$.
5. $|f| \in \mathcal{R}([a, b])$ and $\left| \int_a^b f \right| \leq \int_a^b |f|$.

Proof. Since $f, g \in \mathcal{R}([a, b])$, there exists a sequence of partitions (P_n) satisfying $\lim_{n \rightarrow \infty} [U(f; P_n) - L(f; P_n)] = 0$ with $\int_a^b f = \lim_{n \rightarrow \infty} L(f; P_n) = \lim_{n \rightarrow \infty} U(f; P_n)$ and a sequence of partitions (Q_n) satisfying $\lim_{n \rightarrow \infty} [U(g; Q_n) - L(g; Q_n)] = 0$ with $\int_a^b g = \lim_{n \rightarrow \infty} L(g; Q_n) = \lim_{n \rightarrow \infty} U(g; Q_n)$.

1. Let $S_n = P_n \cup Q_n$ be the common refinement of P_n and Q_n . Recall that

$$M_k(f + g) \leq M_k(f) + M_k(g) \text{ and } m_k(f + g) \geq m_k(f) + m_k(g).$$

It follows that

$$U(f + g; S_n) \leq U(f; S_n) + U(g; S_n) \text{ and } L(f + g; S_n) \geq L(f; S_n) + L(g; S_n). \quad (2.1)$$

Consequently,

$$U(f + g; S_n) - L(f + g; S_n) \leq [U(f; S_n) - L(f; S_n)] + [U(g; S_n) - L(g; S_n)].$$

Then by passing to the limit both sides we obtain $\lim_{n \rightarrow \infty} [U(f+g; S_n) - L(f+g; S_n)] = 0$.

Thus $f+g \in \mathcal{R}([a, b])$, and by passing to the limit in inequalities (2.1) we obtain

$$\int_a^b (f+g) = \int_a^b f + \int_a^b g.$$

2. For $\lambda \geq 0$, we have $M_k(\lambda f) = \lambda M_k(f)$ and $m_k(\lambda f) = \lambda m_k(f)$. It follows

$$U(\lambda f; P_n) = \lambda U(f; P_n) \text{ and } L(\lambda f; P_n) = \lambda L(f; P_n).$$

By passing to the limit we obtain $\lambda f \in \mathcal{R}([a, b])$ and $\int_a^b \lambda f = \lambda \int_a^b f$.

For $\lambda < 0$, we have $M_k(\lambda f) = \lambda m_k(f)$ and $m_k(\lambda f) = \lambda M_k(f)$. It follows

$$U(\lambda f; P_n) = \lambda L(f; P_n) \text{ and } L(\lambda f; P_n) = \lambda U(f; P_n).$$

By passing to the limit we obtain $\lambda f \in \mathcal{R}([a, b])$ and $\int_a^b \lambda f = \lambda \int_a^b f$.

3. Evident.

4. We have $0 \leq (g-f)$, then by 3. we obtain $0 \leq \int_a^b (g-f)$. It follows from 1. that $\int_a^b f \leq \int_a^b g$.

5. Knowing that $M_k(|f|) - m_k(|f|) \leq M_k(f) - m_k(f)$, it follows that $U(|f|; P_n) - L(|f|; P_n) \leq U(f; P_n) - L(f; P_n)$. Hence $\lim_{n \rightarrow \infty} [U(|f|; P_n) - L(|f|; P_n)] = 0$ and so $|f| \in \mathcal{R}([a, b])$.

Since $-|f| \leq f \leq |f|$, then by 4. we obtain $\left| \int_a^b f \right| \leq \int_a^b |f|$.

□

2.17 Theorem. Let $f \in \mathcal{R}([a, b])$. Assume that $m \leq f(x) \leq M$ for all $x \in [a, b]$ and $g : [m, M] \mapsto \mathbb{R}$ is continuous. Then $g \circ f$ is integrable.

Proof. Left to the reader. □

2.18 Corollary. Let $f, g \in \mathcal{R}([a, b])$. Then $|f|, f^2, fg, \max\{f, g\}$ and $\min\{f, g\}$ are integrable.

Proof. Left as exercise. To show that fg is integrable, we may use $fg = \frac{1}{4}[(f+g)^2 - (f-g)^2]$. □

2.3 Integrable Functions.

2.19 Theorem. If $f : [a, b] \mapsto \mathbb{R}$ is bounded monotone, then it is integrable.

Proof. We prove the result for an increasing function. Then $m_k = f(x_{k-1})$ and $M_k = f(x_k)$. Choose the regular subdivision $P_n = \{x_i = a + \frac{i}{n}(b-a); 0 \leq i \leq n\}$. (P_n) is the sequence of partitions of $[a, b]$ that divides it into n subintervals with equal lengths, that is $\Delta x_k = x_k - x_{k-1} = \frac{b-a}{n}$. Then $\lim_{n \rightarrow \infty} [U(f; P_n) - L(f; P_n)] = \lim_{n \rightarrow \infty} \frac{b-a}{n} \sum_{k=1}^n (f(x_k) - f(x_{k-1})) = \lim_{n \rightarrow \infty} \frac{b-a}{n} (f(b) - f(a)) = 0$. Thus $f \in \mathcal{R}([a, b])$. □

2.20 Theorem. *If $f : [a, b] \mapsto \mathbb{R}$ is continuous, then it is integrable.*

Proof. We have f is continuous on a closed and bounded interval, it is uniformly continuous. This means that, given $\varepsilon > 0$, there exists a $\delta > 0$ such that $|x - y| < \delta$ guarantees $|f(x) - f(y)| < \frac{\varepsilon}{(b-a)}$. Let P be a partition of $[a, b]$ such that $\Delta x_k = x_k - x_{k-1} < \delta$. From the extreme value theorem $M_k = f(Y_k)$ and $m_k = f(y_k)$ for some $Y_k, y_k \in [x_{k-1}, x_k]$. Since $|Y_k - y_k| \leq x_k - x_{k-1} < \delta$, then $M_k - m_k = f(Y_k) - f(y_k) < \frac{\varepsilon}{(b-a)}$. It follows that

$$U(f; P) - L(f; P) = \sum_{k=1}^n (M_k - m_k) \Delta x_k < \frac{\varepsilon}{(b-a)} \sum_{k=1}^n \Delta x_k = \varepsilon$$

. Thus f is integrable on $[a, b]$. □

2.21 Theorem. *Let $f : [a, b] \mapsto \mathbb{R}$ be a bounded function.*

1. *If $f \in \mathcal{R}([c, b])$ for all $c \in (a, b)$ then $f \in \mathcal{R}([a, b])$.*
2. *If $f \in \mathcal{R}([a, c])$ for all $c \in (a, b)$ then $f \in \mathcal{R}([a, b])$.*

Proof. We prove the first item. The second one can be done analogously. Let $M > 0$ so that $|f(x)| \leq M$ for all $x \in [a, b]$ and let $P = \{a = x_0, x_1, \dots, x_n = b\}$ be a partition of $[a, b]$. Then

$$U(f; P) - L(f; P) = \sum_{k=1}^n (M_k - m_k) \Delta x_k = (M_1 - m_1)(x_1 - a) + \sum_{k=2}^n (M_k - m_k) \Delta x_k.$$

The idea is that for $\varepsilon > 0$, the first step is to choose x_1 close enough to a so that $(M_1 - m_1)(x_1 - a) < \varepsilon$. This is possible because $(M_1 - m_1) \leq 2M$, so we can pick x_1 so that $x_1 - a \leq \frac{\varepsilon}{4M}$. The second step, since f is integrable on $[x_1, b]$, then there exists a partition P_1 of $[x_1, b]$ so that $U(f; P_1) - L(f; P_1) \leq \frac{\varepsilon}{2}$. Finally we produce $P_2 = P_1 \cup \{a\}$ a partition of $[a, b]$ so that

$$U(f; P_2) - L(f; P_2) = (M_1 - m_1)(x_1 - a) + U(f; P_1) - L(f; P_1) \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

2.22 Examples.

1. *Let $f : [0, 2] \mapsto \mathbb{R}$ defined by $f(x) = \frac{\sin x}{x}$. Note that f (more precisely its extension by continuity) is bounded on $[0, 2]$. Since f is continuous on $[c, 2]$ for all $c \in (0, 2)$, then $f \in \mathcal{R}([c, 2])$ for all $c \in (0, 2)$. It follows from above theorem that $f \in \mathcal{R}([0, 2])$.*
2. *Let $f : (0, 1] \mapsto \mathbb{R}$ defined by $f(x) = \sin \frac{1}{x}$. Note that f can be regarded as bounded function on $[0, 1]$. Since f is continuous on $[c, 1]$ for all $c \in (0, 1)$, then $f \in \mathcal{R}([c, 1])$ for all $c \in (0, 1)$. It follows from above theorem that $f \in \mathcal{R}([0, 1])$.*

2.23 Corollary. Let $f : [a, b] \mapsto \mathbb{R}$ be a bounded function.

1. If f is continuous on $[a, b]$ except one end point, then it is integrable on $[a, b]$.
2. If f is continuous on $[a, b]$ except one point, then it is integrable on $[a, b]$.
3. If f is continuous on $[a, b]$ except a finite number of points, then it is integrable on $[a, b]$.

2.4 Mean Value And Fundamental Theorems Of Calculus.

2.24 Theorem. (First Mean-Value Theorem For Integrals)

Let $f, g \in \mathcal{R}([a, b])$, $m := \inf_{x \in [a, b]} f(x)$ and $M := \sup_{x \in [a, b]} f(x)$. If g is of constant sign on $[a, b]$, then there exists $k \in [m, M]$ such that

$$\int_a^b f(x)g(x) dx = k \int_a^b g(x) dx.$$

If in addition $f \in C[a, b]$, then there exists a point $\xi \in [a, b]$ such that

$$\int_a^b f(x)g(x) dx = f(\xi) \int_a^b g(x) dx.$$

Proof. Assume without loss of generality that $g(x) \geq 0$ for all $x \in [a, b]$. Since $m \leq f(x) \leq M$ and $g(x) \geq 0$ on $[a, b]$, then

$$mg(x) \leq f(x)g(x) \leq Mg(x).$$

Since $mg(x), f \cdot g$, and $Mg(x) \in \mathcal{R}([a, b])$, then

$$m \int_a^b g(x) dx \leq \int_a^b f(x)g(x) dx \leq M \int_a^b g(x) dx.$$

We discuss two cases:

If $\int_a^b g(x) dx = 0$, then $\int_a^b f(x)g(x) dx = 0$. In this case k can be chosen arbitrary in $[m, M]$ so that the equality $\int_a^b f(x)g(x) dx = k \int_a^b g(x) dx$ holds true.

If $\int_a^b g(x) dx > 0$, then $m \leq \frac{\int_a^b f(x)g(x) dx}{\int_a^b g(x) dx} \leq M$. Let $k = \frac{\int_a^b f(x)g(x) dx}{\int_a^b g(x) dx}$, then $k \in [m, M]$ and

the equality $\int_a^b f(x)g(x) dx = k \int_a^b g(x) dx$ holds true.

If in addition $f \in C[a, b]$, then by the intermediate-value theorem there exists a point $\xi \in [a, b]$ such that $k = f(\xi)$. Then it follows that

$$\int_a^b f(x)g(x) dx = f(\xi) \int_a^b g(x) dx.$$

□

The case $g = 1$ is interesting. In this case the first mean-value theorem becomes,

If $f \in \mathcal{R}([a, b])$, then $\int_a^b f(x) dx = k(b - a)$ for some $k \in [m, M]$.

If $f \in C[a, b]$, then $\int_a^b f(x) dx = f(\xi)(b - a)$ for some $\xi \in [a, b]$.

2.25 Theorem. (Second Mean-Value Theorem For Integrals)

If $f \in \mathcal{R}([a, b])$ and g is a monotonic function on $[a, b]$, then there exists a point $\xi \in [a, b]$ such that

$$\int_a^b f(x)g(x) dx = g(a) \int_a^\xi f(x) dx + g(b) \int_\xi^b f(x) dx.$$

Proof. Admitted. □

For the second mean-value theorem we have three interesting particular cases:

1. If $f = 1$ and g is a monotonic function on $[a, b]$, then there exists a point $\xi \in [a, b]$ such that

$$\int_a^b g(x) dx = g(a)(\xi - a) + g(b)(b - \xi).$$

2. If $f \in \mathcal{R}([a, b])$ and g is a decreasing and positive or increasing and negative function on $[a, b]$, then there exists a point $\xi \in [a, b]$ such that

$$\int_a^b f(x)g(x) dx = g(a) \int_a^\xi f(x) dx.$$

This is because we can modify the value of g at the point b so that $g(b) = 0$.

3. If $f \in \mathcal{R}([a, b])$ and g is an increasing and positive or decreasing and negative function on $[a, b]$, then there exists a point $\xi \in [a, b]$ such that

$$\int_a^b f(x)g(x) dx = g(b) \int_\xi^b f(x) dx.$$

This is because we can modify the value of g at the point a so that $g(a) = 0$.

2.26 Property. If $f : [a, b] \mapsto \mathbb{R}$ is integrable, then the function $F(x) = \int_a^x f(t) dt$ is continuous on $[a, b]$.

Proof. Fix $x_0 \in [a, b]$ and let $x \in [a, b]$. Then by the first mean-value theorem, there exists $k \in [m, M]$ such that

$$F(x) - F(x_0) = \int_a^x f(t) dt - \int_a^{x_0} f(t) dt = \int_{x_0}^x f(t) dt = k(x - x_0).$$

It follows that $\lim_{x \rightarrow x_0} F(x) = F(x_0)$ and F is continuous at x_0 . Since x_0 is an arbitrary point of $[a, b]$, then F is continuous on $[a, b]$. □

2.27 Theorem. (Fundamental Theorem Of Calculus I)

If $f \in C([a, b])$, then the function $F(x) = \int_a^x f(t) dt$ satisfies $F \in C^1([a, b])$ and $F' = f$ on $[a, b]$.

Proof. Fix $x_0 \in [a, b]$ and let $x \in [a, b] \setminus \{x_0\}$. Then by the first mean-value theorem, there exists ξ between x_0 and x such that

$$\frac{F(x) - F(x_0)}{x - x_0} = \frac{\int_a^x f(t) dt - \int_a^{x_0} f(t) dt}{x - x_0} = \frac{\int_{x_0}^x f(t) dt}{x - x_0} = \frac{f(\xi)(x - x_0)}{x - x_0} = f(\xi).$$

It follows that $\lim_{x \rightarrow x_0} \frac{F(x) - F(x_0)}{x - x_0} = \lim_{x \rightarrow x_0} f(\xi) = f(x_0)$ and F is differentiable at x_0 with $F'(x_0) = f(x_0)$. Since x_0 is an arbitrary point of $[a, b]$, then F is differentiable on $[a, b]$ and $F' = f$. Since $f \in C([a, b])$, then so $F' \in C([a, b])$. Thus $F \in C^1([a, b])$ and $F' = f$ on $[a, b]$. $\square$

Theorem 2.30 says that, if $f \in C([a, b])$, then F is a primitive of f on $[a, b]$.

2.28 Corollary. If $f : [a, b] \mapsto \mathbb{R}$ is continuous on $[a, b]$ and $u, v : I \mapsto [a, b]$ are differentiable functions on an **interval** $I \subseteq \mathbb{R}$ with values in $[a, b]$. Then the function $G(x) = \int_{v(x)}^{u(x)} f(t) dt$ is differentiable on I and we have

$$G'(x) = u'(x)f(u(x)) - v'(x)f(v(x)), \quad \text{for all } x \in I.$$

2.29 Theorem. (Fundamental Theorem Of Calculus II)

If F is continuous on $[a, b]$ and F is a primitive of f on (a, b) , and if $f \in \mathcal{R}([a, b])$, then $\int_a^b f = F(b) - F(a)$.

Proof. Let P be a partition of $[a, b]$. Then by applying the mean value theorem to F on the interval $[x_{k-1}, x_k]$, we obtain

$$F(x_k) - F(x_{k-1}) = f(c_k)(x_k - x_{k-1}), \quad \text{for some } c_k \in (x_{k-1}, x_k).$$

Since $m_k \leq f(c_k) \leq M_k$, then $L(f; P) \leq \sum_{k=1}^n [F(x_k) - F(x_{k-1})] \leq U(f; P)$. It follows that $L(f; P) \leq$

$F(b) - F(a) \leq U(f; P)$. but $L(f; P) = U(f; P) = \int_a^b f$, then we conclude that $\int_a^b f = F(b) - F(a)$. $\square$

Note that, $\int_a^b g' = g(b) - g(a)$ provided that g is continuous on $[a, b]$, differentiable on (a, b) and g' integrable on $[a, b]$.

2.5 Integration By Parts And Substitution Methods.

2.30 Theorem. (Integration By Parts)

If u and v are continuous functions on $[a, b]$ that are differentiable on (a, b) and if u' and v' are integrable on $[a, b]$, then

$$\int_a^b uv' = [u(b)v(b) - u(a)v(a)] - \int_a^b u'v.$$

Proof. We have $uv' = (uv)' - u'v$. Use the fact that the product of integrable functions is integrable, linearity of integral and Theorem 2.29 to conclude $\int_a^b uv' = [u(b)v(b) - u(a)v(a)] - \int_a^b u'v$. $\square$

2.31 Theorem. (Integration By Substitution)

Let f, ϕ and ϕ' be continuous where $f : [a, b] \mapsto \mathbb{R}$ and $\phi : [\alpha, \beta] \mapsto [a, b]$. Then

$$\int_{\alpha}^{\beta} f(\phi(t))\phi'(t) dt = \int_{\phi(\alpha)}^{\phi(\beta)} f(x) dx.$$

If in addition ϕ possesses an inverse function, then

$$\int_a^b f(x) dx = \int_{\phi^{-1}(a)}^{\phi^{-1}(b)} f(\phi(t))\phi'(t) dt.$$

Proof. Let $F(y) = \int_{\phi(\alpha)}^{\phi(y)} f(x) dx$ and $G(y) = \int_{\alpha}^y f(\phi(t))\phi'(t) dt$. we have $F' = G'$ on $[\alpha, \beta]$. It follows $F(y) - G(y) = F(\alpha) - G(\alpha) = 0$ for all $y \in [\alpha, \beta]$. In particular, we conclude that $F(\beta) = G(\beta)$. $\square$

2.6 Riemann's Original Definition.

In addition to approach of Darboux for integration, there is another equivalent approach due to Riemann, which starts with a direct approximation of the integral by a sum.

2.32 Definition. (Riemann Sum, Mesh)

Let $P = \{a = x_0, x_1, \dots, x_n = b\}$ be a partition of $[a, b]$ and $c_k \in [x_{k-1}, x_k]$.

- The **Riemann** sum of f with respect to P and $(c_k)_{k=1}^n$ is $R(f; P, (c_k)_{k=1}^n) = \sum_{k=1}^n f(c_k)(x_k - x_{k-1})$.
- The **mesh** of P is defined by $\|P\| := \max_{k=1}^n \{x_k - x_{k-1}\}$, that is the length of the largest subinterval $[x_{k-1}, x_k]$.

2.33 Definition. (Riemann's Original Definition)

Let $f : [a, b] \mapsto \mathbb{R}$ be a bounded function. We say that f is **Riemann integrable** on $[a, b]$ if and only if there is a real number I with the following property: For every $\varepsilon > 0$, there exists a $\delta > 0$ such that $\|P\| < \delta$ guarantees $|R(f; P, (c_k)_{k=1}^n) - I| < \varepsilon$ for every possible choice $c_k \in [x_{k-1}, x_k]$.

That is $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n f(c_k)(x_k - x_{k-1}) = I$.

The quantity $I = \int_a^b f$ is called the **Riemann integral** of f .

2.34 Remark. A bounded function f is **Darboux integrable** on $[a, b]$ if and only if it is **Riemann integrable** on $[a, b]$. In this case $\int_a^b f$ in both senses is the same.

Proof. Admitted. □

If we choose the regular partition $P_n = \{x_k = a + k\frac{b-a}{n}; 0 \leq k \leq n\}$ and $c_k \in]x_{k-1}, x_k[$ or $c_k = x_k$ or $c_k = x_{k-1}$ then we obtain the following sequential characterization.

2.35 Corollary. If f is integrable on $[a, b]$, then

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{b-a}{n} \sum_{k=1}^n f(c_k) &= \int_a^b f(x) dx \\ \lim_{n \rightarrow \infty} \frac{b-a}{n} \sum_{k=1}^n f(a + k\frac{b-a}{n}) &= \int_a^b f(x) dx \\ \lim_{n \rightarrow \infty} \frac{b-a}{n} \sum_{k=0}^{n-1} f(a + k\frac{b-a}{n}) &= \int_a^b f(x) dx. \end{aligned}$$

In particular, if f is integrable on $[0, 1]$, then

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f(c_k) &= \int_0^1 f(x) dx \\ \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right) &= \int_0^1 f(x) dx \\ \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} f\left(\frac{k}{n}\right) &= \int_0^1 f(x) dx. \end{aligned}$$

2.36 Examples.

1. Let us study the convergence of the sequence of general term $u_n = \sum_{k=1}^n \frac{n}{n^2+k^2}$. We have

$$u_n = \frac{1}{n} \sum_{k=1}^n \frac{1}{1+\frac{k^2}{n^2}} = \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right),$$

where $f(x) = \frac{1}{1+x^2}$ is continuous, hence integrable on $[0, 1]$. Thus

$$\lim_{n \rightarrow \infty} u_n = \int_0^1 f(x) dx = \arctan x \Big|_0^1 = \frac{\pi}{4}.$$

2. Let us study the convergence of the sequence of general term $u_n = \sum_{k=n}^{2n-1} \frac{n}{k^2}$. We have

$$u_n = \sum_{k=0}^{n-1} \frac{n}{(k+n)^2} = \frac{1}{n} \sum_{k=0}^{n-1} \frac{1}{(1+\frac{k}{n})^2} = \frac{1}{n} \sum_{k=0}^{n-1} f\left(\frac{k}{n}\right),$$

where $f(x) = \frac{1}{(1+x)^2}$ is continuous, hence integrable on $[0, 1]$. Thus

$$\lim_{n \rightarrow \infty} u_n = \int_0^1 f(x) dx = \frac{1}{2}.$$

2.7 Schwarz Inequality For Riemann Integrals.

2.37 Property. Let $f, g \in \mathcal{R}([a, b])$, then

1. $\left(\int_a^b f(t)g(t) dt \right)^2 \leq \int_a^b f^2(t) dt \int_a^b g^2(t) dt.$
2. $\left(\int_a^b f(t)g(t) dt \right)^2 = \int_a^b f^2(t) dt \int_a^b g^2(t) dt \Leftrightarrow f = \lambda g, \lambda \in \mathbb{R}.$

Proof. admitted. □

2.8 Exercises

Exercise 1. Let $f(x) = x^2$ on $[0, b]$.

1. Show that f is integrable on $[0, b]$ by using ε -criteria for integrability.
2. Show, using the sequential criteria, that f is integrable on $[0, b]$ and compute $\int_0^b f(x) dx$.

Hint: you may use $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$.

Exercise 2. Let $f : \mathbb{R} \mapsto \mathbb{R}$ be a continuous T -periodic function, $a, b \in \mathbb{R}$ and $n \in \mathbb{N}$.

1. Show that $\int_{a+T}^{b+T} f(x) dx = \int_a^b f(x) dx$.
2. Prove that $\int_a^{a+T} f(x) dx = \int_0^T f(x) dx$ and $\int_a^{a+nT} f(x) dx = n \int_0^T f(x) dx$.

Exercise 3. Let $f : [-a, a] \mapsto \mathbb{R}$ be a continuous function.

1. Show that if f is an odd function, then $\int_{-a}^a f(x) dx = 0$.
2. Show that if f is an even function, then $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$.
3. Deduce the value of $\int_{-2}^2 x^2 \ln\left(\frac{5-x}{5+x}\right) dx$.

Exercise 4. Let $f : [a, b] \mapsto \mathbb{R}$ be a continuous function and $c \in \mathbb{R}$.

1. Show that

$$\int_a^b f(x) dx = - \int_{c-a}^{c-b} f(c-x) dx \text{ and } \int_a^b f(x) dx = \int_a^b f(a+b-x) dx.$$

2. Deduce the value of $I = \int_0^{\frac{\pi}{4}} \ln(1 + \tan x) dx$. (Hint: Use $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$.)

Exercise 5. Let $f : [a, b] \mapsto \mathbb{R}$ be a continuous function such that $f(a+b-x) = f(x)$ for all $x \in [a, b]$.

1. Show that $\int_a^b x f(x) dx = \frac{a+b}{2} \int_a^b f(x) dx$.
2. Deduce the value of $M = \int_0^{\frac{\pi}{2}} x \frac{\sin(2x)}{1 + \cos^2(2x)} dx$.

Exercise 6. Let $m \in \mathbb{R}$

1. Calculate $I_m = \int_0^{\frac{\pi}{2}} \frac{\sin^m(x)}{\sin^m x + \cos^m x} dx$.
2. Deduce the value of $J = \int_0^{\frac{\pi}{2}} \frac{dx}{1 + \tan^4 x}$.

Exercise 7. Compute the following integrals

$$\begin{aligned} 1) \int_0^{\frac{\pi}{4}} \frac{\cos x [1 + \tan^2 x]}{\sin x + \cos x} dx, \quad 2) \int_0^{\frac{\pi}{2}} \frac{1}{2 + \cos x} dx, \quad 3) \int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin x \cos x} dx, \\ 4) \int_0^{\pi} \frac{\cos^{11} x}{1 + \sin^{17} x} dx, \quad 5) \int_0^1 \frac{1}{\sqrt{1+x^2}} dx, \quad 6) \int_0^{\sqrt{2}} \sqrt{2-x^2} dx. \end{aligned}$$

Exercise 8. Let $f : \mathbb{R} \mapsto \mathbb{R}$ be a continuous function. Calculate $g'(x)$ where g is defined by

$$1) g(x) = \int_{3x}^{x^3} f(t) dt, \quad 2) g(x) = \int_0^x x^3 f(t) dt, \quad 3) g(x) = \int_0^x [x^2 + f(t)]^2 dt.$$

Exercise 9. Compute $g'(x)$ and specify the domain of differentiability of the function g defined by

$$1) g(x) = \int_0^x \ln(1-t) dt, \quad 2) g(x) = \int_{-x}^{x^2} \arctan t dt, \quad 3) g(x) = \int_0^{\sin x} \arg \tanh t dt.$$

Exercise 10. Let $G(x) = \int_{\ln(1-e^x)}^{-\cosh x} \ln(1-e^t) dt$ and $g(x) = \frac{xe^x}{1-e^x} - \sinh x \ln(1-e^{-\cosh x})$.

1. Show that $G' = g$ on an interval J to be determined.

2. Are the domain of definition of G' and g the same? Justify.

Exercise 11. Find the limit of the following sequences

$$\begin{aligned} 1) u_n = \sum_{k=0}^{n-1} \frac{n}{n^2+k^2}, \quad 2) u_n = \sum_{k=1}^n \frac{k}{n^2+k^2}, \quad 3) u_n = \sum_{k=1}^{n+1} \frac{k^2}{n^2\sqrt[3]{n^3+k^3}}, \quad 4) u_n = \frac{1}{n} \left(\frac{(2n)!}{n!} \right)^{\frac{1}{n}}, \\ 5) u_n = \sum_{k=1}^n \frac{k}{n^2} \cos \frac{k}{n}, \quad 6) u_n = \sum_{k=1}^n \frac{k}{n^2} e^{-\frac{k\pi}{2n}}, \quad 7) u_n = \sum_{k=1}^n \frac{1}{\sqrt{n^2+2kn}}, \quad 8) u_n = \frac{1}{n} \sqrt[n]{\prod_{k=1}^n (n+k)}. \end{aligned}$$

Exercise 12. Consider the sequence $I_n = \int_0^1 \frac{(1-x)^n e^x}{n!} dx$.

1. Show that $\lim_{n \rightarrow \infty} I_n = 0$.

2. Find a relation between I_n and I_{n+1} .

3. Deduce that $\lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{1}{k!} = e$.

Exercise 13. Let f be a continuous function on $] -2, 2[$. Find $\lim_{x \rightarrow 0} \frac{1}{x} \int_0^x f(t) dt$.

Exercise 14. Let f be a continuous function on $[0, 1]$. Find the following limits

$$1) \lim_{x \rightarrow 0^+} \frac{1}{x^3} \int_{\sin x}^x f(t) dt, \quad 2) \lim_{n \rightarrow \infty} \int_0^{\frac{1}{n}} \frac{nf(t)}{1+(nt)^2} dt.$$

Exercise 15. Find the following limits

$$\begin{aligned} 1) \lim_{x \rightarrow \infty} \int_x^{2x} \frac{1}{\ln t} dt, \quad 2) \lim_{x \rightarrow 0^+} \int_x^{2x} \frac{e^t}{t} dt, \quad 3) \lim_{x \rightarrow \infty} \int_x^{2x} \frac{\sin t}{t} dt, \\ 4) \lim_{x \rightarrow \infty} \int_x^{2x} \frac{\cos \frac{1}{t}}{t} dt, \quad 5) \lim_{x \rightarrow 1^+} \int_x^{x^2} \frac{\sin \frac{\pi t}{1}}{t \ln t} dt, \quad 6) \lim_{x \rightarrow 1^+} \int_x^{x^3} \frac{1}{\ln t} dt. \end{aligned}$$

Exercise 16. Let $f[0, \infty[\mapsto \mathbb{R}$ be a continuous function such that $\lim_{x \rightarrow \infty} f(x) = \ell \in \mathbb{R}$. Find the limit of the following sequences

$$1) \int_n^{n+2} f(x) dx, \quad 2) \int_n^{n+\sqrt{n}} \frac{f(x)}{\sqrt{x}} dx, \quad 3) \int_0^n \frac{f(x)}{n} dx, \quad 4) \int_{n^2+1}^{n^2+n+1} \frac{\arctan x^2}{\sqrt{x}} dx.$$

Exercise 17. Let $f : [a, b] \mapsto \mathbb{R}$ be a continuous function.

1. Show that if $f \geq 0$ on $[a, b]$ and $\int_a^b f(x) dx = 0$, then $f(x) = 0$ for all $x \in [a, b]$.
2. Assume $\int_a^b (f(x))^2 dx = \int_a^b (f(x))^3 dx = \int_a^b (f(x))^4 dx$. Show that $f = 0$ or $f = 1$ on $[a, b]$.
Hint: Expand $(f^2(x) - f(x))^2$.
3. Show that $\left| \int_a^b f(x) dx \right| = \int_a^b |f(x)| dx$ if and only if $f \geq 0$ or $f \leq 0$ on $[a, b]$.

Exercise 18. Let $f : [0, 1] \mapsto \mathbb{R}$ be a differentiable function such that f' is continuous and $f(1) = 0$.

1. Show that $\int_0^1 f(x) dx = - \int_0^1 x f'(x) dx$
2. Prove that $\left(\int_0^1 f(x) dx \right)^2 \leq \frac{1}{3} \int_0^1 (f'(x))^2 dx$. Under which conditions the equality holds.

Exercise 19. Let f be a function of class C^2 on $[0, 1]$ such that $f(0) = f'(1) = 0$ and $I = \int_0^1 f(x) dx$.

1. Show that $I = \frac{1}{2} \int_0^1 (x^2 - 2x) f''(x) dx$.
2. Deduce that $|I| \leq \frac{1}{3} \sup_{x \in [0, 1]} |f''(x)|$.

Exercise 20. Let $a \in]0, 1[$ and $I_n(a) = \int_a^1 (\ln x)^n dx$.

1. Find a relation between $I_{n-1}(a)$ and $I_n(a)$ for $n \geq 1$.
2. Show that $\lim_{a \rightarrow 0^+} I_n(a) = (-1)^n n!$.

Riemann integrals have been defined for bounded functions over closed bounded intervals. In this chapter we extend the concept of Riemann integral to functions defined on **unbounded intervals** or to **unbounded functions**. Such integrals are called **improper** and they are defined in terms of limits of ordinary integrals.

In this chapter I will denote a nonempty **semi-open** or **open** possibly unbounded interval, $[a, b)$, $(a, b]$ or (a, b) , of the real line $\mathbb{R}$.

3.1 Definitions And Examples.

3.1 Definition. Let $f(t)$ be a function defined on I . We say that:

- f is **locally integrable on** $[a, b)$ if f is integrable on every closed interval $[a, c] \subseteq [a, b)$.
- f is **locally integrable on** $(a, b]$ if f is integrable on every closed interval $[c, b] \subseteq (a, b]$.
- f is **locally integrable on** (a, b) if f is integrable on every closed interval $[c, d] \subseteq (a, b)$.

Let $\mathcal{R}_{loc}(I)$ denotes the set of locally integrable functions on I .

3.2 Remark. If f is a continuous, bounded continuous except on a finite number of points, or monotone function on I then it is respectively continuous, bounded continuous except on a finite number of points or monotone on every closed interval of I , hence locally integrable on I .

The three basic types of improper integrals are as follows:

Improper Integrals Of Type I; (Unbounded domains of integration.)

3.3 Definition. Let $a, b \in \mathbb{R}$, $f \in \mathcal{R}_{loc}([a, \infty))$ and $g \in \mathcal{R}_{loc}((-\infty, b])$. We formally set

$$\int_a^{+\infty} f(t) dt = \lim_{c \rightarrow +\infty} \int_a^c f(t) dt, \quad \int_{-\infty}^b g(t) dt = \lim_{d \rightarrow -\infty} \int_d^b g(t) dt.$$

- If the limit exists and is finite, we say that the improper integral converges.
- If the limit is infinite or does not exist, we say that the improper integral diverges to ∞ or diverges, respectively.

3.4 Examples. Let f denotes the integrands of the following improper integrals.

1. Consider $\int_0^\infty e^{-t} dt$. Since $f \in \mathcal{C}([0, \infty))$, then $f \in \mathcal{R}_{loc}([0, \infty))$ and we have,

$$\int_0^\infty e^{-t} dt = \lim_{c \rightarrow +\infty} \int_0^c e^{-t} dt = - \lim_{c \rightarrow +\infty} [e^{-t}]_0^c = - \lim_{c \rightarrow +\infty} (e^{-c} - e^0) = 1.$$
Hence the improper integral converges.
2. Consider $\int_{-\infty}^0 e^{-t} dt$. Since $f \in \mathcal{C}((-\infty, 0])$, then $f \in \mathcal{R}_{loc}((-\infty, 0])$ and we have,

$$\int_{-\infty}^0 e^{-t} dt = \lim_{d \rightarrow -\infty} \int_d^0 e^{-t} dt = - \lim_{d \rightarrow -\infty} [e^{-t}]_d^0 = - \lim_{d \rightarrow -\infty} (e^0 - e^{-d}) = +\infty.$$
Hence the improper integral diverges to ∞ .
3. Consider $\int_0^\infty \sin t dt$. Since $f \in \mathcal{C}([0, \infty))$, then $f \in \mathcal{R}_{loc}([0, \infty))$ and we have,

$$\int_0^\infty \sin t dt = \lim_{c \rightarrow +\infty} \int_0^c \sin t dt = - \lim_{c \rightarrow +\infty} [\cos t]_0^c = - \lim_{c \rightarrow +\infty} (\cos c - \cos 0), \nexists.$$
Hence the improper integral diverges.

Improper Integrals Of Type II;(Unbounded integrands.)

3.5 Definition. Let $a, b \in \mathbb{R}$, $f \in \mathcal{R}_{loc}([a, b))$ and $g \in \mathcal{R}_{loc}((a, b])$. We formally set

$$\int_a^b f(t) dt = \lim_{c \rightarrow b^-} \int_a^c f(t) dt, \quad \int_a^b g(t) dt = \lim_{d \rightarrow a^+} \int_d^b g(t) dt.$$

- If the limit exists and is finite, we say that the improper integral converges.
- If the limit is infinite or does not exist, we say that the improper integral diverges to ∞ or diverges, respectively.

3.6 Examples.

1. Consider $\int_0^1 \frac{1}{t-1} dt$. Since $f \in \mathcal{C}([0, 1))$, then $f \in \mathcal{R}_{loc}([0, 1))$ and we have,

$$\int_0^1 \frac{1}{t-1} dt = \lim_{c \rightarrow 1^-} \int_0^c \frac{1}{t-1} dt = \lim_{c \rightarrow 1^-} [\ln |t-1|]_0^c = \lim_{c \rightarrow 1^-} \ln |c-1| = -\infty.$$
Hence the improper integral diverges to $-\infty$.
2. Consider $\int_1^2 \frac{1}{t-1} dt$. Since $f \in \mathcal{C}((1, 2])$, then $f \in \mathcal{R}_{loc}((1, 2])$ and we have,

$$\int_1^2 \frac{1}{t-1} dt = \lim_{d \rightarrow 1^+} \int_d^2 \frac{1}{t-1} dt = \lim_{d \rightarrow 1^+} [\ln |t-1|]_d^2 = \lim_{d \rightarrow 1^+} -\ln |d-1| = \infty.$$
Hence the improper integral diverges to ∞ .
3. Consider $\int_0^1 \frac{1}{(1-t)^{\frac{1}{2}}} dt$. Since $f \in \mathcal{C}([0, 1))$, then $f \in \mathcal{R}_{loc}([0, 1))$ and we have,

$$\int_0^1 \frac{1}{(1-t)^{\frac{1}{2}}} dt = \lim_{c \rightarrow 1^-} \int_0^c \frac{1}{(1-t)^{\frac{1}{2}}} dt = -2 \lim_{c \rightarrow 1^-} [(1-t)^{\frac{1}{2}}]_0^c = -2 \lim_{c \rightarrow 1^-} ((1-c)^{\frac{1}{2}} - 1) = 2.$$
Hence the improper integral converges.

4. Consider $\int_0^1 \frac{1}{(t-1)^2} \sin\left(\frac{1}{t-1}\right) dt$. Since $f \in \mathcal{C}([0, 1))$, then $f \in \mathcal{R}_{loc}([0, 1))$ and we have,
- $$\int_0^1 \frac{1}{(t-1)^2} \sin\left(\frac{1}{t-1}\right) dt = \lim_{c \rightarrow 1^-} \int_0^c \frac{1}{(t-1)^2} \sin\left(\frac{1}{t-1}\right) dt = \lim_{c \rightarrow 1^-} \left[\cos\left(\frac{1}{t-1}\right) \right]_0^c = \lim_{c \rightarrow 1^-} \left(\cos\left(\frac{1}{c-1}\right) - \cos 1 \right), \nexists. \text{ Hence the improper integral diverges.}$$

Improper Integrals Of Type III; (Build up as sum of Type I and Type II.)

- First, one can consider improper integrals $\int_a^b f(t) dt$ where a difficulty occurs at both end points of the interval of integration, either because the endpoints is at infinity or because the integrand is unbounded there. The trick here is to pick an intermediate point $c \in (a, b)$ and write $\int_a^b = \int_a^c + \int_c^b$, thus reducing the integral to a sum of two integrals that are each of type I or II; The original integral is said to be convergent if and only if each of the two sub-integrals is convergent.
- Second, one can consider improper integrals $\int_a^b f(t) dt$ where a difficulty occurs at an interior point c of the interval of integration, because f has no finite limit at c^+ or c^- . The trick here is to write formally $\int_a^b = \int_a^c + \int_c^b$, thus reducing the integral to a sum of two integrals that are each of type II; The original integral is said to be convergent if and only if each of the two sub-integrals is convergent.

3.7 Examples.

1. Consider $\int_{-\infty}^{\infty} e^{-t} dt$. Since $f \in \mathcal{C}((-\infty, \infty))$, then $f \in \mathcal{R}_{loc}((-\infty, \infty))$ and we have,
- $$\int_{-\infty}^{\infty} e^{-t} dt = \int_{-\infty}^0 e^{-t} dt + \int_0^{\infty} e^{-t} dt. \text{ Hence the improper integral diverges, because one of the improper integrals to the right diverges.}$$
2. Consider $\int_1^3 \frac{1}{\sqrt{(t-1)(3-t)}} dt$. Since $f \in \mathcal{C}((1, 3))$, then $f \in \mathcal{R}_{loc}((1, 3))$ and we have,
- $$\int_1^3 \frac{1}{\sqrt{(t-1)(3-t)}} dt = \int_1^2 \frac{1}{\sqrt{(t-1)(3-t)}} dt + \int_2^3 \frac{1}{\sqrt{(t-1)(3-t)}} dt := I_1 + I_2. \text{ we have;}$$
- $$I_1 = \int_1^2 \frac{1}{\sqrt{-(t-2)^2 - 1}} dt = \lim_{d \rightarrow 1^+} \left[\arcsin(t-2) \right]_d^2 = \frac{\pi}{2} \text{ is convergent and}$$
- $$I_2 = \int_2^3 \frac{1}{\sqrt{-(t-2)^2 - 1}} dt = \lim_{c \rightarrow 3^-} \left[\arcsin(t-2) \right]_2^c = \frac{\pi}{2} \text{ is convergent. Thus the improper integral converges and we have,}$$
- $$\int_1^3 \frac{1}{\sqrt{(t-1)(3-t)}} dt = \int_1^2 \frac{1}{\sqrt{(t-1)(3-t)}} dt + \int_2^3 \frac{1}{\sqrt{(t-1)(3-t)}} dt = \frac{\pi}{2} + \frac{\pi}{2} = \pi.$$
3. Consider $\int_0^2 \frac{1}{t-1} dt$. Since $f \in \mathcal{C}([0, 1)) \cap \mathcal{C}((1, 2])$, then $f \in \mathcal{R}_{loc}([0, 1)) \cap \mathcal{R}_{loc}((1, 2])$ and we have,
- $$\int_0^2 \frac{1}{t-1} dt = \int_0^1 \frac{1}{t-1} dt + \int_1^2 \frac{1}{t-1} dt. \text{ Hence the improper integral diverges, because one of the improper integrals to the right diverges.}$$

3.8 Remark. If f locally integrable on (a, b) , that is a difficulty occurs at both end points of the interval of integration, then the improper integral $\int_a^b f(t) dt$ is convergent if $\lim_{d \rightarrow a^+} \left(\lim_{c \rightarrow b^-} \int_d^c f(t) dt \right)$ exists. The order in which limits are taken does not matter.

3.9 Remark. Let $f \in \mathcal{R}_{loc}((-a, a))$, $g \in \mathcal{R}_{loc}([a, c)) \cap \mathcal{R}_{loc}((c, b])$ and $k \in \mathcal{R}_{loc}((a, b))$ then the **(Cauchy) principal value**

$$\begin{aligned} PV \int_{-a}^a f(t) dt &:= \lim_{c \rightarrow a^+} \int_{-c}^c f(t) dt = \lim_{h \rightarrow 0^+} \int_{-a+h}^{a-h} f(t) dt, \\ PV \int_a^b g(t) dt &:= \lim_{h \rightarrow 0^+} \left(\int_a^{c-h} g(t) dt + \int_{c+h}^b g(t) dt \right), \\ PV \int_a^b k(t) dt &:= \lim_{h \rightarrow 0^+} \int_{a+h}^{b-h} k(t) dt. \end{aligned}$$

It may happen that the principal value exists although the improper integral is divergent. For example; the improper integrals $\int_{-\infty}^{\infty} t dt$, $\int_{-1}^1 \frac{t}{1-t^2} dt$, $\int_{-1}^1 \frac{1}{t} dt$ and $\int_{-2}^3 \frac{1}{t} dt$ are divergent, has no sense, while their principal values exist and $PV \int_{-\infty}^{\infty} t dt = 0$, $PV \int_{-1}^1 \frac{t}{1-t^2} dt = 0$, $PV \int_{-1}^1 \frac{1}{t} dt = 0$ and $PV \int_{-2}^3 \frac{1}{t} dt = \ln(\frac{3}{2})$.

One can easily see that; If the improper integral exists, then the principal value also exists and equal to the improper integral. Thus the principal value of an improper integral constitutes a proper generalization of the improper integral. When we know a-priori that the improper integral exists, we can evaluate it by just computing its principal value.

3.10 Remark. Recall that, if f locally integrable on (a, b) and bounded on $]a, b[$, then f is integrable on $[a, b]$. For example; The integral $\int_0^1 \sin \frac{1}{t} dt$ is not of Type I nor of Type II near zero. The integrand $f(t) = \sin \frac{1}{t}$ is locally integrable on $(0, 1]$ and can be regarded as bounded function on $[0, 1]$, Hence integrable on $[0, 1]$.

3.2 Basic Properties Of Improper Integrals.

The basic properties of Riemann integral can be generalized to improper Riemann integral by a limit process.

3.11 Property. (Chasles Relation)

Let $f \in \mathcal{R}_{loc}([a, b))$, and $c \in (a, b)$. Then $\int_a^b f(t) dt$ and $\int_c^b f(t) dt$ have same nature and if they

are convergent, we have

$$\int_a^b f(t) dt = \int_a^c f(t) dt + \int_c^b f(t) dt.$$

Proof. Let $x \in (a, b)$. Then by Chasles relation for Riemann integral we have

$$\int_a^x f(t) dt = \int_a^c f(t) dt + \int_c^x f(t) dt.$$

Pass to the limit as $x \rightarrow b^-$, we obtain

$$\lim_{x \rightarrow b^-} \int_a^x f(t) dt = \int_a^c f(t) dt + \lim_{x \rightarrow b^-} \int_c^x f(t) dt \text{ with } \int_a^c f(t) dt \text{ is constant.}$$

Therefor $\int_a^b f(t) dt$ and $\int_c^b f(t) dt$ have same nature and if they are convergent, we have

$$\int_a^b f(t) dt = \int_a^c f(t) dt + \int_c^b f(t) dt.$$

□

3.12 Property. (Change Of Variable)

Let $u : [a, b) \mapsto \mathbb{R}$ be of class $\mathcal{C}^1$ on $[a, b)$, $b \in \overline{\mathbb{R}}$ and f be continuous function on $u([a, b))$. Then for $\beta \in (a, b)$ we have

$$\int_a^\beta f(u(t))u'(t) dt = \int_{u(a)}^{u(\beta)} f(x) dx.$$

Thus, if the integral to the right has finite limit as $\beta \rightarrow b^-$, then $\int_a^\beta f(u(t))u'(t) dt$ is convergent and we have

$$\int_a^b f(u(t))u'(t) dt = \lim_{\beta \rightarrow b^-} \int_{u(a)}^{u(\beta)} f(x) dx.$$

3.13 Example. $\int_0^1 \frac{\cos t}{\sqrt{t}} dt = 2 \int_0^1 \frac{\cos(\sqrt{t})^2}{2\sqrt{t}} dt$ formally. Let $\beta \in (0, 1)$, then $\int_\beta^1 \frac{\cos t}{\sqrt{t}} dt = 2 \int_{\sqrt{\beta}}^1 \cos x^2 dx \xrightarrow{\beta \rightarrow 0^+} 2 \int_0^1 \cos x^2 dx$ proper integral. Thus the improper integral $\int_0^1 \frac{\cos t}{\sqrt{t}} dt$ is convergent and we have $\int_0^1 \frac{\cos t}{\sqrt{t}} dt = 2 \int_0^1 \cos x^2 dx$.

3.14 Property. (Change Of Variable Bijection)

Let $a, b, c, d \in \overline{\mathbb{R}}$, $u : (a, b) \mapsto (c, d)$ of class $\mathcal{C}^1$ and strictly monotone on (a, b) and let $f : (c, d) \mapsto \mathbb{R}$ be a continuous function on (c, d) . Then for $\alpha, \beta \in (c, d)$ with $\alpha < \beta$ we have

$$\int_\alpha^\beta f(x) dx = \int_{u^{-1}(\alpha)}^{u^{-1}(\beta)} f(u(t))u'(t) dt \text{ with } u^{-1}(\alpha), u^{-1}(\beta) \in (a, b).$$

Thus, if the integral to the right has finite limit as $(\alpha, \beta) \rightarrow (c^+, d^-)$, then $\int_c^d f(x) dx$ is convergent and we have:

$$\int_c^d f(x) dx = \int_a^b f(u(t))u'(t) dt \text{ if } u \text{ is strictly increasing,}$$

$$\int_c^d f(x) dx = \int_b^a f(u(t))u'(t) dt \quad \text{if } u \text{ is strictly decreasing.}$$

3.15 Example. Consider the improper integral $\int_0^\infty \frac{1}{e^x} \sin \frac{1}{e^x} dx$. For $x = u(t) = \ln t$, we have u is of class $\mathcal{C}^1$ and strictly increasing on $[1, \infty)$. Then $\int_0^\infty \frac{1}{e^x} \sin \frac{1}{e^x} dx = \int_1^\infty \sin\left(\frac{1}{t}\right) \frac{1}{t^2} dt = \cos\left(\frac{1}{t}\right) \Big|_1^\infty = 1 - \cos 1$.

3.16 Property. (Integration By Parts)

Let u and v be two functions of class $\mathcal{C}^1$ on $[a, b)$, $b \in \overline{\mathbb{R}}$. If $\lim_{t \rightarrow b^-} u(t)v(t)$ is finite, then $\int_a^b u(t)v'(t) dt$ and $\int_a^b u'(t)v(t) dt$ have same nature; and in case of convergence, we have

$$\int_a^b u(t)v'(t) dt = u(t)v(t) \Big|_a^{b^-} - \int_a^b u'(t)v(t) dt,$$

where $u(t)v(t) \Big|_a^{b^-} = \lim_{t \rightarrow b^-} u(t)v(t) - u(a)v(a)$.

Proof. Let $\beta \in (a, b)$. Then

$$\int_a^\beta u(t)v'(t) dt = [u(\beta)v(\beta) - u(a)v(a)] - \int_a^\beta u'(t)v(t) dt.$$

Since $\lim_{\beta \rightarrow b^-} u(\beta)v(\beta)$ is finite, then $\int_a^b u(t)v'(t) dt$ and $\int_a^b u'(t)v(t) dt$ have same nature; and in case of convergence, by passing to the limit as $\beta \rightarrow b^-$ in the above identity we obtain

$$\int_a^b u(t)v'(t) dt = u(t)v(t) \Big|_a^{b^-} - \int_a^b u'(t)v(t) dt.$$

□

The **integration by parts formula works for** differentiable functions u and v on $[a, b)$, $b \in \overline{\mathbb{R}}$ such that $u', v' \in \mathcal{R}_{loc}([a, b))$.

3.17 Property. (Linearity Of Integral)

Let $f, g \in \mathcal{R}_{loc}((a, b))$, $a, b \in \overline{\mathbb{R}}$, and $\lambda \in \mathbb{R}$. If $\int_a^b f(t) dt$ and $\int_a^b g(t) dt$ are convergent, then

1. $\int_a^b (f(t) + g(t)) dt$ is convergent and $\int_a^b (f(t) + g(t)) dt = \int_a^b f(t) dt + \int_a^b g(t) dt$.
2. $\int_a^b \lambda f(t) dt$ is convergent and $\int_a^b \lambda f(t) dt = \lambda \int_a^b f(t) dt$.

Proof. Let $\alpha, \beta \in (a, b)$ such that $\alpha < \beta$. For both items we look for the proper integrals

$$\int_{\alpha}^{\beta} (f(t) + g(t)) dt = \int_{\alpha}^{\beta} f(t) dt + \int_{\alpha}^{\beta} g(t) dt \text{ and } \int_{\alpha}^{\beta} \lambda f(t) dt = \lambda \int_{\alpha}^{\beta} f(t) dt,$$

then take the limit both sides as $(\alpha, \beta) \rightarrow (a, b)$ to obtain the result. $\square$

Note that, from passing the limit in the proper integrals $\int_{\alpha}^{\beta} (f(t) + g(t)) dt = \int_{\alpha}^{\beta} f(t) dt + \int_{\alpha}^{\beta} g(t) dt$, we have the right to write $\int_a^b (f(t) + g(t)) dt = \int_a^b f(t) dt + \int_a^b g(t) dt$ if both integrals converge, diverge to $+\infty$, diverge to $-\infty$ or if one integral converges and the other diverges to $\pm\infty$. If both integrals are divergent, then $\int_a^b (f(t) + g(t)) dt$ may converges, diverges or diverges to ∞ . For example, the integrals $\int_0^{\infty} x dx$ and $\int_0^{\infty} -x dx$ are both divergent, but $\int_0^{\infty} (x - x) dx = 0$ and $\int_0^{\infty} (x + x) dx = \infty$.

If one integral is convergent and the other is divergent, then the integral $\int_a^b (f(t) + g(t)) dt$ is divergent.

3.3 Convergence Tests.

3.18 Property. (Examples Of Riemann)

Let $a, b, \alpha \in \mathbb{R}$ with $a < b$, then

1.

$$\int_1^{\infty} \frac{1}{x^{\alpha}} dx \text{ converges } \Leftrightarrow \alpha > 1.$$

2.

$$\int_{-\infty}^{-1} \frac{1}{|x|^{\alpha}} dx \text{ converges } \Leftrightarrow \alpha > 1.$$

3.

$$\int_0^1 \frac{1}{x^{\alpha}} dx, \int_a^b \frac{1}{(x-a)^{\alpha}} dx \text{ and } \int_a^b \frac{1}{(b-x)^{\alpha}} dx \text{ converge } \Leftrightarrow \alpha < 1.$$

Proof. 1. Let $c \in (1, \infty)$. Then, $\int_1^c \frac{1}{x^{\alpha}} dx = \begin{cases} \frac{x^{1-\alpha}}{1-\alpha} \Big|_1^c & \text{if } \alpha \neq 1, \\ \ln x \Big|_1^c & \text{if } \alpha = 1 \end{cases} = \begin{cases} \frac{c^{1-\alpha}-1}{1-\alpha} & \text{if } \alpha \neq 1, \\ \ln c & \text{if } \alpha = 1. \end{cases}$

Therefore, $\int_1^{+\infty} \frac{1}{x^{\alpha}} dx = \begin{cases} \frac{1}{\alpha-1} & \text{if } \alpha > 1, \\ \infty & \text{if } \alpha \leq 1. \end{cases}$

2. For $x = u(t) = -t$, we have u is of class $\mathcal{C}^1$ and strictly decreasing on $[1, \infty)$. Then $\int_{-\infty}^{-1} \frac{1}{|x|^{\alpha}} dt = \int_{\infty}^1 \frac{1}{t^{\alpha}} dt$. Hence the convergence occur if and only if $\alpha > 1$.

3. For $x = u(t) = \frac{1}{t}$, we have u is of class $\mathcal{C}^1$ and strictly decreasing on $[1, \infty)$. Then, $\int_0^1 \frac{1}{x^{\alpha}} dx = - \int_{+\infty}^1 \frac{1}{t^{-\alpha+2}} dt$. Hence the convergence occur if and only if $\alpha < 1$. For the other integrals we take respectively $x - a = u(t) = \frac{1}{t}$ and $b - x = u(t) = \frac{1}{t}$. $\square$

3.19 Property. Let $f \in \mathcal{R}_{loc}([a, b))$, $b \in \overline{\mathbb{R}}$. If $\int_a^b f(t) dt$ is convergent, then $\lim_{x \rightarrow b^-} \int_x^b f(t) dt = 0$.

Proof. Since the integral is convergent, then by Chasles relation and for $x \in (a, b)$ we have

$$\int_a^b f(t) dt = \int_a^x f(t) dt + \int_x^b f(t) dt.$$

Then by passing to the limit as $x \rightarrow b^-$ we obtain the desired result

$$\int_a^b f(t) dt = \int_a^b f(t) dt + \lim_{x \rightarrow b^-} \int_x^b f(t) dt \Leftrightarrow 0 = \lim_{x \rightarrow b^-} \int_x^b f(t) dt.$$

□

3.20 Property. Let $f \in \mathcal{R}_{loc}([a, b))$, $b \in \overline{\mathbb{R}}$ be such that $f(x) \geq 0$ for all $x \in [a, b)$. Then the function $F(x) = \int_a^x f(t) dt$ is an increasing function on $[a, b)$.

Proof. let $x, y \in [a, b)$ so that $x < y$. Then

$$F(x) - F(y) = \int_a^x f(t) dt - \int_a^y f(t) dt = \int_y^a f(t) dt + \int_a^x f(t) dt = \int_y^x f(t) dt = - \int_x^y f(t) dt \leq 0.$$

□

3.21 Corollary. Let $f \in \mathcal{R}_{loc}([a, b))$, $b \in \overline{\mathbb{R}}$ be such that $f(x) \geq 0$ for all $x \in [a, b)$. Then the integral $\int_a^b f(t) dt = \lim_{x \rightarrow b^-} F(x)$ is convergent if F is bounded above and diverges to $+\infty$ elsewhere.

3.22 Property. (Comparison Test)

Let $f, g \in \mathcal{R}_{loc}([a, b))$, $b \in \overline{\mathbb{R}}$ be such that $0 \leq f(t) \leq g(t)$ for all $t \in [a, b)$. Then

1. If $\int_a^b g(t) dt$ converges, then $\int_a^b f(t) dt$ converges and we have $0 \leq \int_a^b f(t) dt \leq \int_a^b g(t) dt$.
2. If $\int_a^b f(t) dt$ diverges, then $\int_a^b g(t) dt$ diverges to $+\infty$.

Proof. 1. We have $0 \leq f(t) \leq g(t)$ for all $t \in [a, b)$, then

$$0 \leq F(x) = \int_a^x f(t) dt \leq G(x) = \int_a^x g(t) dt.$$

Since $\int_a^b g(t) dt$ is convergent, then $G(x)$ is bounded and so $F(x)$ is bounded above. Therefore $\int_a^b f(t) dt$ converges and by passing to the limit as $x \rightarrow b^-$ of both sides in the above inequality, we obtain $0 \leq \int_a^b f(t) dt \leq \int_a^b g(t) dt$.

2. Suppose to the contrary that $\int_a^b g(t) dt$ converges, then by first item we obtain $\int_a^b f(t) dt$ converges. But we have $\int_a^b f(t) dt$ diverges to ∞ . Thus $\int_a^b g(t) dt$ diverges to ∞ . □

3.23 Examples.

1. Consider $I := \int_1^\infty \frac{(\sin t)^2}{1 + \cos t + t^2} dt$. Since the integrand $f \in \mathcal{C}([1, \infty))$, then $f \in \mathcal{R}_{loc}([1, \infty))$ and we have $0 \leq f(t) \leq \frac{1}{t^2}$ for all $t \in [1, \infty)$. We have $\int_1^\infty \frac{1}{t^2} dt$ is convergent (Riemann Example), then by comparison test I is convergent and we have

$$0 \leq I := \int_1^\infty \frac{(\sin t)^2}{1 + \cos t + t^2} dt \leq \int_1^\infty \frac{1}{t^2} dt = 1.$$

2. Consider $J := \int_0^\infty \frac{(\sin t)^2}{1 + \cos t + t^2} dt$. Since the integrand $f \in \mathcal{C}([0, \infty))$, then $f \in \mathcal{R}_{loc}([0, \infty))$. Since I converges, then by Chasles Relation J also converges.
3. Consider $K := \int_2^\infty \frac{1}{\ln t} dt$. Since the integrand $f \in \mathcal{C}([2, \infty))$, then $f \in \mathcal{R}_{loc}([2, \infty))$ and we have $0 \leq \frac{1}{t} \leq \frac{1}{\ln t}$ for all $t \in [2, \infty)$. We have $\int_1^\infty \frac{1}{t} dt$ is divergent (Riemann Example), then by comparison test K is divergent to ∞ .

3.24 Definition. (Absolute And Conditional Convergence)

Let $f \in \mathcal{R}_{loc}([a, b))$, $b \in \overline{\mathbb{R}}$. The improper integral $\int_a^b f(t) dt$ converges absolutely if the integral $\int_a^b |f(t)| dt$ converges. However, if $\int_a^b f(t) dt$ converges and $\int_a^b |f(t)| dt$ diverges, then the integral $\int_a^b f(t) dt$ is said to be **Conditionally convergent**.

The notion of absolute convergence is useful functions with arbitrary sign.

3.25 Property. (Absolute Convergence Test)

Let $f \in \mathcal{R}_{loc}([a, b))$, $b \in \overline{\mathbb{R}}$. If $\int_a^b f(t) dt$ converges absolutely then it also converges and $\int_a^b f(t) dt \leq \int_a^b |f(t)| dt$.

Proof. Note that $0 \leq f(t) + |f(t)| \leq 2|f(t)|$ for any $t \in [a, b)$. Since $\int_a^b |f(t)| dt$ converges, then the comparison test tell us that $\int_a^b [f(t) + |f(t)|] dt$ converges. Thus, due to linearity of integral, $\int_a^b f(t) dt = \int_a^b ([f(t) + |f(t)|] - |f(t)|) dt$ is convergent. Since $0 \leq f(t) + |f(t)| \leq 2|f(t)|$ for any $t \in [a, b)$, then by comparison test and linearity of integral we obtain $\int_a^b f(t) dt \leq \int_a^b |f(t)| dt$. $\square$

3.26 Remark. (Modified Comparison Test)

Let $f, g \in \mathcal{R}_{loc}([a, b))$, $b \in \overline{\mathbb{R}}$ such that $|f(t)| \leq g(t)$ for all $t \in [a, b)$ or near b^- . If $\int_a^b g(t) dt$ converges, then $\int_a^b f(t) dt$ is absolutely convergent and hence convergent.

3.27 Example. Consider $I := \int_1^\infty \frac{\sin t}{t^{\alpha+1}} dt$, $0 < \alpha$. We have $\left| \frac{\sin t}{t^{\alpha+1}} \right| = \frac{|\sin t|}{t^{\alpha+1}} \leq \frac{1}{t^{\alpha+1}}$. Since $\int_a^b \frac{1}{t^{\alpha+1}} dt$ converges, Riemann Examples $\alpha + 1 > 1$, then by Comparison Test I is absolutely convergent and hence convergent.

3.28 Property. (Cauchy Criterion)

Let $f \in \mathcal{R}_{loc}([a, b))$, $b \in \overline{\mathbb{R}}$. Then

$$\int_a^b f(t) dt \text{ converges} \Leftrightarrow \lim_{(X_1, X_2) \rightarrow (b^-, b^-)} \int_{X_1}^{X_2} f(t) dt = 0. \text{ Equivalently;}$$

$$\int_a^b f(t) dt \text{ converges} \Leftrightarrow \forall \varepsilon > 0, \exists x_0 \in [a, b) \text{ such that } \left| \int_{X_1}^{X_2} f(t) dt \right| < \varepsilon \text{ holds } \forall X_1, X_2 \in [x_0, b).$$

Proof. Let $F(x) = \int_a^x f(t) dt$ for all $x \in [a, b)$. By definition, $\int_a^b f(t) dt$ converges if and only if $\lim_{x \rightarrow b^-} F(x)$ exists. We have

$$\int_{X_1}^{X_2} f(t) dt = \int_a^{X_2} f(t) dt - \int_a^{X_1} f(t) dt = F(X_2) - F(X_1).$$

Therefore the equivalence is simply the Cauchy criterion for the existence of $\lim_{x \rightarrow b^-} F(x)$. $\square$

It is useful to note that the following inequality $\left| \int_{X_1}^{X_2} f(t) dt \right| \leq \left| \int_{X_1}^{X_2} |f(t)| dt \right|$ holds for proper integrals. Hence, Cauchy Criterion gives that absolute convergence implies convergence.

3.29 Example. Consider $I := \int_1^\infty \frac{\cos t}{t^\alpha} dt$, $0 < \alpha < 1$.

- I is convergent by Cauchy Criterion;

By integration by parts for proper integrals and Cauchy Criterion we have

$$\int_{X_1}^{X_2} \frac{1}{t^\alpha} \cos t dt = \frac{\sin X_2}{X_2^\alpha} - \frac{\sin X_1}{X_1^\alpha} + \alpha \int_{X_1}^{X_2} \frac{\sin t}{t^{\alpha+1}} dt \xrightarrow{(X_1, X_2) \rightarrow (\infty, \infty)} 0.$$

Thus, again by Cauchy Criterion I is convergent.

- I is convergent by Integration By Parts For Improper Integrals;

$$\int_1^\infty \frac{1}{t^\alpha} \cos t \, dt = \left. \frac{\sin t}{t^\alpha} \right|_1^\infty + \alpha \int_1^\infty \frac{\sin t}{t^{\alpha+1}} \, dt = -\sin 1 + \alpha \int_1^\infty \frac{\sin t}{t^{\alpha+1}} \, dt.$$

Since the integral to the right side is convergent, then I is convergent.

- I is not absolutely convergent;

We have, $\left| \frac{\cos t}{t^\alpha} \right| \geq \frac{\cos^2 t}{t^\alpha} = \frac{1 + \cos(2t)}{2t^\alpha} = \frac{1}{2t^\alpha} + \frac{\cos(2t)}{2t^\alpha}$. Since $\int_1^\infty \frac{1}{2t^\alpha} \, dt$ is divergent and $\int_1^\infty \frac{\cos(2t)}{2t^\alpha} \, dt$ is convergent, then $\int_1^\infty \frac{\cos^2 t}{t^\alpha} \, dt$ is divergent. Consequently, and by comparison test we obtain that I is not absolutely convergent.

3.30 Definition. (Landau Notations)

Let f and g be two functions defined in a neighborhood of $a \in \overline{\mathbb{R}}$. We say that

1. f is dominated by g in a neighborhood of a , and we write $f = O_{x \rightarrow a}(g)$, if there exist a function b defined in a neighborhood of a such that
 - $f(x) = b(x)g(x)$ in a neighborhood of a .
 - b is bounded in a neighborhood of a .
2. f is negligible with respect to g in a neighborhood of a , and we write $f = o_{x \rightarrow a}(g)$, if there exist function ε defined in a neighborhood of a such that
 - $f(x) = \varepsilon(x)g(x)$ in a neighborhood of a .
 - $\varepsilon(x) \xrightarrow{x \rightarrow a} 0$.
3. f is equivalent to g in a neighborhood of a , and we write $f \sim_{x \rightarrow a}(g)$, if there exist function ε defined in a neighborhood of a such that
 - $f(x) = (\varepsilon(x) + 1)g(x)$ in a neighborhood of a .
 - $\varepsilon(x) \xrightarrow{x \rightarrow a} 0$.

Note that $f \sim_{x \rightarrow a} g \Leftrightarrow f - g = o_{x \rightarrow a}(g)$.

3.31 Property. Let f and g be two functions defined in a neighborhood N of $a \in \overline{\mathbb{R}}$. If g never vanishes on $N \setminus \{a\}$, then:

1. $f = O_{x \rightarrow a}(g) \Leftrightarrow$ the function $\frac{f}{g}$ is bounded in a neighborhood of a .
2. $f = o_{x \rightarrow a}(g) \Leftrightarrow \frac{f(x)}{g(x)} \xrightarrow{x \rightarrow a} 0$.
3. $f \sim_{x \rightarrow a}(g) \Leftrightarrow \frac{f(x)}{g(x)} \xrightarrow{x \rightarrow a} 1$.

3.32 Property. Let $f, g \in \mathcal{R}_{loc}([a, b))$, $b \in \overline{\mathbb{R}}$. Then

1. If $f = O_{x \rightarrow b}(g)$ and $\int_a^b g(t) dt$ converges absolutely, then $\int_a^b f(t) dt$ converges absolutely.
2. If $f = o_{x \rightarrow b}(g)$ and $\int_a^b g(t) dt$ converges absolutely, then $\int_a^b f(t) dt$ converges absolutely.
3. (Equivalent test)
If $f \sim_{x \rightarrow b} g$, then $\int_a^b |f(t)| dt$ and $\int_a^b |g(t)| dt$ have same nature.

Proof. 1. Assume $f = O_{x \rightarrow b}(g)$, there exist a bounded function B defined in a neighborhood of b such that $f(x) = B(x)g(x)$ in a neighborhood of b . Hence there exists $M > 0$ such that $|f(x)| \leq M|g(x)|$ in a neighborhood of b . Since $\int_a^b |g(t)| dt$ converges, then by comparison test $\int_a^b f(t) dt$ converges absolutely.

2. Evident. We use, $f = o_{x \rightarrow b}(g) \Rightarrow f = O_{x \rightarrow b}(g)$.

3. Evident. We use, $f \sim_{x \rightarrow b} g \Rightarrow (f = O_{x \rightarrow b}(g) \wedge g = O_{x \rightarrow b}(f))$.

□

3.33 Examples.

1. Consider $I = \int_1^\infty \ln(1 + \frac{\sin t}{t}) dt$. Since the integrand $f \in \mathcal{C}([1, \infty))$, then $f \in \mathcal{R}_{loc}([1, \infty))$ and we have $f(t) = \ln(1 + \frac{\sin t}{t}) = \frac{\sin t}{t} - \frac{\sin^2 t}{2t^2} + o(\frac{\sin^2 t}{t^2})$. Since $\int_1^\infty \frac{\sin t}{t} dt$ is convergent, integration by parts, and $\int_1^\infty \frac{\sin^2 t}{t^2} dt$ is convergent, by comparison test and Riemann examples, then I is convergent. Note that we cannot use equivalent test here because $\frac{\sin t}{t}$ does not have constant sign near ∞ .
2. Consider $J = \int_1^\infty \ln(1 + \frac{\sin^2 t}{t}) dt$. Since the integrand $f \in \mathcal{C}([1, \infty))$, then $f \in \mathcal{R}_{loc}([1, \infty))$ and we have $\ln(1 + \frac{\sin^2 t}{t}) \sim_{x \rightarrow \infty} \frac{\sin^2 t}{t} \geq 0$ near ∞ . Since $\int_1^\infty \frac{\sin^2 t}{t} dt = \int_1^\infty \frac{1 - \cos(2t)}{2t} dt$ is divergent, then J is divergent.

3.34 Theorem. (Limit Form Of Comparison Test)

Let $f, g \in \mathcal{R}_{loc}([a, b))$, $b \in \overline{\mathbb{R}}$ be such that $f(x) \geq 0$ and $g(x) > 0$ in neighborhood of b .

1. If $\lim_{x \rightarrow b^-} \frac{f(x)}{g(x)} = \ell > 0$, then $\int_a^b g(t) dt$ and $\int_a^b f(t) dt$ have same nature.
2. If $\lim_{x \rightarrow b^-} \frac{f(x)}{g(x)} = 0$ and $\int_a^b g(t) dt$ converges, then $\int_a^b f(t) dt$ converges.

3. If $\lim_{x \rightarrow b^-} \frac{f(x)}{g(x)} = \infty$ and $\int_a^b g(t) dt$ diverges, then $\int_a^b f(t) dt$ diverges to ∞ .

Proof. 1. We use, $\lim_{x \rightarrow b^-} \frac{f(x)}{g(x)} = \ell > 0 \Rightarrow f \underset{x \rightarrow b}{\sim} \ell g$.

2. We use, $\lim_{x \rightarrow b^-} \frac{f(x)}{g(x)} = 0 \Rightarrow f = o_{x \rightarrow b}(g)$.

3. We use, $\lim_{x \rightarrow b^-} \frac{f(x)}{g(x)} = \infty \Rightarrow \frac{f(x)}{g(x)} > M \Rightarrow f(x) > Mg(x)$ for some positive M in a neighborhood of b . □

3.35 Corollary. (x^α Test)

• Let $f \in \mathcal{R}_{loc}([1, \infty))$ and $\alpha \in \mathbb{R}$.

1. If $x^\alpha f(x) \xrightarrow{x \rightarrow \infty} \ell \neq 0$, then $\int_1^\infty |f(t)| dt$ converges for $\alpha > 1$ and diverges for $\alpha \leq 1$.

2. If $x^\alpha f(x) \xrightarrow{x \rightarrow \infty} 0$ and $\alpha > 1$, then $\int_1^\infty |f(t)| dt$ converges.

3. If $x^\alpha f(x) \xrightarrow{x \rightarrow \infty} \infty$ and $\alpha \leq 1$, then $\int_1^\infty f(t) dt$ diverges.

• Let $f \in \mathcal{R}_{loc}([a, b))$ and $a, b, \alpha \in \mathbb{R}$.

1. If $(b-x)^\alpha f(x) \xrightarrow{x \rightarrow b} \ell \neq 0$, then $\int_a^b |f(t)| dt$ converges for $\alpha < 1$ and diverges for $\alpha \geq 1$.

2. If $(b-x)^\alpha f(x) \xrightarrow{x \rightarrow b} 0$ and $\alpha < 1$, then $\int_a^b |f(t)| dt$ converges.

3. If $(b-x)^\alpha f(x) \xrightarrow{x \rightarrow b} \infty$ and $\alpha \geq 1$, then $\int_a^b f(t) dt$ diverges.

• Let $f \in \mathcal{R}_{loc}((a, b])$ and $a, b, \alpha \in \mathbb{R}$.

1. If $(x-a)^\alpha f(x) \xrightarrow{x \rightarrow a} \ell \neq 0$, then $\int_a^b |f(t)| dt$ converges for $\alpha < 1$ and diverges for $\alpha \geq 1$.

2. If $(x-a)^\alpha f(x) \xrightarrow{x \rightarrow a} 0$ and $\alpha < 1$, then $\int_a^b |f(t)| dt$ converges.

3. If $(x-a)^\alpha f(x) \xrightarrow{x \rightarrow a} \infty$ and $\alpha \geq 1$, then $\int_a^b f(t) dt$ diverges.

3.36 Property. (Bertrand Examples)

Let $\alpha, \beta \in \mathbb{R}$. Then

1. $I(\alpha, \beta) = \int_e^\infty \frac{1}{x^\alpha \ln^\beta x} dx$ converges $\Leftrightarrow (\alpha > 1)$ or $(\alpha = 1 \text{ and } \beta > 1)$.

$$2. J(\alpha, \beta) = \int_0^{\frac{1}{e}} \frac{1}{x^\alpha |\ln x|^\beta} dx \text{ converges} \Leftrightarrow (\alpha < 1) \text{ or } (\alpha = 1 \text{ and } \beta > 1).$$

Proof. 1. We discuss three cases:

- Case one $\alpha > 1$: Choose $\mu \in (1, \alpha)$ for example $\mu = \frac{1+\alpha}{2}$. Since $\lim_{x \rightarrow \infty} \frac{x^\mu}{x^\alpha \ln^\beta x} = 0$ and $\mu > 1$, then $I(\alpha, \beta)$ converges.
- Case two $\alpha < 1$: Choose $\mu \in (\alpha, 1)$ for example $\mu = \frac{1+\alpha}{2}$. Since $\lim_{x \rightarrow \infty} \frac{x^\mu}{x^\alpha \ln^\beta x} = \infty$ and $\mu < 1$, then $I(\alpha, \beta)$ diverges.
- Case three $\alpha = 1$: We perform the change of variable $x = u(t) = e^t$ which is of class $\mathcal{C}^1$ and strictly increasing on $[1, \infty)$. Then $\int_e^\infty \frac{1}{x \ln^\beta x} dx = \int_1^\infty \frac{1}{t^\beta} dt$ converges if $\beta > 1$. Thus we conclude that $I(\alpha, \beta)$ converges if and only if $(\alpha > 1)$ or $(\alpha = 1 \text{ and } \beta > 1)$.

2. We perform the the change of variable $x = u(t) = \frac{1}{t}$ which is of class $\mathcal{C}^1$ and strictly decreasing on $[e, \infty)$. Then $J(\alpha, \beta) = \int_0^{\frac{1}{e}} \frac{1}{x^\alpha |\ln x|^\beta} dx = \int_e^\infty \frac{1}{t^{-\alpha+2} \ln^\beta t} dt$. Hence by first item;

$$J(\alpha, \beta) \text{ converges} \Leftrightarrow (-\alpha+2 > 1) \text{ or } (-\alpha+2 = 1 \text{ and } \beta > 1) \Leftrightarrow (\alpha < 1) \text{ or } (\alpha = 1 \text{ and } \beta > 1).$$

□

There are improper integrals which are convergent but not absolutely convergent. These are often quite subtle to handle. We give Abel-Dirichlet test which is useful to deal with such integrals. The basic tool for this test is the second mean value theorem and hence essentially the integration by parts formula.

3.37 Property. (Abel-Dirichlet Test)

Let $f, g \in \mathcal{R}_{loc}([a, b))$, $b \in \overline{\mathbb{R}}$ be such that g is monotonic. Then $\int_a^b f(t)g(t) dt$ converges if one of the following pairs of conditions hold:

C_1 : g is bounded and $\int_a^b f(t) dt$ converges. Or

C_2 : $g(t) \xrightarrow[t \rightarrow b^-]{} 0$ and $F(x) = \int_a^x f(t) dt$ is bounded on $[a, b)$.

Proof. Let $X_1, X_2 \in [a, b)$. Then by the second mean-value theorem, there exists $\xi \in [X_1, X_2]$ such that

$$\int_{X_1}^{X_2} f(t)g(t) dt = g(X_1) \int_{X_1}^\xi f(t) dt + g(X_2) \int_\xi^{X_2} f(t) dt.$$

Hence by the Cauchy Criterion, we conclude that the integral converges if either of the two pairs of conditions holds. □

3.4 Schwarz Inequality For Improper Integrals.

We generalize Schwarz inequality for Riemann integrals to improper integrals by a limit process.

3.38 Property. Let $f, g \in \mathcal{R}_{loc}([a, b))$, $b \in \overline{\mathbb{R}}$. If $\int_a^b f^2(t) dt$ and $\int_a^b g^2(t) dt$ are convergent, then $\int_a^b f(t)g(t) dt$ is convergent and we have

1. $\left(\int_a^b f(t)g(t) dt \right)^2 \leq \int_a^b f^2(t) dt \int_a^b g^2(t) dt.$
2. $\left(\int_a^b f(t)g(t) dt \right)^2 = \int_a^b f^2(t) dt \int_a^b g^2(t) dt \Leftrightarrow f = \lambda g, \lambda \in \mathbb{R}.$

Proof. The inequality $|fg| \leq \frac{(f^2 + g^2)}{2}$ ensures the absolute convergence of $\int_a^b f(t)g(t) dt$.

1. Schwarz inequality for Riemann integral gives

$$\forall x \in [a, b), \left(\int_a^x f(t)g(t) dt \right)^2 \leq \int_a^x f^2(t) dt \int_a^x g^2(t) dt.$$

By passing to the limit as $x \rightarrow b^-$ we obtain

$$\left(\int_a^b f(t)g(t) dt \right)^2 \leq \int_a^b f^2(t) dt \int_a^b g^2(t) dt.$$

2. Schwarz equality for Riemann integral gives

$$\forall x \in [a, b), \left(\int_a^x f(t)g(t) dt \right)^2 = \int_a^x f^2(t) dt \int_a^x g^2(t) dt \Leftrightarrow f = \lambda g, \lambda \in \mathbb{R}.$$

By passing to the limit as $x \rightarrow b^-$ we obtain

$$\left(\int_a^b f(t)g(t) dt \right)^2 = \int_a^b f^2(t) dt \int_a^b g^2(t) dt \Leftrightarrow f = \lambda g, \lambda \in \mathbb{R}.$$

□

3.5 Integration Of Comparison Relations.

3.39 Property. Let $f, g \in \mathcal{R}_{loc}([a, b))$ and g positive on $[a, b)$, $b \in \overline{\mathbb{R}}$. Assume that $\int_a^b g(t) dt$ is convergent, then for x tends to b^- we have

1. If $f = O_{t \rightarrow b}(g)$ then, $\int_x^b f(t) dt = \int_x^b O_{t \rightarrow b}(g) dt = O_{x \rightarrow b} \left(\int_x^b g(t) dt \right).$
2. If $f = o_{t \rightarrow b}(g)$ then, $\int_x^b f(t) dt = \int_x^b o_{t \rightarrow b}(g) dt = o_{x \rightarrow b} \left(\int_x^b g(t) dt \right).$
3. If $f \sim_{t \rightarrow b} g$, then $\int_x^b f(t) dt \sim_{x \rightarrow b} \int_x^b g(t) dt.$

Proof. Exercise. □

3.40 Corollary. Let $f \in \mathcal{R}_{loc}((a, b))$, $a, b \in \overline{\mathbb{R}}$ and $x_0 \in (a, b)$. Assume that f has finite expansion up to order n at x_0 , $f(x) = \sum_{i=0}^n a_i x^i + o_{x \rightarrow x_0}(x - x_0)^n$. Then the function $x \mapsto \int_{x_0}^x f(t) dt$ has finite expansion up to order $n + 1$ at x_0 and we have $\int_{x_0}^x f(t) dt = \int_{x_0}^x \sum_{i=0}^n a_i t^i dt + o_{x \rightarrow x_0}(x - x_0)^{n+1}$.

3.41 Property. Let $f, g \in \mathcal{R}_{loc}([a, b))$ and f, g be positive on $[a, b)$, $b \in \overline{\mathbb{R}}$. Assume that $\int_a^b g(t) dt$ is divergent, then for x tends to b^- we have

1. If $f = O_{t \rightarrow b}(g)$ then, $\int_a^x f(t) dt = O_{x \rightarrow b}\left(\int_a^x g(t) dt\right)$.

2. If $f = o_{t \rightarrow b}(g)$ then, $\int_a^x f(t) dt = o_{x \rightarrow b}\left(\int_a^x g(t) dt\right)$.

3. If $f \sim_{t \rightarrow b} g$, then $\int_a^x f(t) dt \sim_{x \rightarrow b} \int_a^x g(t) dt$.

3.42 Example.

$$\frac{1}{\ln(1+t)} \underset{t \rightarrow 0^+}{\sim} \frac{1}{t} \Rightarrow \int_x^1 \frac{1}{\ln(1+t)} dt \underset{x \rightarrow 0^+}{\sim} \ln \frac{1}{x}.$$

3.6 Exercises

Exercise 1. Study the nature of the following improper integrals

$$\begin{aligned} 1) & \int_0^\infty e^x dx, & 2) & \int_0^\infty e^{-x} dx, & 3) & \int_{-\infty}^\infty x dx, & 4) & \int_1^\infty \frac{1}{\sqrt{x^3}} dx, \\ 5) & \int_0^\infty \sin(5x) dx, & 6) & \int_e^\infty \frac{1}{x \ln x} dx, & 7) & \int_e^\infty \frac{1}{x(\ln x)^2} dx, & 8) & \int_0^\infty \frac{\cos x}{1+x^2} dx, \\ 9) & \int_0^\infty \frac{x+1}{\sqrt{x^4+1}} dx, & 10) & \int_1^\infty \frac{\ln x}{\sqrt{x}} dx, & 11) & \int_0^\infty \cos x^2 dx, & 12) & \int_0^\infty \sin(x^2+x+1) dx. \end{aligned}$$

Exercise 2. Study the nature of the following improper integrals

$$\begin{aligned} 1) & \int_{-1}^1 \frac{1}{x^2} dx, & 2) & \int_0^1 \frac{1}{\sqrt{x^3}} dx, & 3) & \int_{-2}^0 \frac{1}{(x+1)^2} dx, & 4) & \int_0^2 \frac{1}{\sqrt{2x}} dx, \\ 5) & \int_0^2 \frac{1}{\sqrt{2x-x}} dx, & 6) & \int_0^{\frac{\pi}{2}} \frac{1}{1+(\sin x)^2} dx, & 7) & \int_0^{\frac{\pi}{2}} \sqrt{\tan x} dx, & 8) & \int_a^b \frac{1}{\sqrt{(x-a)(b-x)}} dx, \\ 9) & \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \ln(1+\sin x) dx, & 10) & \int_0^\infty \frac{e^{-\sqrt{x^2+x+1}}}{\sqrt{x}} dx, & 11) & \int_0^1 \frac{\ln x}{\sqrt{1-x}} dx, & 12) & \int_0^1 \frac{\ln(1-x^2)}{x^2} dx. \end{aligned}$$

Exercise 3. Prove that the following integrals are semi-convergent

$$1) \int_1^\infty \frac{\cos x}{\sqrt{x}} dx, \quad 2) \int_{-2}^\infty \cos x^2 dx, \quad 3) \int_1^\infty x^2 \sin x^4 dx.$$

Exercise 4.

1. Show that $\int_1^\infty \frac{x-1}{(x+1)(x^2-x+1)} dx$ is convergent and determine its value.
2. Study the nature of $\int_1^\infty \frac{x+2}{(x-1)(x^2+x+1)} dx$.

Exercise 5. Let $x \in [0, 1[$ and $I(x) = \int_0^x t \ln\left(\frac{1+t}{1-t}\right) dt$. Calculate $I(x)$ and find $\lim_{x \rightarrow 1^-} I(x)$.

Exercise 6. Let $I = \int_0^1 \frac{\ln x}{1+x^2}$ and $J = \int_1^\infty \frac{\ln x}{1+x^2}$.

1. Show that I and J are convergent. Calculate $I + J$.
2. Let $a \in \mathbb{R}_+^*$. Show that $\int_0^\infty \frac{\ln x}{1+a^2x^2} = -\frac{\pi \ln a}{2a}$.

Exercise 7. Let $p, q \in \mathbb{R}$. Study the nature of $I(p, q) = \int_{-2}^3 (x+2)^p (3-x)^{q-1} dx$.

Exercise 8. Let $p \in \mathbb{R}$ and $\Gamma(p) = \int_0^\infty x^{p-1} e^{-x} dx$.

1. Study the nature of $\Gamma(p)$.
2. Find a relation between $\Gamma(p+1)$ and $\Gamma(p)$.
3. Deduce that $\Gamma(n+1) = n!$ for all $n \in \mathbb{N}$.

Exercise 9. Study according to the values of α the nature of $\int_0^1 \frac{x^\alpha + x^{1-\alpha}}{1+x} dx$.

Exercise 10. Let $n \in \mathbb{Z}$ and $a \in]0, 1[$.

1. Calculate $I_n(a) = \int_a^1 x^n \ln x dx$ in terms of n and a .
2. Establish for what values of n the integral $\int_0^1 x^n \ln x dx$ is convergent and give its value.

Exercise 11. Consider the sequence $u_n = \int_0^1 \frac{x^n}{\sqrt{x(1-x)}} dx$.

1. Show that u_n is well defined.
2. Show that $\{u_n\}_n$ is decreasing and bounded below. Deduce the nature of $\{u_n\}_n$.
3. Compute u_0, u_1 and show that $2nu_n = (2n-1)u_{n-1}$ for $n \geq 1$.
4. Determine the value of u_n in terms of n .

Exercise 12. Study according to the values of $\alpha \in \mathbb{R}$ the nature of the following integrals

$$\begin{aligned} 1) \int_0^\infty \frac{x^{\alpha+1} + x^{3-\alpha}}{x^4 + \sqrt{x^3}} dx, \quad 2) \int_0^\infty \frac{\sin^2 x}{x^\alpha} dx, \quad 3) \int_0^\infty \frac{1}{x^\alpha [1 + (\ln x)^2]} dx, \\ 4) \int_0^1 \frac{\cos^\alpha(\frac{\pi x}{2})}{\sqrt{x(1-x)}} dx, \quad 5) \int_0^\infty \frac{x^{2\alpha-2} + x^{\alpha+1}}{x^3 \sqrt{x^3} (3+x)^{\alpha+\frac{1}{2}}} dx, \quad 6) \int_0^\infty \frac{x \ln x}{(1+x^2)^\alpha} dx. \end{aligned}$$

Exercise 13. Let $x \in \mathbb{R}_+^*$.

1. Prove that $\int_0^1 \frac{x \sin t}{t(t+x)} dt$ and $\int_1^\infty \frac{x \sin t}{t(t+x)} dt$ are convergent.
2. Let $u(x) = \int_0^1 \frac{x \sin t}{t(t+x)} dt$ and $v(x) = \int_1^\infty \frac{x \sin t}{t(t+x)} dt$.
 - (a) Show that $0 \leq u(x) \leq x \ln \left(\frac{x+1}{x} \right)$.
 - (b) Show that $\lim_{x \rightarrow 0^+} v(x) = 0$.
3. Deduce that $\lim_{x \rightarrow 0^+} \int_0^\infty \frac{x \sin t}{t(t+x)} dt = 0$.

Exercise 14. Let $f : [1, \infty[\rightarrow \mathbb{R}$ be a continuous function such that $\int_1^\infty f(x) dx$ is convergent.

Show that $\int_1^\infty \frac{f(x)}{x^\alpha} dx$ is convergent for $\alpha > 0$.

Exercise 15. Find $\lim_{x \rightarrow \infty} \int_0^\infty \frac{\sin t}{t+x} dt$.

Exercise 16. Let $f : [0, \infty[\rightarrow \mathbb{R}$ be a continuous function such that $\int_1^\infty \frac{f(t)}{t} dt$ is convergent.

Assume that $0 < \varepsilon < X$ and $0 < a < b$.

1. Show that $\int_\varepsilon^X \frac{f(at) - f(bt)}{t} dt = \int_{a\varepsilon}^{b\varepsilon} \frac{f(t)}{t} dt - \int_{aX}^{bX} \frac{f(t)}{t} dt$.
2. Deduce that $\int_0^\infty \frac{f(at) - f(bt)}{t} dt$ is convergent to $f(0) \ln \left(\frac{b}{a} \right)$.
3. Calculate $\int_0^\infty \frac{\cos(at) - \cos(bt)}{t} dt$ and $\int_0^\infty \frac{e^{-at} - e^{-bt}}{t} dt$.

4.1 Numerical Series.

The sum of finitely many real numbers $a_0, a_1, \dots, a_n$ is a given real number $\sum_{k=0}^n a_k$. Sometimes we wish to form the sum of an infinite sequence of real numbers $a_0 + a_1 + \dots + a_n + \dots = \sum_{k=0}^{\infty} a_k$. For example, if a man walks halfway across a road of length two units and repeats this process indefinitely, then one can think the total distance he will travel is an infinite sum

$$2 = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n} + \dots = \sum_{k=0}^{\infty} \frac{1}{2^k}.$$

In $(n+1)$ steps he will travel n -th partial sum units

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n} = \sum_{k=0}^n \frac{1}{2^k} = 2 - \frac{1}{2^{n-1}} \text{ units.}$$

Thus he will get closer and closer to the other side of the road as $n \rightarrow \infty$ and we have

$$\lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{1}{2^k} = 2 = \sum_{k=0}^{\infty} \frac{1}{2^k},$$

the limit of the n -th partial sum equals to the infinite sum. This infinite sum can be defined using sequences and leads to the notion of series.

Given a sequence $\{a_k\}_{k \geq 0}$ we can form another **sequence of partial sums** $\{s_n\}_{n \geq 0}$ by

$$s_n = \sum_{k=0}^n a_k = a_0 + a_1 + \dots + a_{n-1} + a_n.$$

Let us (formally) define

$$\sum_{k=0}^{\infty} a_k = \lim_{n \rightarrow \infty} \sum_{k=0}^n a_k = \lim_{n \rightarrow \infty} s_n.$$

The symbol $\sum_{k=0}^{\infty} a_k$ (or $\sum_{k \geq 0} a_k$) is called **series**, $a_k = s_k - s_{k-1}$ is the **general term of the series** and k is called the **index**. Note that the index is a dummy variable

$$a_0 + a_1 + \dots + a_n + \dots = \sum_{k=0}^{\infty} a_k = \sum_{i=0}^{\infty} a_i = \sum_{n=0}^{\infty} a_n,$$

and it may start from some $m \in \mathbb{N}$. In this case, formally

$$\sum_{k=m}^{\infty} a_k = \lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \sum_{k=m}^n a_k = \lim_{n \rightarrow \infty} \sum_{k=m}^{n+m} a_k.$$

4.1 Definition. Given a sequence $\{a_k\}_{k \geq 0}$ and $s_n = \sum_{k=0}^n a_k$. Consider the limit $\lim_{n \rightarrow \infty} s_n$.

- The series is said to **converge** to a real number s if the sequence of partial sums converges to s , and one writes

$$s = \lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \sum_{k=0}^n a_k = \sum_{k=0}^{\infty} a_k.$$

- The series is said to **diverge** or **diverge to ∞** if the sequence of partial sums diverges, or diverges to ∞ , respectively.

In the first case we say that the series is **convergent** and in the other case we say that the series is **divergent**.

4.2 Property. (Necessary But Not Sufficient Condition For Convergence)

If the series $\sum_{n=0}^{\infty} a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$. Equivalently;

If $\lim_{n \rightarrow \infty} a_n \neq 0$, then the series $\sum_{n=0}^{\infty} a_n$ is divergent.

Proof. Suppose $s = \lim_{n \rightarrow \infty} s_n$. Since $a_n = s_n - s_{n-1}$, then $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (s_n - s_{n-1}) = s - s = 0$. $\square$

4.3 Examples.

1. The series $\sum_{k=0}^{\infty} \frac{1}{2^k}$ converges to 2.

Using the identity $1 + x + x^2 + \cdots + x^n = \frac{1 - x^{n+1}}{1 - x}$ for $x \neq 1$, then we obtain

$$s_n = \sum_{k=0}^n \frac{1}{2^k} = \frac{1 - (\frac{1}{2})^{n+1}}{1 - \frac{1}{2}} = 2 - \frac{1}{2^n}.$$

Since $\lim_{n \rightarrow \infty} s_n = 2$, then the series $\sum_{k=0}^{\infty} \frac{1}{2^k}$ converges to 2 and we have $\sum_{k=0}^{\infty} \frac{1}{2^k} = 2$.

2. The series $\sum_{k=0}^{\infty} (-1)^k$ diverges.

Since $s_{2n} = 1$ and $s_{2n+1} = 0$, then $\lim_{n \rightarrow \infty} s_n$ does not exist. So the series $\sum_{k=0}^{\infty} (-1)^k$ diverges. Or, since $\lim_{n \rightarrow \infty} (-1)^n \neq 0$, then the series is divergent.

3. The series $\sum_{k=0}^{\infty} (1)^k$ diverges to ∞ .

Note that $s_n = \sum_{k=0}^n 1^n = 1 + 1 + \cdots + 1 = n + 1$. Hence $\lim_{n \rightarrow \infty} s_n = \infty$. So the series diverges to ∞ . Or, since $\lim_{n \rightarrow \infty} 1^n \neq 0$, then the series is divergent.

The above three examples are special cases of geometric series.

4.4 Example. (Geometric Series)

The *geometric series* $\sum_{k=0}^{\infty} x^k$

- converges if $|x| < 1$ and $\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$.
- Diverges to ∞ if $x \geq 1$ and diverges if $x \leq -1$.

Solution. If $x = 1$, then the series $\sum_{k=0}^{\infty} (1)^n$ diverges to ∞ .

Suppose $x \neq 1$, then $s_n = 1 + x + x^2 + \cdots + x^n = \frac{1 - x^{n+1}}{1 - x}$. Hence

$$\lim_{n \rightarrow \infty} s_n \begin{cases} = \frac{1}{1-x}, & \text{if } |x| < 1; \\ = \infty, & \text{if } x > 1; \\ \text{does not exist,} & \text{if } x \leq -1. \end{cases}$$

Another way to show the geometric series is divergent for $|x| \geq 1$ is that $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} x^n \neq 0$.

4.5 Remark. (Geometric Series)

In general, for any $m \in \mathbb{N}$, the series $\sum_{k=m}^{\infty} x^n$ converges if and only if $|x| < 1$ and

$$\sum_{k=m}^{\infty} x^n = \lim_{n \rightarrow \infty} \sum_{k=m}^{n+m} x^n = \lim_{n \rightarrow \infty} (x^m + x^{m+1} + \cdots + x^{m+n}) = x^m \lim_{n \rightarrow \infty} \sum_{k=0}^n x^n = x^m \frac{1}{1-x}.$$

4.6 Property. (Telescoping Series)

The series $\sum_{k=m}^{\infty} a_k$ is called **telescoping** if $a_k = b_{k+1} - b_k$ for a suitable sequence $\{b_k\}_{k \geq m}$ and

$$\sum_{k=m}^{\infty} a_k = \sum_{k=m}^{\infty} (b_{k+1} - b_k) = \lim_{n \rightarrow \infty} b_{n+1} - b_m.$$

Proof. we have

$$s_n = \sum_{k=m}^n a_k = a_m + a_{m+1} + \cdots + a_n = (b_{m+1} - b_m) + (b_{m+2} - b_{m+1}) + \cdots + (b_{n+1} - b_n) = b_{n+1} - b_m.$$

$$\text{Then } \sum_{k=m}^{\infty} a_k = \lim_{n \rightarrow \infty} b_{n+1} - b_m. \quad \square$$

This property show how to construct series that converge to any given number or diverge to ∞ .

4.7 Examples.

1. A typical example.

$$\sum_{k=2}^{\infty} \frac{1}{(k-1)k} = - \sum_{k=2}^{\infty} \left(\frac{1}{k} - \frac{1}{k-1} \right) = - \left(\lim_{n \rightarrow \infty} \frac{1}{n} - \frac{1}{2-1} \right) = 1.$$

2. The condition $\lim_{k \rightarrow \infty} a_k = 0$ is necessary but not sufficient condition for convergence.

$$\sum_{k=1}^{\infty} \ln(1 + \frac{1}{k}) = \sum_{k=1}^{\infty} (\ln(k+1) - \ln k) = \lim_{n \rightarrow \infty} \ln(n+1) - \ln 1 = \infty.$$

4.8 Property. (Linearity Of Series)

Let $\alpha \in \mathbb{R}$, $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ be convergent series. Then

1. $\sum_{n=0}^{\infty} \alpha a_n$ is a convergent series and $\sum_{n=0}^{\infty} \alpha a_n = \alpha \sum_{n=0}^{\infty} a_n$.
2. $\sum_{n=0}^{\infty} (a_n + b_n)$ is a convergent series and $\sum_{n=0}^{\infty} (a_n + b_n) = \sum_{n=0}^{\infty} a_n + \sum_{n=0}^{\infty} b_n$.

Proof. For both items we look at the partial sums

$$\sum_{k=0}^n \alpha a_k = \alpha \sum_{k=0}^n a_k \text{ and } \sum_{k=0}^n (a_k + b_k) = \sum_{k=0}^n a_k + \sum_{k=0}^n b_k,$$

then take the limit of both sides to obtain the result. $\square$

Note that, from passing the limit in $\sum_{k=0}^n (a_k + b_k) = \sum_{k=0}^n a_k + \sum_{k=0}^n b_k$, we have the right to write $\sum_{n=0}^{\infty} (a_n + b_n) = \sum_{n=0}^{\infty} a_n + \sum_{n=0}^{\infty} b_n$ if both series converge, diverge to $+\infty$, diverge to $-\infty$ or if one series converges and the other diverges to $\pm\infty$.

If both series are divergent, then $\sum_{n=0}^{\infty} (a_n + b_n)$ may converges, diverges or diverges to ∞ . For example, the series $\sum_{n=0}^{\infty} (-1)^n$ and $\sum_{n=0}^{\infty} -(-1)^n$ are both divergent, but $\sum_{n=0}^{\infty} ((-1)^n - (-1)^n) = 0$ and $\sum_{n=0}^{\infty} ((-1)^n + (-1)^n)$ diverges.

If one series is convergent and the other is divergent, then the series $\sum_{n=0}^{\infty} (a_n + b_n)$ is divergent.

4.9 Property. If $\sum_{k=0}^{\infty} a_k$ is a convergent series and $\sum_{k=0}^{\infty} b_k$ any series such that $b_k = a_k$ except for at most finitely-many k , then $\sum_{k=0}^{\infty} b_k$ is also convergent.

Proof. Let $s_n = \sum_{k=0}^n a_k$ and $\sigma_n = \sum_{k=0}^n b_k$. Note that $\sigma_n = \sum_{k=0}^n (b_k - a_k) + \sum_{k=0}^n a_k = \sum_{k=0}^{n_0} (b_k - a_k) + s_n$ for some fixed $n_0 \in \mathbb{N}$. Take the limit of both sides to obtain the result. $\square$

From this property one can deduce that a series behavior does not change by adding, modifying or eliminating a finite number of terms.

4.10 Property. If $\sum_{k=0}^{\infty} a_k$ is a **positive term series**, $a_k \geq 0$ for all $k \in \mathbb{N}$, the sequence of partial sum $\{s_n\}_{n \geq 0}$ is an increasing sequence and so the series is either converges (when the sequence of partial sum $\{s_n\}_{n \geq 0}$ is bounded above) or diverges to ∞ .

Proof. We have $s_{n+1} - s_n = a_{n+1} \geq 0$. Then the sequence $\{s_n\}_{n \geq 0}$ is an increasing sequence. $\square$

Note that, to show a positive-term series is convergent, it is sufficient to show the sequence of partial sum is bounded above.

4.11 Definition. The infinite sum $r_n = \sum_{k=n+1}^{\infty} a_k$ is called the ***nth remainder*** or the ***tail*** of the series . If the series $\sum_{k=0}^{\infty} a_k$ converges to s , then $r_n = s - s_n$.

Here is another necessary condition for convergence.

4.12 Property. If $\sum_{k=0}^{\infty} a_k$ is a convergent series, then $\lim_{n \rightarrow \infty} r_n = 0$. Equivalently;

If $\lim_{n \rightarrow \infty} r_n \neq 0$, then the series $\sum_{k=0}^{\infty} a_k$ is divergent.

Proof. Suppose $\sum_{k=0}^{\infty} a_k = s$. Then $\lim_{n \rightarrow \infty} r_n = \lim_{n \rightarrow \infty} (s - s_n) = s - s = 0$. $\square$

4.13 Example. The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges to $+\infty$.

Since $r_n = \sum_{k=n+1}^{\infty} \frac{1}{k} = \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{n+n} + \cdots \geq n \cdot \frac{1}{n+n} \geq \frac{1}{2}$, then $\lim_{n \rightarrow \infty} r_n \neq 0$. So the positive term series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges to $+\infty$.

4.14 Theorem. (Comparison Test)

Let $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ be positive-term series. suppose that there exists $N \in \mathbb{N}$ such that $0 \leq a_k \leq b_k$ for all $k \geq N$.

1. If $\sum_{k=0}^{\infty} b_k$ converges, then $\sum_{k=0}^{\infty} a_k$ converges and $\sum_{k=N}^{\infty} a_k \leq \sum_{k=N}^{\infty} b_k$.
2. If $\sum_{k=0}^{\infty} a_k$ diverges to ∞ then $\sum_{k=0}^{\infty} b_k$ diverges to ∞ .

Proof. 1. We have $a_k \leq b_k$ for all $k \geq N$, then $s_n = \sum_{k=N}^n a_k \leq \sigma_n = \sum_{k=N}^n b_k$. Since σ_n is convergent, then it is bounded and so s_n is bounded above. Therefore the series $\sum_{k=N}^{\infty} a_k$

converges and by taking the limit of both sides in the above inequality, we obtain $\sum_{k=N}^{\infty} a_k \leq \sum_{k=N}^{\infty} b_k$. Since a finite number of terms has no effect on the convergence of a series, then $\sum_{k=0}^{\infty} a_k$ converges.

2. Suppose to the contrary that $\sum_{k=0}^{\infty} b_k$ converges, then by first item we obtain $\sum_{k=0}^{\infty} a_k$ converges.

But we have $\sum_{k=0}^{\infty} a_k$ diverges to ∞ . Thus $\sum_{k=0}^{\infty} b_k$ diverges to ∞ .

□

4.15 Examples.

1. The harmonic series $\sum_{k=1}^{\infty} \frac{1}{k} = \infty$.

Since $\ln(1+x) \leq x$ for any $x > -1$, it follows $0 \leq \ln(1 + \frac{1}{k}) \leq \frac{1}{k}$ for any $k \geq 1$. But from the divergence of the telescoping series $\sum_{k=1}^{\infty} \ln(1 + \frac{1}{k})$ we obtain that the harmonic series diverges to ∞ .

2. The series $\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges and $\sum_{k=1}^{\infty} \frac{1}{k^2} \leq 2$.

Since $0 \leq \frac{1}{k^2} \leq \frac{1}{(k-1)(k)}$ for any $k \geq 2$, and $\sum_{k=2}^{\infty} \frac{1}{(k-1)(k)} = 1$, then the series $\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges and $\sum_{k=1}^{\infty} \frac{1}{k^2} = 1 + \sum_{k=2}^{\infty} \frac{1}{k^2} \leq 1 + \sum_{k=2}^{\infty} \frac{1}{(k-1)(k)} \leq 2$.

3. The comparison test applies only to series with positive terms.

We have $a_k = -\frac{1}{k} \leq b_k = \frac{1}{k^2}$, and the series $\sum_{k=1}^{\infty} b_k$ converges while $\sum_{k=1}^{\infty} a_k$ diverges to $-\infty$.

4. The series $\sum_{k=1}^{\infty} \frac{1}{k^p}$ converges if and only if $p > 1$.

- If $p \leq 0$, then $\lim_{k \rightarrow \infty} \frac{1}{k^p} \neq 0$. Hence the series diverges to ∞ .
- If $0 < p < 1$, then $0 \leq \frac{1}{k} < \frac{1}{k^p}$. Hence the series diverges to ∞ .
- If $p > 2$, then $0 \leq \frac{1}{k^p} < \frac{1}{k^2}$. Hence the series is convergent.
- The case $1 < p < 2$ will be examined later by Cauchy's condensation test or integral test.

4.16 Definition. (Absolute And Conditional Convergence)

The series $\sum_{k=0}^{\infty} a_k$ is said to be **absolutely convergent** if the positive term series $\sum_{k=0}^{\infty} |a_k|$ converges.

However, if $\sum_{k=0}^{\infty} a_k$ converges and $\sum_{k=0}^{\infty} |a_k|$ diverges, then the series $\sum_{k=0}^{\infty} a_k$ is said to be **Conditionally convergent**.

The notion of absolute convergence is useful for series with arbitrary sign.

4.17 Property. (Absolute Convergence Test)

If $\sum_{k=0}^{\infty} a_k$ converges absolutely then it also converges and $\sum_{k=0}^{\infty} a_k \leq \sum_{k=0}^{\infty} |a_k|$.

Proof. Note that $0 \leq a_k + |a_k| \leq 2|a_k|$ for any $k \geq 0$. Since $\sum_{k=0}^{\infty} |a_k|$ converges, then the comparison test tell us that $\sum_{k=0}^{\infty} (a_k + |a_k|)$ converges. Thus, due to linearity of series, $\sum_{k=0}^{\infty} a_k = \sum_{k=0}^{\infty} ((a_k + |a_k|) - |a_k|)$ is convergent. Since $0 \leq a_k + |a_k| \leq 2|a_k|$ for any $k \geq 0$, then by comparison test we obtain $\sum_{k=0}^{\infty} a_k \leq \sum_{k=0}^{\infty} |a_k|$. $\square$

4.18 Remark. (Modified Comparison Test)

Let $\sum_{k=0}^{\infty} b_k$ be a convergent positive-term series. If there exists $N \in \mathbb{N}$ such that $|a_k| \leq b_k$ for all $k \geq N$, then the series (with arbitrary sign) $\sum_{k=0}^{\infty} a_k$ is absolutely convergent and hence convergent.

4.19 Theorem. (Limit Form Of Comparison Test)

Let $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ be positive-term series with $b_n > 0$ for sufficiently large n .

1. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = l > 0$, then the series both converge or both diverge.
2. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum_{k=0}^{\infty} b_k$ converges, then $\sum_{k=0}^{\infty} a_k$ converges.
3. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ and $\sum_{k=0}^{\infty} b_k$ diverges, then $\sum_{k=0}^{\infty} a_k$ diverges to ∞ .

Proof. 1. Choose $0 < \varepsilon = \frac{l}{2} < l$. By definition of limit we have, $\exists n_0 \in \mathbb{N}$ such that,

$$n \geq n_0 \Rightarrow \left| \frac{a_n}{b_n} - l \right| < \varepsilon.$$

Then $0 < \frac{l}{2}b_n < a_n < \frac{3l}{2}b_n$ for $n \geq n_0$. We conclude from comparison test the series both converge or both diverge.

2. Choose $\varepsilon = 1$. By definition of limit we have, $\exists n_0 \in \mathbb{N}$ such that,

$$n \geq n_0 \Rightarrow \left| \frac{a_n}{b_n} \right| < \varepsilon.$$

Then $a_n < b_n$ for $n \geq n_0$. We conclude from comparison test that the convergence of $\sum_{k=0}^{\infty} b_k$ implies the convergence of $\sum_{k=0}^{\infty} a_k$.

3. Choose $M = 1$. By definition of limit we have, $\exists n_0 \in \mathbb{N}$ such that,

$$n \geq n_0 \Rightarrow \frac{a_n}{b_n} > M.$$

Then $a_n > b_n$ for $n \geq n_0$. We conclude from comparison test that the divergence of $\sum_{k=0}^{\infty} b_k$ implies the divergence of $\sum_{k=0}^{\infty} a_k$. $\square$

4.20 Definition. (Landau Notations)

Let $\{a_n\}$ and $\{c_n\}$ be two sequences. We say that

1. $\{a_n\}$ is dominated by $\{c_n\}$, and we write $a_n = O(c_n)$, if there exist a bounded sequence $\{b_n\}$ and $N \in \mathbb{N}$ such that $a_n = b_n c_n$ for any $n \geq N$.
2. $\{a_n\}$ is negligible with respect to $\{c_n\}$, and we write $a_n = o(c_n)$, if there exist a sequence $\{\varepsilon_n\}$ goes to zero as n goes to infinity and an $N \in \mathbb{N}$ such that $a_n = \varepsilon_n c_n$ for any $n \geq N$.
3. $\{a_n\}$ is equivalent to $\{c_n\}$, and we write $a_n \sim c_n$, if there exist a sequence $\{\varepsilon_n\}$ goes to zero as n goes to infinity and an $N \in \mathbb{N}$ such that $a_n = (1 + \varepsilon_n)c_n$ for any $n \geq N$. Equivalently; $a_n \sim c_n$ if and only if $a_n - c_n = o(c_n)$.

Note that $a_n = O(c_n)$ respectively ($a_n = o(c_n)$, $a_n \sim c_n$), if and only if there exist a sequence $\{\mu_n\}$ such that $a_n = \mu_n c_n$ eventually (after a certain rank) and $\{\mu_n\}$ is bounded respectively ($\mu_n \xrightarrow{n \rightarrow \infty} 0$, $\mu_n \xrightarrow{n \rightarrow \infty} 1$).

4.21 Property. Let $\{a_n\}$ and $\{c_n\}$ be two sequences. If $\{c_n\}$ eventually (after a certain rank) never vanishes, then:

1. $a_n = O(c_n) \Leftrightarrow \{\frac{a_n}{c_n}\}$ is bounded.
2. $a_n = o(c_n) \Leftrightarrow \frac{a_n}{c_n} \xrightarrow{n \rightarrow \infty} 0$.
3. $a_n \sim c_n \Leftrightarrow \frac{a_n}{c_n} \xrightarrow{n \rightarrow \infty} 1$.

4.22 Property. Let $\{a_n\}$ and $\{c_n\}$ be two sequences.

1. If $a_n = O(c_n)$ and $\sum_{k=0}^{\infty} c_k$ converges absolutely, then $\sum_{k=0}^{\infty} a_k$ converges absolutely.
2. If $a_n = o(c_n)$ and $\sum_{k=0}^{\infty} c_k$ converges absolutely, then $\sum_{k=0}^{\infty} a_k$ converges absolutely.
3. (Equivalent test)
If $a_n \sim c_n$, then $\sum_{k=0}^{\infty} |a_k|$ and $\sum_{k=0}^{\infty} |c_k|$ have same nature.

Proof. 1. Assume $a_n = O(c_n)$, there exist a bounded sequence $\{b_n\}$ and $N \in \mathbb{N}$ such that $a_n = b_n c_n$ for any $n \geq N$. Hence there exists $M > 0$ such that $|a_n| \leq M|c_n|$ eventually. Since $\sum_{k=0}^{\infty} |c_k|$ converges, then by comparison test $\sum_{k=0}^{\infty} a_k$ converges absolutely.

2. Evident. We use, $a_n = o(c_n) \Rightarrow a_n = O(c_n)$.

3. Evident. We use $a_n \sim c_n \Rightarrow [a_n = O(c_n) \wedge c_n = O(a_n)]$.

□

We give some examples after alternating series test.

4.23 Property. (Cauchy's Condensation Test)

Let $\{a_n\}_{n \geq 1}$ be a positive decreasing sequence of real numbers, the series $\sum_{k=1}^{\infty} a_k$ converges if and only if $\sum_{k=0}^{\infty} 2^k a_{2^k} = a_1 + 2a_2 + 4a_4 + 8a_8 + \cdots$ converges.

Proof. Let $s_n = \sum_{k=1}^n a_k$ and $\sigma_n = \sum_{k=0}^n 2^k a_{2^k}$. Since

$$\begin{aligned} a_2 &\leq a_2 \leq a_1, \\ 2a_4 &\leq a_3 + a_4 \leq 2a_2, \\ 4a_8 &\leq a_5 + a_6 + a_7 + a_8 \leq 4a_4, \\ &\dots\dots\dots \\ 2^n a_{2^{n+1}} &\leq a_{2^n+1} + \cdots + a_{2^{n+1}} \leq 2^n a_{2^n}, \end{aligned}$$

then by adding the inequalities, we obtain

$$\frac{1}{2}(\sigma_{n+1} - a_1) \leq s_{2^{n+1}} - a_1 \leq \sigma_n.$$

We can conclude that the increasing sequences s_n and σ_n are either both bounded above or both unbounded above. Then the two series converge or diverge together. $\square$

One can easily see that $s_n \leq \sigma_n$ for any $n \in \mathbb{N}$. Hence in the proof of the property we may obtain a better inequalities

$$\frac{1}{2}(\sigma_{n+1} - a_1) \leq s_{2^{n+1}} - a_1 \leq \sigma_{2^{n+1}} - a_1.$$

By passing the limit in case of convergence, we obtain $\frac{1}{2}\sigma + \frac{1}{2}a_1 \leq s \leq \sigma$, where $s = \sum_{k=1}^{\infty} a_k$ and $\sigma = \sum_{k=0}^{\infty} 2^k a_{2^k}$.

4.24 Corollary. (Riemann Series)

The series $\sum_{k=1}^{\infty} \frac{1}{k^p}$ converges for $p > 1$ and diverges for $p \leq 1$.

Proof. We discuss two cases according to the values of p .

1. If $p > 0$, then $\{\frac{1}{n^p}\}_{n \geq 1}$ is a positive decreasing sequence. Hence the series $\sum_{k=1}^{\infty} \frac{1}{k^p}$ and the condensed series $\sum_{k=0}^{\infty} 2^k \frac{1}{(2^k)^p} = \sum_{k=0}^{\infty} \frac{1}{(2^{p-1})^k}$ have same nature. But the geometric series $\sum_{k=0}^{\infty} \frac{1}{(2^{p-1})^k}$ converges if and only if $2^{p-1} < 1$, that is, $p > 1$.

2. If $p \leq 0$, then $\lim_{k \rightarrow \infty} \frac{1}{k^p} \neq 0$. So the Riemann series diverges to ∞ .

Thus $\sum_{k=1}^{\infty} \frac{1}{k^p} = \infty$ for $p \leq 1$ and converges for $p > 1$.

□

In case of convergence, that is $p > 1$, the sum is written $\zeta(p) = \sum_{n=1}^{\infty} \frac{1}{n^p}$, and known as Riemann's zeta function. Also the inequalities $\frac{1}{2}\sigma + \frac{1}{2}a_1 \leq s \leq \sigma$ ensures that $\frac{1}{2} \frac{1}{1-1/2^{p-1}} + \frac{1}{2} \leq \zeta(p) \leq \frac{1}{1-1/2^{p-1}}$. For example $1 \leq \zeta(2) \leq 2$ and $\frac{7}{6} \leq \zeta(3) \leq \frac{4}{3}$. One can show that $\zeta(2) = \frac{\pi^2}{6}$, but no closed expression for $\zeta(3)$ is known.

4.25 Corollary. (n^α -Test)

Consider the series $\sum_{k=1}^{\infty} a_k$ with terms of arbitrary sign.

1. If $n^\alpha a_n \xrightarrow{n \rightarrow \infty} \ell \neq 0$, then $\sum_{k=1}^{\infty} a_k$ converges absolutely if and only if $\alpha > 1$.
2. If $n^\alpha a_n \xrightarrow{n \rightarrow \infty} 0$ and $\alpha > 1$, then $\sum_{k=1}^{\infty} a_k$ converges absolutely.
3. If $n^\alpha a_n \xrightarrow{n \rightarrow \infty} \infty$ and $\alpha \leq 1$, then $\sum_{k=1}^{\infty} a_k$ diverges.

Proof. Evident.

1. $n^\alpha a_n \xrightarrow{n \rightarrow \infty} \ell \neq 0 \Rightarrow |a_n| \sim \frac{|\ell|}{n^\alpha}$.
2. $n^\alpha a_n \xrightarrow{n \rightarrow \infty} 0 \Rightarrow a_n = o(\frac{1}{n^\alpha})$.
3. $n^\alpha a_n \xrightarrow{n \rightarrow \infty} \infty \Rightarrow a_n > \frac{1}{n^\alpha}$ eventually, by the definition of limit.

□

4.26 Corollary. (Bertrand Series)

The series $\sum_{k=2}^{\infty} \frac{1}{k^p \ln^q k}$ converges if and only if $\left((p > 1) \text{ or } (p = 1 \text{ and } q > 1) \right)$.

Proof. We discuss three cases according to the values of p .

1. If $p > 1$, take $\alpha \in]1, p[$ for example $\alpha = \frac{1+p}{2}$. Hence $\lim_{n \rightarrow \infty} n^\alpha \frac{1}{n^p \ln^q n} = \lim_{n \rightarrow \infty} \frac{1}{n^{(p-\alpha)} \ln^q n} = 0$, for any $q \in \mathbb{R}$. Hence, by the n^α -test, the series converges.
2. If $p < 1$, take $\alpha \in]p, 1[$ for example $\alpha = \frac{1+p}{2}$. Hence $\lim_{n \rightarrow \infty} n^\alpha \frac{1}{n^p \ln^q n} = \lim_{n \rightarrow \infty} \frac{1}{n^{(p-\alpha)} \ln^q n} = \infty$, for any $q \in \mathbb{R}$. Hence, by the n^α -test, the series diverges.
3. If $p = 1$, we have $\left(\frac{1}{x \ln^q x} \right)' = \frac{-\ln^{q-1}(x)(q+\ln x)}{(x \ln^q x)^2} < 0$ for sufficiently large x . Hence, in this case the general term of Bertrand series is a positive decreasing sequence for sufficiently large n and so the condensed series $\sum_{k=0}^{\infty} \frac{1}{k^q \ln^q 2} = \frac{1}{\ln^q 2} \sum_{k=0}^{\infty} \frac{1}{k^q}$ have the same nature as Bertrand series $\sum_{k=2}^{\infty} \frac{1}{k \ln^q k}$. Thus, for $p = 1$, Bertrand series converges if and only if $q > 1$.

□

4.27 Theorem. (Integral Test)

Let f be continuous, positive and decreasing on $[k_0, \infty)$, for $k_0 \in \mathbb{N}$. Then the integral $\int_{k_0}^{\infty} f(x) dx$ is convergent if and only if the series $\sum_{k=k_0}^{\infty} f(k)$ is convergent. Moreover, we have

$$\sum_{k=k_0+1}^{\infty} f(k) \leq \int_{k_0}^{\infty} f(x) dx \leq \sum_{k=k_0}^{\infty} f(k).$$

Proof. Since f is decreasing on $[k_0, \infty)$, for $k \geq k_0$ we have

$$f(k) \leq f(x) \leq f(k-1), \quad \forall x \in [k-1, k],$$

and as the function f is continuous on $[k-1, k]$, hence integrable on $[k-1, k]$,

$$f(k) \leq \int_{k-1}^k f(x) dx \leq f(k-1).$$

Then for all $n \in \mathbb{N}$ with $n > k_0$ we obtain

$$\sum_{k=k_0+1}^n f(k) \leq b_n := \int_{k_0}^n f(x) dx \leq \sum_{k=k_0+1}^n f(k-1) = \sum_{k=k_0}^{n-1} f(k).$$

If the integral converges, then the sequence $\{b_n\}$ is bounded. So the n -th partial sum $\{s_n\}$ is increasing, bounded above and thus the series converges.

If the integral diverges, then from the second inequality we obtain that the series diverges. □

Typical applications of the integral test are **Riemann series** and **Bertrand series**.

Note that, under the assumptions of the above theorem, if $\sum_{k=k_0}^{\infty} f(k)$ converges then for all $n > k_0$,

$$\int_{n+1}^{\infty} f(x) dx \leq r_n \leq \int_n^{\infty} f(x) dx, \text{ and } s_n + \int_{n+1}^{\infty} f(x) dx \leq s := s_n + r_n \leq s_n + \int_n^{\infty} f(x) dx,$$

4.28 Example. Suppose we want to estimate the precision of the sum of $\sum_{k=1}^{\infty} \frac{1}{k^3}$ computed up to the first 10 terms. Since $s_{10} = 1.197532 \dots$, and $\int_n^{\infty} \frac{1}{x^3} dx = \frac{1}{2n^2}$, then $s_{10} + \int_{11}^{\infty} f(x) dx \leq s := s_{10} + R_{10} \leq s_{10} + \int_{10}^{\infty} f(x) dx$, gives $1.201664 \leq s \leq 1.202532$. The exact value for s is $1.202057 \dots$

4.2 Alternating Series.**4.29 Theorem. (Leibniz's Alternating Series Test)**

An Alternating series $\sum_{k=0}^{\infty} (-1)^k b_k$ with $b_k > 0$ converges if the sequence $\{b_k\}_{k \geq 0}$ is decreasing and $\lim_{k \rightarrow \infty} b_k = 0$. Moreover, for all $n \geq 0$ one has

$$|r_n| = |s - s_n| \leq b_{n+1} \quad \text{and} \quad s_{2n+1} \leq s \leq s_{2n}.$$

Proof. Since the sequence $\{b_k\}_{k \geq 0}$ is a decreasing sequence,

$$s_{2n} = s_{2n-2} - b_{2n-1} + b_{2n} \leq s_{2n-2} \quad \text{and} \quad s_{2n+1} = s_{2n-1} + b_{2n} - b_{2n+1} \geq s_{2n-1}.$$

Moreover,

$$s_{2n} = s_{2n-1} + b_{2n} \geq s_{2n-1} \geq \cdots \geq s_1 \quad \text{and} \quad s_{2n+1} = s_{2n} - b_{2n+1} \leq s_{2n} \leq \cdots \leq s_0.$$

Therefore $\{s_{2n}\}_{n \geq 0}$ and $\{s_{2n+1}\}_{n \geq 0}$ are monotone bounded sequences. Thus

$$\lim_{n \rightarrow \infty} s_{2n} = \inf_{n \geq 0} s_{2n} := r \quad \text{and} \quad \lim_{n \rightarrow \infty} s_{2n+1} = \sup_{n \geq 0} s_{2n+1} := l.$$

As, $r - l = \lim_{n \rightarrow \infty} (s_{2n} - s_{2n+1}) = \lim_{n \rightarrow \infty} b_{2n+1} = 0$, then the series converges and

$$s_{2n+1} \leq s = r = l \leq s_{2n} \quad \text{for all } n \geq 0.$$

In addition,

$$0 \leq s - s_{2n+1} \leq s_{2n+2} - s_{2n+1} = b_{2n+2} \quad \text{and} \quad 0 \leq s_{2n} - s \leq s_{2n} - s_{2n+1} = b_{2n+1}.$$

So $|r_n| = |s - s_n| \leq b_{n+1}$ for all $n \geq 0$. □

The series $\sum_{k=0}^{\infty} (-1)^k b_k$ with $b_k > 0$ is called an alternating series because the signs of the terms alternate between $+$ and $-$.

4.30 Examples.

1. The series $\sum_{k=1}^{\infty} \frac{(-1)^k}{k^p}$ converges for $p > 0$ and diverges for $p \leq 0$.
2. The series $\sum_{k=1}^{\infty} \frac{(-1)^k}{k}$ is conditionally convergent.
3. Let us approximate the sum $\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}$ to the third digit, that means with a margin less than 10^{-3} . The series converges by Leibniz's Test. As $b_7 = \frac{1}{5040} < 10^{-3}$, then $|s - s_6| \leq b_7 < 10^{-3}$. But $s_6 = 0.368056 \cdots$, the estimate $s \approx 0.368$ is correct up to third place.
4. The condition the sequence $\{b_k\}_{k \geq 0}$ is decreasing in Leibniz's test is only sufficient. In fact, for $k \geq 2$ let

$$b_k = \begin{cases} \frac{1}{k}, & k \text{ even;} \\ \frac{k-1}{k^2}, & k \text{ odd.} \end{cases}$$

We have $b_k > 0$, $\lim_{k \rightarrow \infty} b_k = 0$. The series $\sum_{k=2}^{\infty} (-1)^k b_k = \sum_{k=2}^{\infty} (-1)^k \frac{1}{k} + \sum_{\substack{k \geq 3 \\ k \text{ odd}}} \frac{1}{k^2}$ converges, for

the two series to the right converge. But the sequence is not decreasing since $b_k > b_{k+1}$ for k even and $b_k < b_{k+1}$ for k odd.

5. Consider the series $\sum_{n=1}^{\infty} \sin(\frac{(-1)^n}{n})$. We have $\sin(\frac{(-1)^n}{n}) = \frac{(-1)^n}{n} + o(\frac{1}{n^2})$. Since $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is convergent, by alternating series test, and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is absolutely convergent, by Riemann examples, then the series $\sum_{n=1}^{\infty} \sin(\frac{(-1)^n}{n})$ is convergent. Note that we cannot use equivalent test because $\frac{(-1)^n}{n}$ does not have constant sign eventually.
6. Consider the series $\sum_{n=1}^{\infty} \sin(\frac{1}{n})$. We have $\sin(\frac{1}{n}) \sim \frac{1}{n} \geq 0$ eventually. Since $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent, then $\sum_{n=1}^{\infty} \sin(\frac{1}{n})$ is divergent

4.3 Ratio And Root Test.

4.31 Property. (Ratio Test)

Let a_n be a sequence such that $a_n \neq 0$ eventually.

1. If there exists a real $r < 1$ such that $\left| \frac{a_{n+1}}{a_n} \right| \leq r$ eventually, then $\sum_{k=k_0}^{\infty} a_k$ converges absolutely.
2. If $\left| \frac{a_{n+1}}{a_n} \right| \geq 1$ eventually, then $\sum_{k=k_0}^{\infty} a_k$ diverges.

Proof. Note that, if u_n and v_n are eventually strictly positive and $\frac{u_{n+1}}{u_n} \leq \frac{v_{n+1}}{v_n}$, then $u_n = O(v_n)$. Because the sequence $\frac{u_{n+1}}{v_{n+1}}$ is decreasing and bounded below by zero, hence convergent and bounded.

1. Evident. We use, $\left(\left| \frac{a_{n+1}}{a_n} \right| \leq r = \frac{r^{n+1}}{r^n} \right) \Rightarrow |a_n| = O(r^n)$.
2. Evident. We use, $\left| \frac{a_{n+1}}{a_n} \right| \geq \frac{1^{n+1}}{1^n} \Rightarrow 1 = O(|a_n|) \Rightarrow \lim_{n \rightarrow \infty} a_n \neq 0$.

□

4.32 Theorem. (d'Alembert's Ratio Test)

Let a_n be a sequence such that $a_n \neq 0$ eventually and $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \ell$ exists, finite or infinite.

1. If $\ell < 1$, the series $\sum_{k=k_0}^{\infty} a_k$ converges absolutely.
2. If $\ell > 1$, the series $\sum_{k=k_0}^{\infty} a_k$ diverges.
3. If $\ell = 1$, the test gives no information.

Proof. 1. If $\ell < 1$, then choose $\ell < r < 1$. Since $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \ell$, there exists an integer N such that $\left| \frac{a_{n+1}}{a_n} \right| < r$ for all $n \geq N$. So we conclude from previous property that the series $\sum_{k=k_0}^{\infty} a_k$ converges absolutely.

2. If $\ell > 1$, there exists an integer N such that $\left| \frac{a_{n+1}}{a_n} \right| > 1$ for all $n \geq N$. So we conclude from previous property the the series $\sum_{k=k_0}^{\infty} a_k$ diverges.

3. The Riemann series for all values of p yields the limit $\ell = 1$.

□

4.33 Theorem. (Cauchy's Root Test)

Let a_n be a sequence such that $a_n \neq 0$ eventually and $\lim_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} = \ell$ exists, finite or infinite.

1. If $\ell < 1$, the series $\sum_{k=k_0}^{\infty} a_k$ converges absolutely.
2. If $\ell > 1$, the series $\sum_{k=k_0}^{\infty} a_k$ diverges.
3. If $\ell = 1$, the test gives no information.

Proof. 1. If $\ell < 1$, then choose $\ell < r < 1$. Since $\lim_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} = \ell$, there exists an integer N such that $|a_n| < r^n$ for all $n \geq N$. Hence by comparing with the geometric series $\sum_{k=k_0}^{\infty} r^k$, the absolute convergence of $\sum_{k=k_0}^{\infty} a_k$ follows.

2. If $\ell > 1$, there exists an integer N such that $|a_n| > 1$ for all $n \geq N$ and hence the n -th term does not approach zero.

3. The Riemann series for $p = 1$ and $p = 2$ yields the limit $\ell = 1$. □

4.4 Abel's Summation By Parts.

There are numerical series which are convergent but not absolutely convergent. These are often quite subtle to handle. We give some tests which are useful to deal with such series. The basic tool for these tests is the following Abel's summation by parts formula.

4.34 Theorem. (Abel's Summation By Parts Formula)

Let a_n and b_n be two sequences and $A_n := \sum_{k=1}^n a_k$. We then have the identity

$$\sum_{k=m+1}^n a_k b_k = A_n b_{n+1} - A_m b_{m+1} - \sum_{k=m+1}^n A_k (b_{k+1} - b_k).$$

It follows that $\sum_{k=m+1}^n a_k b_k$ converges if the sequences $\{A_n b_{n+1}\}$ and $\sum_{k=m+1}^n A_k (b_{k+1} - b_k)$ converge.

Comments; Use the observation $\Delta A_k = A_{k+1} - A_k = a_k$, the identity reads

$$\sum_{k=m+1}^n (\Delta A_k) b_k = A_n b_{n+1} - A_m b_{m+1} - \sum_{k=m+1}^n A_k (\Delta b_k).$$

A particular case $m = 0$ of the summation by parts formula reads

$$\sum_{k=1}^n (\Delta A_k) b_k = A_n b_{n+1} - \sum_{k=1}^n A_k (\Delta b_k).$$

Proof. Let $A_0 = 0$. We have

$$\sum_{k=m+1}^n a_k b_k = \sum_{k=m+1}^n (A_k - A_{k-1}) b_k = \sum_{k=m+1}^n A_k b_k - \sum_{k=m+1}^n A_{k-1} b_k.$$

By shifting index for the second summation on the right,

$$\sum_{k=m+1}^n A_{k-1}b_k = \sum_{k=m}^{n-1} A_k b_{k+1} = A_m b_{m+1} - A_n b_{n+1} + \sum_{k=m+1}^n A_k b_{k+1},$$

the summation by parts follows. $\square$

4.35 Theorem. (Dirichlet's Test)

The series $\sum_{k=1}^{\infty} a_k b_k$ is convergent if the sequence $\{A_n\}$ where $A_n := \sum_{k=1}^n a_k$ is bounded and $\{b_n\}$ is decreasing to zero.

Proof. In view of the summation by parts formula, it is sufficient to show $\{A_n b_{n+1}\}$ and $\sum_{k=1}^n A_k(b_{k+1} - b_k)$ converge. The first sequence $\{A_n b_{n+1}\}$ converges to zero, since it is the product of bounded sequence by another which converges to zero. For the second one $\sum_{k=1}^n A_k(b_{k+1} - b_k)$; Assume $|A_n| \leq M$ for all n and for some $M > 0$. Then $|A_k(b_{k+1} - b_k)| \leq M(b_{k+1} - b_k)$. The series $\sum_{k=1}^{\infty} (b_{k+1} - b_k) = -b_1$ is telescoping, then the convergence of $\sum_{k=1}^n A_k(b_{k+1} - b_k)$ follows from the comparison test. $\square$

4.36 Theorem. (Dedekind's Test)

The series $\sum_{k=1}^{\infty} a_k b_k$ is convergent if the sequence $\{A_n\}$ where $A_n := \sum_{k=1}^n a_k$ is bounded, $b_n \xrightarrow{n \rightarrow \infty} 0$ and $\sum_{k=1}^{\infty} |b_{k+1} - b_k|$ is convergent.

Proof. In view of the summation by parts formula, it is sufficient to show $\{A_n b_{n+1}\}$ and $\sum_{k=1}^n A_k(b_{k+1} - b_k)$ converge. The first sequence $\{A_n b_{n+1}\}$ converges to zero, since it is the product of bounded sequence by another which converges to zero. For the second one $\sum_{k=1}^n A_k(b_{k+1} - b_k)$; Assume $|A_n| \leq M$ for all n and for some $M > 0$. Then $|A_k(b_{k+1} - b_k)| \leq M|b_{k+1} - b_k|$. Since the series $\sum_{k=1}^{\infty} |b_{k+1} - b_k|$ is convergent, then the convergence of $\sum_{k=1}^n A_k(b_{k+1} - b_k)$ follows from the comparison test. $\square$

4.37 Theorem. (Abel's Test)

The series $\sum_{k=1}^{\infty} a_k b_k$ is convergent if $\sum_{k=1}^{\infty} a_k$ is convergent and $\{b_n\}$ is monotone and bounded.

Proof. In view of the summation by parts formula, it is sufficient to show $\{A_n b_{n+1}\}$ and $\sum_{k=1}^n A_k(b_{k+1} - b_k)$ converge. The first sequence $\{A_n b_{n+1}\}$ converges, since it is the product of convergent sequences. For the second one $\sum_{k=1}^n A_k(b_{k+1} - b_k)$; Assume $|A_n| \leq M$ for all n and for some $M > 0$ and $\{b_n\}$

is monotone increasing. Then $|A_k(b_{k+1} - b_k)| \leq M(b_{k+1} - b_k)$. The series $\sum_{k=1}^{\infty} (b_{k+1} - b_k) = \lim_{n \rightarrow \infty} b_{n+1} - b_1$ is telescoping, then the convergence of $\sum_{k=1}^n A_k(b_{k+1} - b_k)$ follows from the comparison test. $\square$

4.5 Rearrangements

The purpose of this section is to expose one of the benefits of working with absolutely convergent series.

4.38 Definition. If $\sum_{k=1}^{\infty} a_k$ is a series. Then $\sum_{k=1}^{\infty} b_k$ is a rearrangement of $\sum_{k=1}^{\infty} a_k$, if there is one-to-one and onto function $\phi : \mathbb{N} \mapsto \mathbb{N}$ such that $b_n = a_{\phi(n)}$. That is $b_n = a_{\phi(n)}$, where $\phi(1), \phi(2), \dots$ is any sequence of positive integers in which every positive integer appears once and only once.

We want to show that the sum of an absolutely convergent series can be rearranged in an arbitrary way without changing the sum.

4.39 Theorem. If $\sum_{k=1}^{\infty} a_k$ is absolutely convergent, then every rearrangement $\sum_{k=1}^{\infty} b_k$ is also absolutely convergent. Moreover, $\sum_{k=1}^{\infty} a_k = \sum_{k=1}^{\infty} b_k$ and $\sum_{k=1}^{\infty} |a_k| = \sum_{k=1}^{\infty} |b_k|$.

Proof. Let $s := \sum_{k=1}^{\infty} a_k$. Let $\varepsilon > 0$ and $n_0 \in \mathbb{N}$ such that $\sum_{k=n_0}^{\infty} |a_k| < \frac{\varepsilon}{2}$. Let $N = \max\{\phi(1), \dots, \phi(n_0)\}$, then $\{a_1, a_2, \dots, a_{n_0}\} \subset \{b_1, b_2, \dots, b_N\}$. Hence for $n > N$,

$$\left| \sum_{k=1}^n a_k - \sum_{k=1}^n b_k \right| \leq \sum_{k=n_0}^{\infty} |a_k| < \frac{\varepsilon}{2}.$$

Consequently,

$$\begin{aligned} \left| \sum_{k=1}^{\infty} a_k - \sum_{k=1}^n b_k \right| &\leq \left| \sum_{k=1}^{\infty} a_k - \sum_{k=1}^n a_k \right| + \left| \sum_{k=1}^n a_k - \sum_{k=1}^n b_k \right| \\ &< \sum_{k=n+1}^{\infty} |a_k| + \frac{\varepsilon}{2} \leq \sum_{k=n_0}^{\infty} |a_k| + \frac{\varepsilon}{2} \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Thus $\sum_{k=1}^{\infty} b_k$ is convergent and $\sum_{k=1}^{\infty} a_k = \sum_{k=1}^{\infty} b_k$. Similarly, we can obtain $\sum_{k=1}^{\infty} |b_k|$ is convergent and $\sum_{k=1}^{\infty} |a_k| = \sum_{k=1}^{\infty} |b_k|$. $\square$

This theorem is quite useful in finding the sum of a series which is absolutely convergent.

4.40 Example. The series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ is absolutely convergent and $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$. Then by rearranging the terms of the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$, we obtain

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \sum_{n=1}^{\infty} \frac{1}{n^2} - 2 \sum_{n=1}^{\infty} \frac{1}{(2n)^2} = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{12}.$$

It can be shown that, if every rearrangement of $\sum_{k=1}^{\infty} a_k$ is convergent, then $\sum_{k=1}^{\infty} a_k$ is absolutely convergent.

4.41 Theorem. (Riemann Rearrangement Theorem)

If $\sum_{k=1}^{\infty} a_k$ is conditionally convergent, then $\sum_{k=1}^{\infty} a_k$ can be rearranged to have any real number as its sum, to be divergent to ∞ , and to be divergent to $-\infty$.

4.6 Cauchy Product.

Consider the geometric series, $\sum_{k=1}^{\infty} \frac{1}{2^k} = 2$, $\sum_{k=1}^{\infty} \frac{1}{3^k} = \frac{3}{2}$ and $\sum_{k=1}^{\infty} \frac{1}{6^k} = \frac{6}{5}$. Then

$$\sum_{k=1}^{\infty} \frac{1}{2^k} \frac{1}{3^k} \neq \sum_{k=1}^{\infty} \frac{1}{2^k} \sum_{k=1}^{\infty} \frac{1}{3^k}.$$

One way to multiply series and preserve the above property is the so-called **Cauchy product**, formally

$$\begin{aligned} \sum_{n=0}^{\infty} a_n \sum_{n=0}^{\infty} b_n &= (a_0 + a_1 + a_2 + \cdots)(b_0 + b_1 + b_2 + \cdots) \\ &= (a_0 b_0) + (a_0 b_1 + a_1 b_0) + (a_0 b_2 + a_1 b_1 + a_2 b_0) + \cdots \\ &= \sum_{n=0}^{\infty} c_n, \end{aligned}$$

where,

$$c_n = a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0 = \sum_{k=0}^n a_k b_{n-k}.$$

4.42 Definition. Let $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ be two series. The series $\sum_{n=0}^{\infty} c_n$, where $c_n := \sum_{k=0}^n a_k b_{n-k}$, is their **Cauchy product**.

In general the Cauchy product of two convergent series may not be convergent. Take $a_n = b_n = \frac{(-1)^n}{\sqrt{n+1}}$. Then the n -th term of the Cauchy product is $c_n = (-1)^n \sum_{k=0}^n \frac{1}{\sqrt{(n-k+1)(k+1)}}$.

For $k \leq n$, we have $(n-k+1)(k+1) = (\frac{n}{2} + 1)^2 - (\frac{n}{2} - k)^2 \leq (\frac{n}{2} + 1)^2$. Hence $|c_n| \geq \frac{2(n+1)}{n+2} \rightarrow 2$. It follows that the Cauchy product is not convergent.

4.43 Theorem. (Mertens Theorem)

Let $\sum_{n=0}^{\infty} a_n$ be absolutely convergent, $\sum_{n=0}^{\infty} b_n$ be convergent and define $c_n := \sum_{k=0}^n a_k b_{n-k}$. Then $\sum_{n=0}^{\infty} c_n$ is convergent and we have

$$\sum_{n=0}^{\infty} a_n \sum_{n=0}^{\infty} b_n = \sum_{n=0}^{\infty} c_n.$$

Proof. Let $A = \sum_{n=0}^{\infty} a_n$, $B = \sum_{n=0}^{\infty} b_n$, $A_n = \sum_{k=0}^n a_k$, $B_n = \sum_{k=0}^n b_k$, $C_n = \sum_{k=0}^n c_k$ and $D_n = B - B_n$. Note that,

$$\begin{aligned} C_n &= a_0 B_n + a_1 B_{n-1} + \cdots + a_n B_0 \\ &= a_0(-B + B_n) + a_1(-B + B_{n-1}) + \cdots + a_n(-B + B_0) + B \left(\sum_{k=0}^n a_k \right) \\ &= A_n B - R_n, \end{aligned}$$

where $R_n = a_0 D_n + a_1 D_{n-1} + \cdots + a_n D_0$. Let $\alpha = \sum_{n=0}^{\infty} |a_n|$. Since $D_n \rightarrow 0$, then $\{D_n\}$ is bounded, say $|D_n| < D$ for all n . Now, for $\varepsilon > 0$ given, there exists $N \in \mathbb{N}$ such that $\alpha = \sum_{n=N}^{\infty} |a_n| < \varepsilon$ and $|D_n| < \varepsilon$ for all $n > N$. Then, for $n \geq 2N$ we have

$$|R_n| \leq (|a_0| + \cdots + |a_{n-N}|)\varepsilon + (|a_{n-N+1}| + \cdots + |a_n|)D \leq \alpha\varepsilon + D\varepsilon.$$

Hence $R_n \rightarrow 0$. Since $A_n B \rightarrow AB$, then the result follows from passing to the limit in $C_n = A_n B - R_n$. $\square$

4.7 Exercises

Exercise 1. Express $u_k = \frac{4}{k(k+2)}$ in partial fractions and hence find the sum

$$S_n = \sum_{k=1}^n u_k.$$

Deduce the nature of the series $\sum_{n=0}^{\infty} u_n$.

Exercise 2. Find the value to which each of the following series converges:

$$a) \sum_{n=0}^{\infty} \frac{2}{4^n}; \quad b) \sum_{n=1}^{\infty} \frac{(-1)^n 3^{n-1}}{4^n}; \quad c) \sum_{n=2}^{\infty} \frac{1}{n^2 - n}.$$

Exercise 3. Apply the comparison test, find the nature of the series $\sum_{n=1}^{\infty} u_n$, where

$$u_n = \begin{cases} \frac{1}{n_1} & \text{for } n \text{ even} \\ \frac{1}{\sqrt{n}} & \text{for } n \text{ odd} \end{cases}$$

Exercise 4. Determine the nature of the following series:

$$a) \sum_{n=1}^{\infty} \frac{1}{n^n}; \quad b) \sum_{n=1}^{\infty} \left(\frac{n+1}{n}\right)^n; \quad c) \sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^n.$$

Exercise 5. Apply, if possible, the integral test on the following series. If the test cannot be applied, state why.

$$a) \sum_{n=1}^{\infty} \frac{n}{n+2}; \quad b) \sum_{n=1}^{\infty} \frac{e^n}{(e^n + 1)^2}; \quad c) \sum_{n=1}^{\infty} \sin\left(\frac{\pi n}{2}\right); \quad d) \sum_{n=0}^{\infty} \frac{\ln(n+2)}{n+2}.$$

Exercise 6. Let $p \in \mathbb{N}$ and $a \in \mathbb{R}$ satisfies $0 < a < 1$. Use the ratio test, prove that the series

$$\sum_{n=0}^{\infty} n^p a^n$$

converges. Deduce the value of the following limit

$$\lim_{n \rightarrow \infty} n^p a^n.$$

Exercise 7. Consider the series $\sum_{n=1}^{\infty} u_n$ with positive terms $u_n \geq 0$ and $\alpha > \frac{1}{2}$. Suppose that

$\sum_{n=1}^{\infty} u_n$ converges.

(i) Prove that the series $\sum_{n=1}^{\infty} (u_n)^2$ is convergent.

(ii) Prove that the series $\sum_{n=1}^{\infty} \left(u_n - \frac{1}{n^\alpha}\right)^2$ is convergent

(iii) Deduce the nature of the series $\sum_{n=1}^{\infty} \frac{u_n}{n^\alpha}$.

Exercise 8. Let $u_n = \ln \left(\frac{n+2}{n} \right)$. Find the n -th partial sum

$$S_n = \sum_{k=1}^n u_k.$$

Deduce the nature of the series $\sum_{n=1}^{\infty} u_n$.

Exercise 9. Determine the nature of the following series:

$$a) \sum_{n=1}^{\infty} \frac{\arctan n}{n}; \quad b) \sum_{n=1}^{\infty} \frac{\ln n}{\sqrt{n}}; \quad c) \sum_{n=0}^{\infty} \frac{3^n - 4^n}{5^n}; \quad d) \sum_{n=1}^{\infty} \frac{3^{n+1} - n}{n3^n}.$$

Exercise 10. Same question for the series $\sum_{n=1}^{\infty} u_n$ with terms

$$a) u_n = \frac{2^n}{n!}; \quad b) u_n = \frac{n^n}{n!}; \quad c) u_n = \frac{n!}{(2n)^n}; \quad d) u_n = \left(\frac{3n+1}{4n-3} \right)^n;$$

$$e) u_n = n^{\frac{1}{n}}; \quad f) u_n = \frac{1}{\sqrt{n+1} - \sqrt{n}}; \quad g) u_n = \frac{1}{\sqrt{n} + n};$$

$$h) u_n = 1 - \frac{(-1)^n}{n}; \quad i) u_n = \sin \left(\frac{\pi}{2n^{\frac{3}{2}}} \right); \quad j) u_n = \ln n e^{-n};$$

$$k) u_n = \frac{1}{\ln(\ln(n+1))}; \quad l) u_n = \frac{1}{n \left(1 + \frac{1}{\ln(n+1)} \right)}; \quad m) u_n = \left(\frac{n+1}{2n} \right)^{n^2}.$$

Exercise 11. Determine whether each of the following series converges conditionally, converges absolutely, or diverges.

$$a) \sum_{n=1}^{\infty} (-1)^n \frac{2^n}{e^n - 1}; \quad b) \sum_{n=1}^{\infty} (-1)^n \frac{n!}{(2n)!}; \quad c) \sum_{n=1}^{\infty} \frac{(-1)^n \ln n}{n};$$

$$d) \sum_{n=2}^{\infty} (-1)^n \frac{n}{(\ln n)^2}; \quad e) \sum_{n=1}^{\infty} \frac{(-1)^n}{(\sin n)^2}; \quad f) \sum_{n=1}^{\infty} (-1)^n \frac{e^{\frac{1}{n}}}{\arctan n}.$$

Exercise 12. Euler proved that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Deduce the values of the following series:

$$a) \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}; \quad b) \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}.$$

Exercise 13. Study the nature of the following series:

$$\sum_{n=0}^{\infty} (-1)^n \frac{\sqrt{n}}{n+1}.$$

Exercise 14. Same question for the series

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \cos\left(\frac{2\pi n}{3}\right).$$

Exercise 15. Establish for what values of $\alpha \in \mathbb{R}$, the series

$$\sum_{n=0}^{\infty} \frac{\alpha^{n+1} 2^{-n}}{2n+1}$$

is convergent.